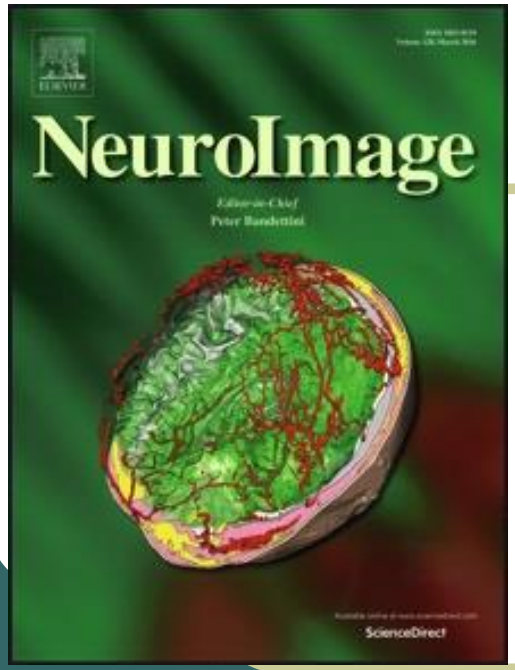
Introduction to Neuroengineering and Neuroinformatics

神經工程暨腦資訊概論



Li-Fen Chen

2026.10.13



What is Neuroengineering

- Neuroengineering (or neural engineering) is an interdisciplinary branch of biomedical engineering that combines *engineering* principles with *neuroscience* to *understand*, *repair*, *replace*, *enhance*, or *exploit* the properties of neural systems.
- Core Purpose
  - **Bridge Living and Artificial Worlds:** It designs direct interfaces between electronic, mechanical, or computational devices and living neural tissue (the brain, spinal cord, and peripheral nerves).
  - **Quantitative Focus:** Unlike traditional neuroscience (which focuses on observation and theory), neuroengineering applies quantitative methods, signal processing, and mathematical modeling to manipulate and control neural activity.

(From Google AI 摘要)

Key Application Areas

- **Brain-Computer Interfaces (BCIs):**
  - Creating direct communication pathways between the brain and external devices, such as letting a paralyzed patient control a robotic arm or computer cursor.
- **Neuromodulation:**
  - Using implantable or non-invasive energy (like electricity, magnetism, or ultrasound) to alter nerve activity. A common example is **deep brain stimulation (DBS)**, which treats movement disorders like Parkinson's disease.
- **Neuroprosthetics:**
  - Developing artificial devices to replace lost sensory or motor functions, such as cochlear implants for hearing, retinal implants for sight, or mind-controlled prosthetic limbs.
- **Neuroimaging & Data Analysis:**
  - Building hardware and software systems to map, record, and decode brain activity using tools like EEG, fMRI, and machine learning.
- **Neural Tissue Engineering:**
  - Investigating ways to physically repair or regrow damaged nerves and brain tissue following a traumatic injury or stroke.

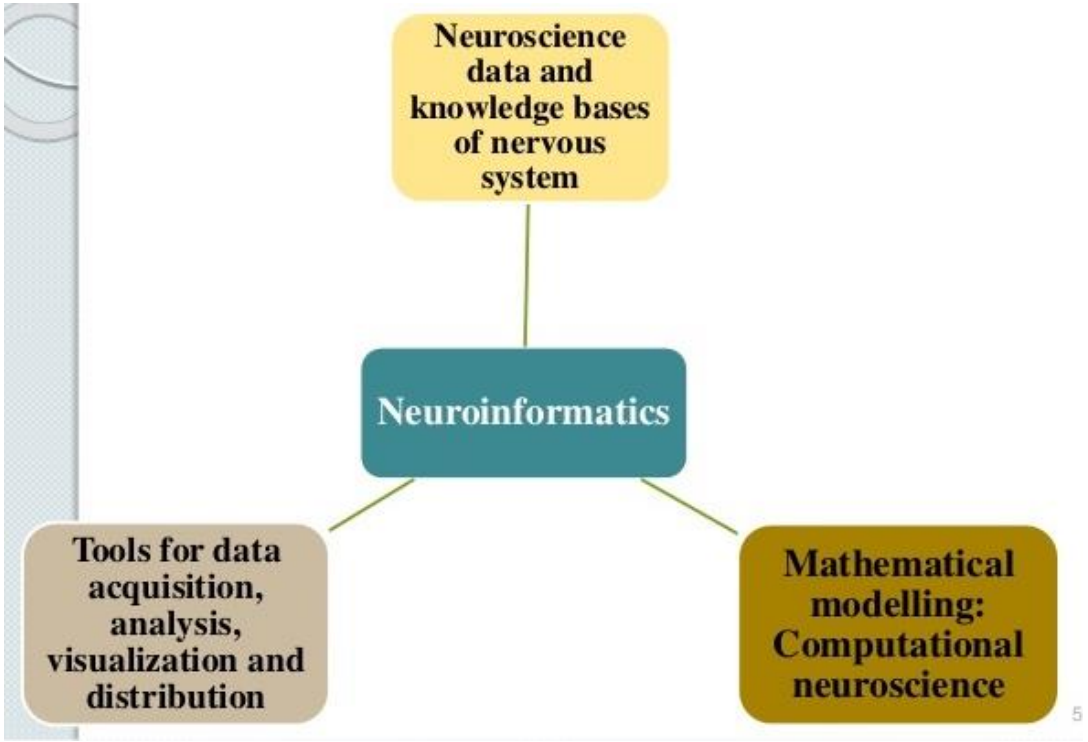
(From Google AI 摘要)

Example

- *Hugh Herr: The new bionics that let us run, climb and dance*
  - https://www.youtube.com/watch?v=CDsNZJTWw0w
- 2026 諾貝爾生醫獎
  - **Controlling the Brain with Light (Karl Deisseroth, Stanford University)**
  - https://www.youtube.com/watch?v=C8bPbHuOZXg
  - https://www.youtube.com/watch?v=bggKUbIN0Ww

What is Neuroinformatics

- **Neuroinformatics** is an interdisciplinary field that merges neuroscience with computer science, information technology, and data science to organize, analyze, model, and share complex data about the nervous system.



Fundamental goals of neuroinformatics

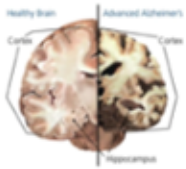
- **Neuroinformatics** is the mean to connect neuroscience, medical science, information technology and computer science (ICT).

ICT-based brain research – aims and implications

• organise and integrate neuroscience data



• accelerate our quest for understanding the brain



• support neuromedicine, understanding brain diseases

• develop brain-inspired future computing technologies, brain-like intelligent systems



• promote education, deliver multi-disciplinary training.

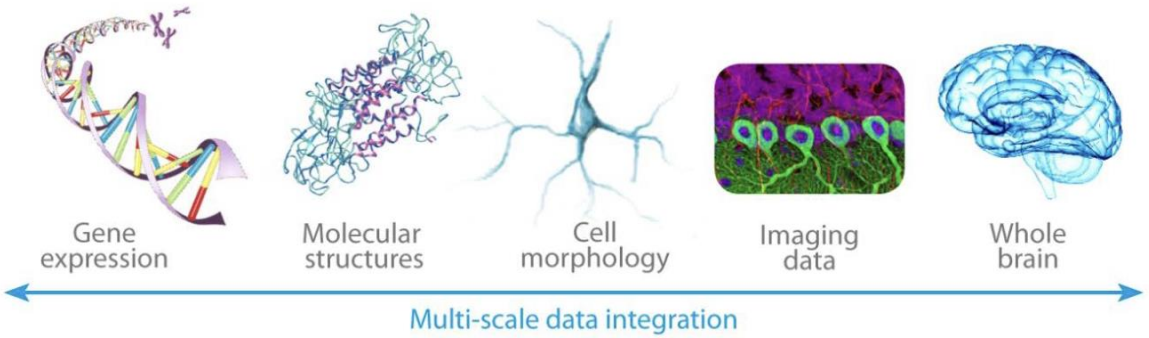


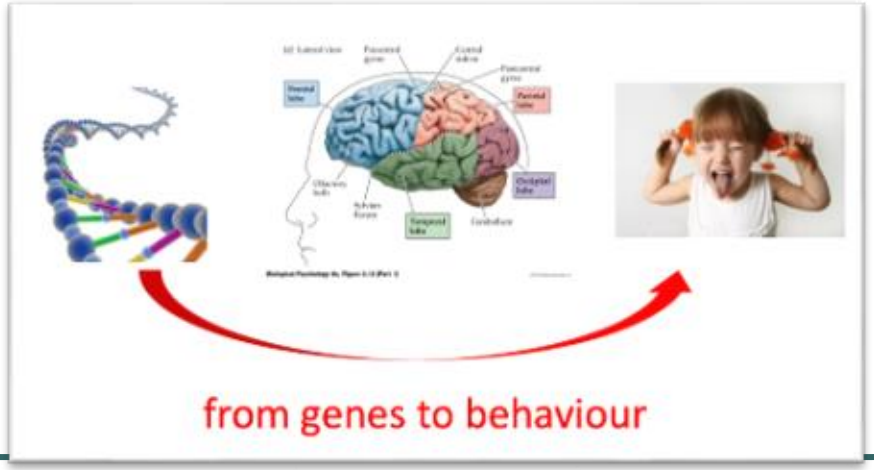
(From Pawel Herman)

ICT-based brain research – fundamental goals

- organise and integrate multi-level data

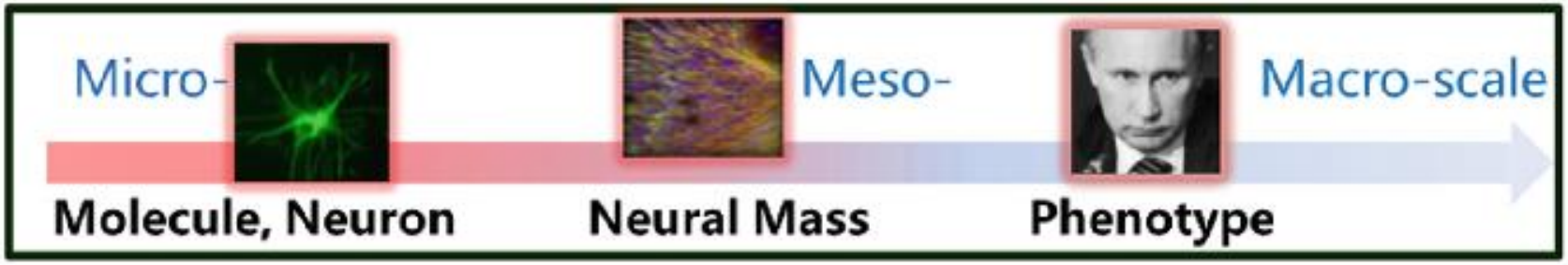
gather existing data across **scales and levels**, identify missing data

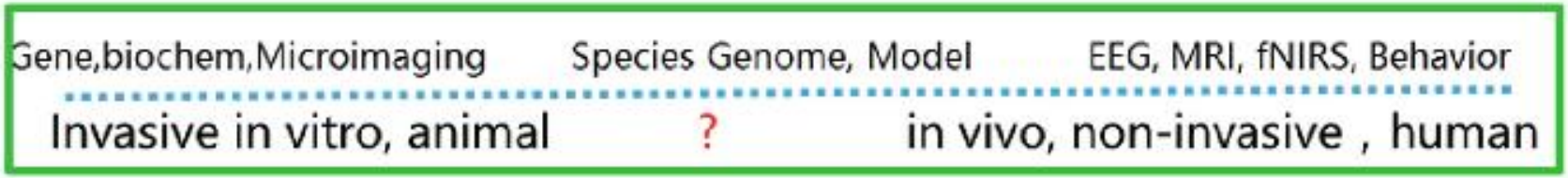


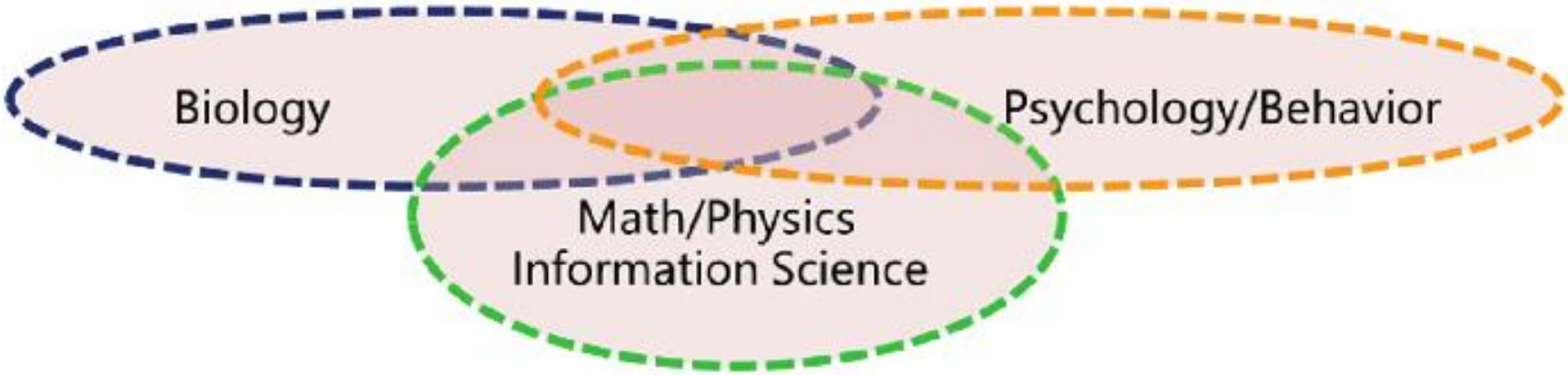


(From Pawel Herman)

From micro- to macro-scale scope





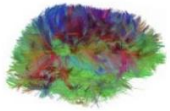


Focus for neuroinformatics – data, theory

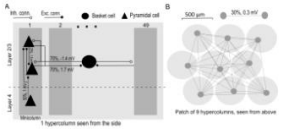
organise and integrate multi-level data

gather existing data at multiple levels, identify missing data

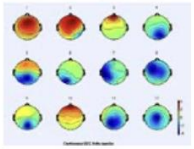
develop tools for storing, visualising and sharing information



build models, simulate, develop theories



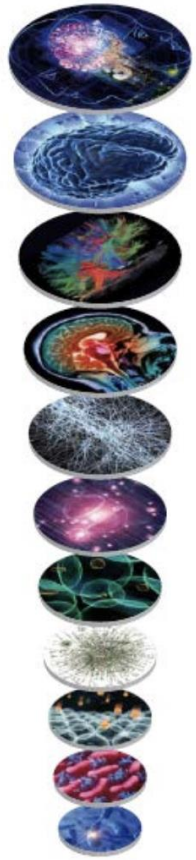
devise new approaches to data analysis



build databases, brain atlases

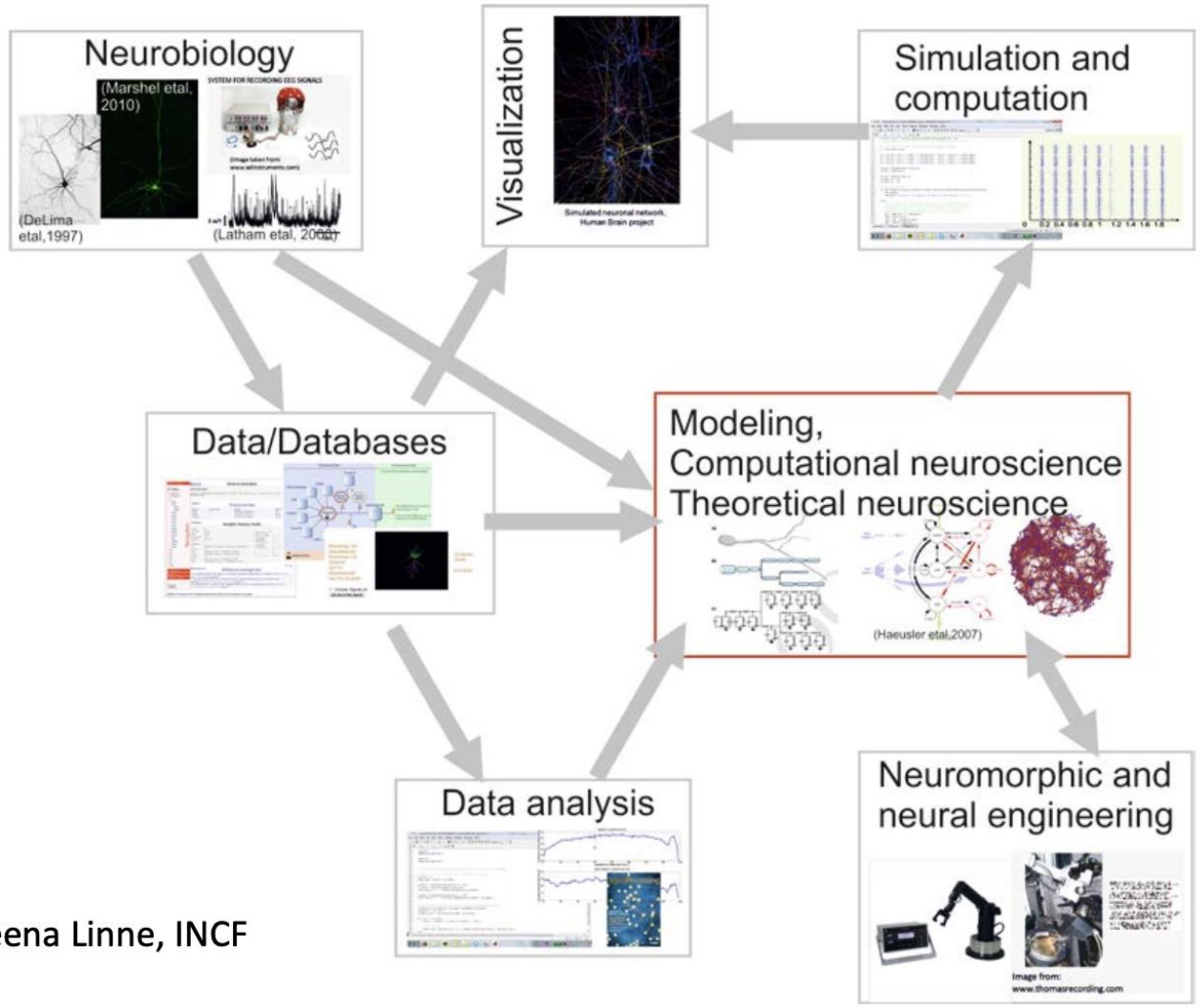






(From Pawel Herman)

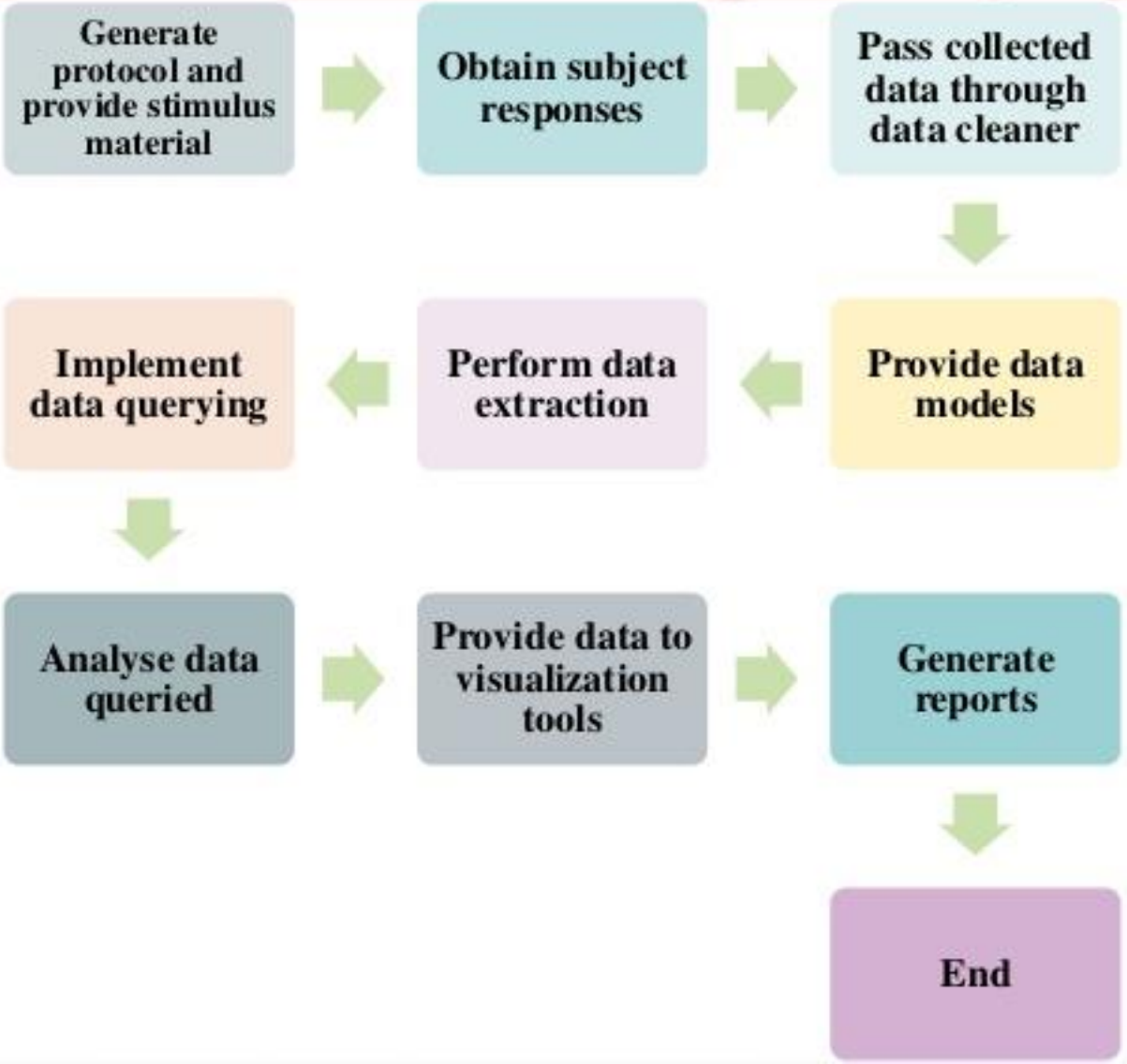
Computational neuroscience and neuroinformatics



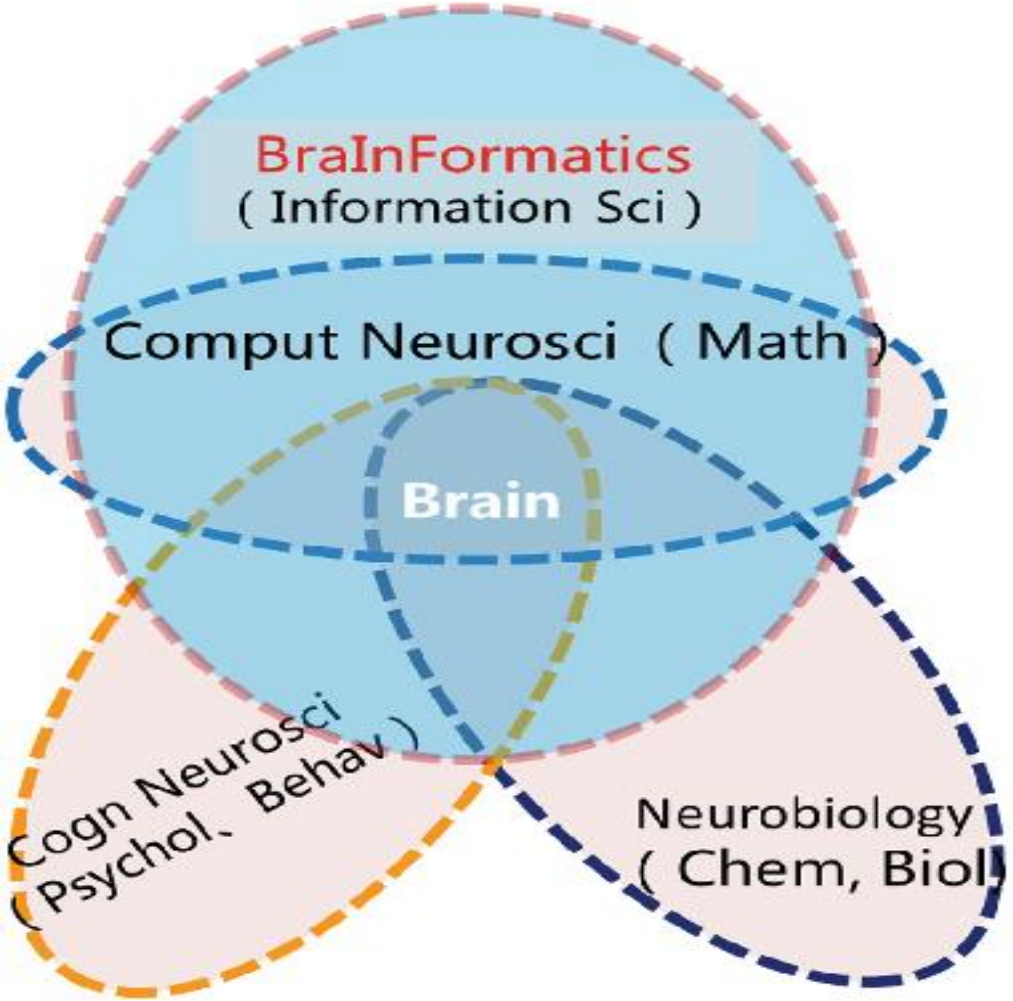
Marja-Leena Linne, INCF

(From Pawel Herman)

Neuroinformatics Repository System



Brainformatics



Neuroinformatics versus Open Science

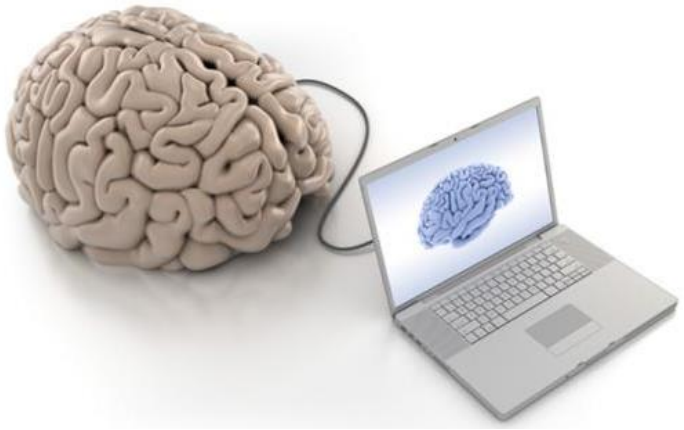
- **Neuroinformatics**
  - Focuses on the *technical* and *structural framework*—such as brain atlases, data standards (e.g., BIDS), and simulation software (e.g., NEURON, Brian)—needed to handle multi-scale neural data.
- **Open Science**
  - Focuses on *philosophy*, *policy*, and *equity*—advocating for reproducibility, open-access publishing, pre-registration, and the removal of paywalls or barriers to scientific discovery.

課程大綱與評分方式

(1) 115/10/13 (二) 9am-12pm @圖資大樓839
陳麗芬老師 Introduction to neuroimaging and neuroinformatics
(2) 115/10/27(二) 9am-12pm @圖資大樓839
陳麗芬老師 Neuroimage techniques and analysis
(3) 115/11/03(二) 9-12pm @圖資大樓839
鄭州閔講師 Introduction to MRI basics and images
(4) 115/11/10(二) 9-12pm @圖資大樓839
林嘉澍老師：Neuroimage meta-analysis
(5) 115/11/14 (六) 15:00-18:00 @榮總 參訪 EEG & MRI
鄭州閔講師 MRI practice
(6) 115/11/17 (二) 9-12pm 期末考，實體考試，可攜帶一張A4紙的筆記。

- 學期評分方式：
  - 期末考（60%）：申論題與選擇題（陳/林老師的部分）
  - MRI-related（40%）：11/03(35%)、11/14（5%）、Bonus (extra 5%)

How to Study Brain?



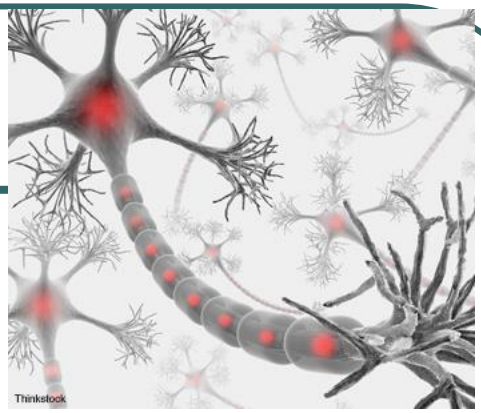
- Q1: How to measure the brain?
  - imaging techniques
- Q2: How to represent the brain?
  - data format
- Q3: How to extract/mine structural and functional information of the brain?
  - analysis techniques
- Q4: How to understand and apply for clinical diseases?
  - diagnosis and treatment

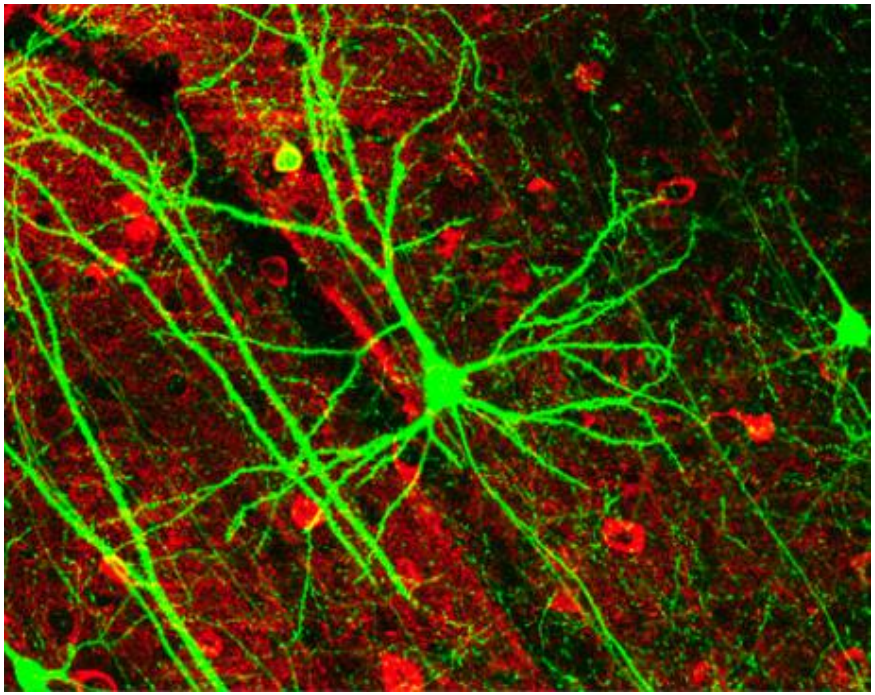
Aim:

- To help human beings for better life quality

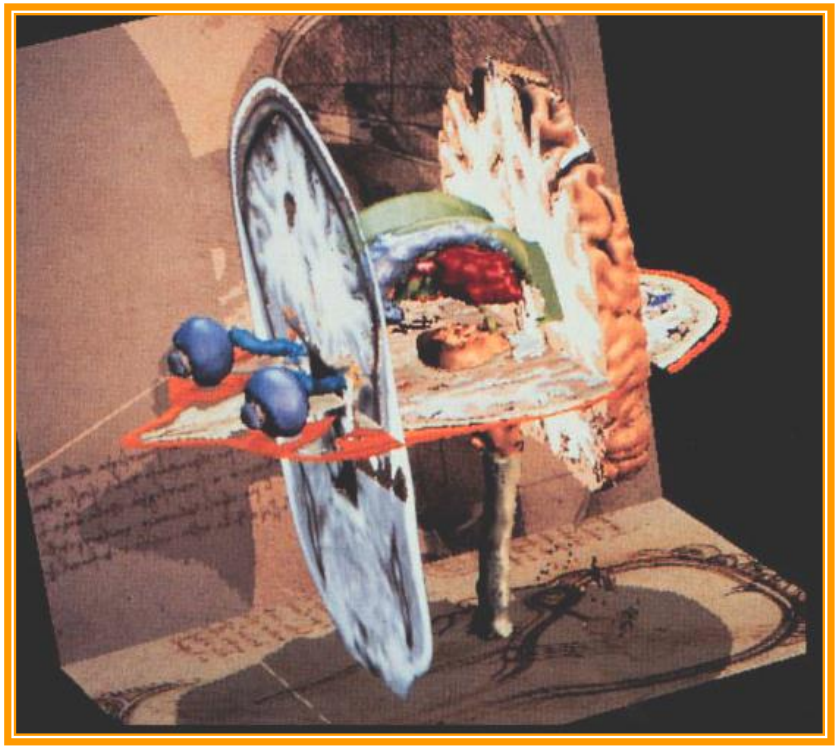
Structure vs. Function

- Microscopic vs. Macroscopic

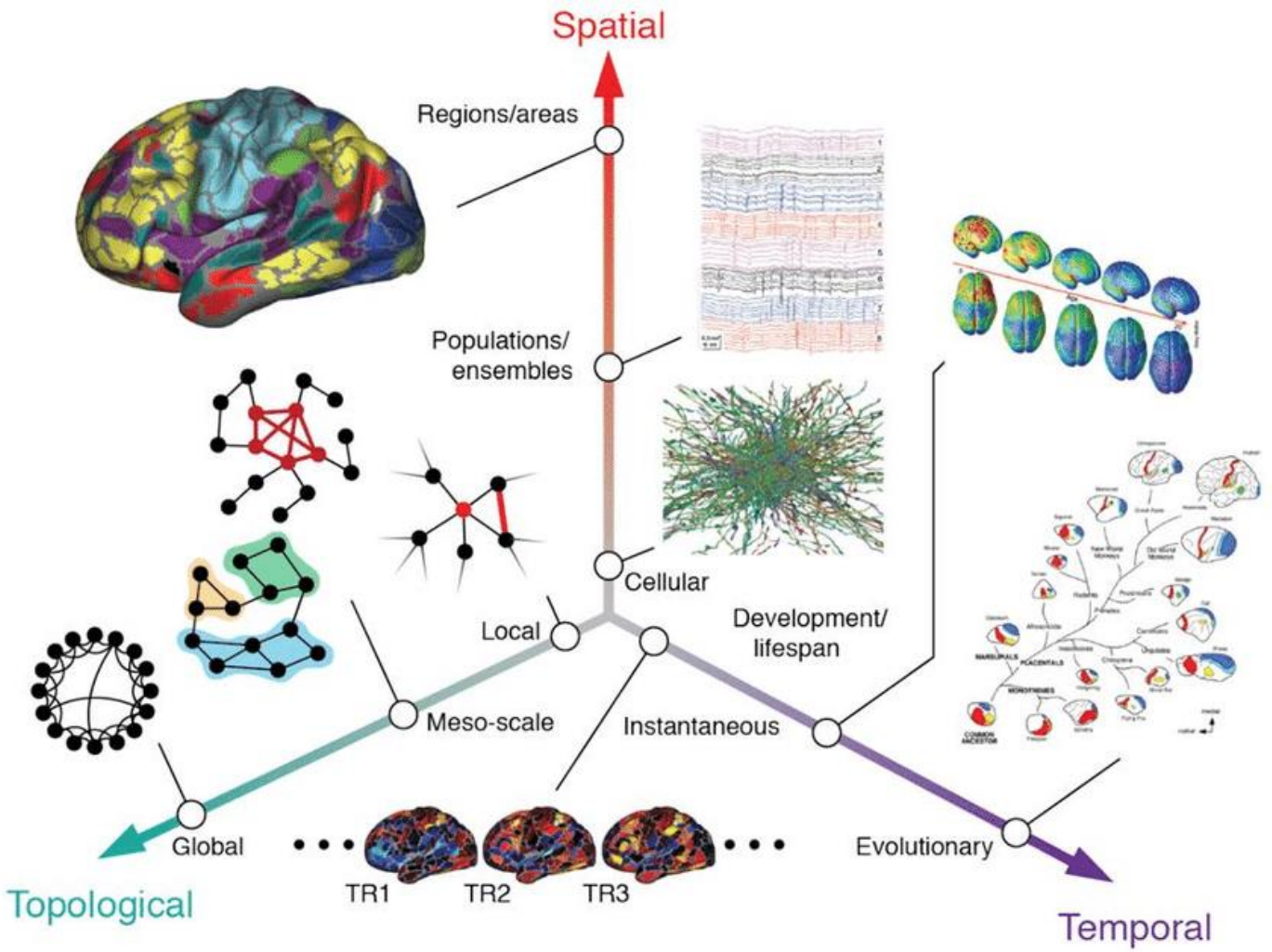




(http://upload.wikimedia.org/wikipedia/commons/0/0d/GFPneuron.png)



Multiple spatiotemporal-scale network analysis



Q1: How to measure?



電腦斷層掃描(CT)



正子斷層掃描(PET)



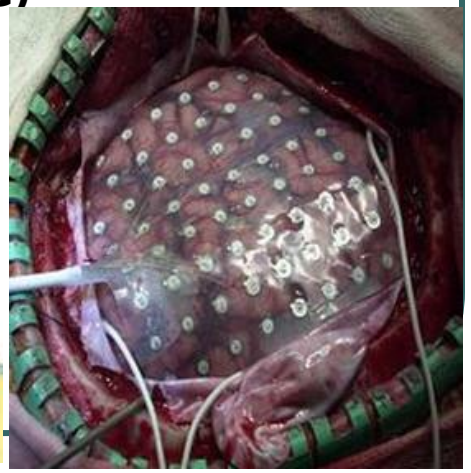
腦磁儀(MEG)



磁振造影(MRI)

腦電波(EEG)





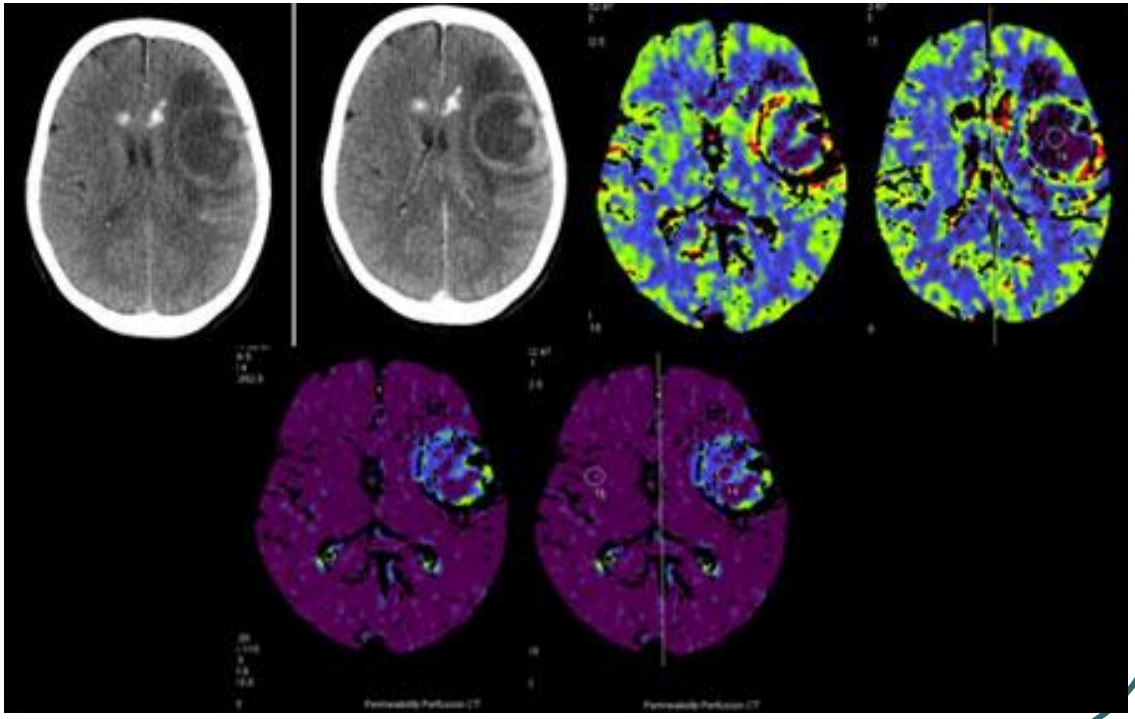
電腦斷層掃描(Computerized Tomography)

- 不同角度X光穿透人體，因不同密度造成訊號的差異，經電腦計算後成像。
  - 範例:急性腦缺血CT灌注影像，根據顯影劑在腦內分佈狀況可判別腦血管阻塞程度。
  - 資料庫內容:

    腫瘤大小、位置、

    嚴重程度等等

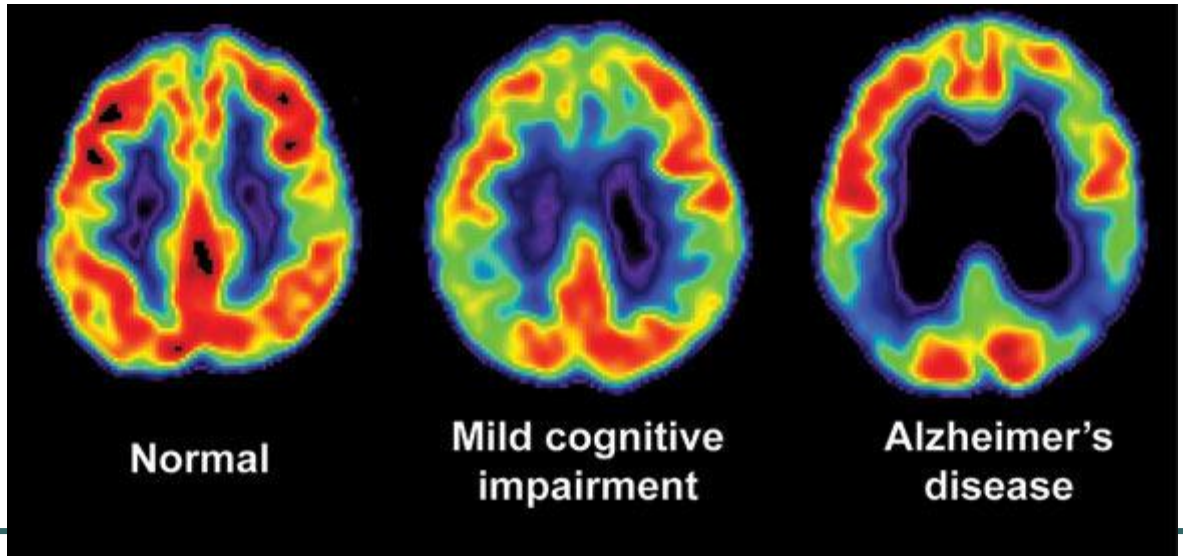




正子斷層掃描(Positron Emission Tomography)

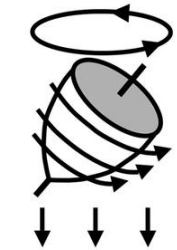
- 一種核子醫學檢查，透過靜脈注射放射性正子藥物（如葡萄糖），經人體吸收代謝後，進入掃描器造影，再由電腦運算重組成立體的影像。
  - 範例:右邊為阿茲海默重症(老年性癡呆症Alzheimer's Disease, AD)、中間為輕度認知功能障礙(mild cognitive impairment)患者的腦部攝影、左邊對照組健康人的腦部攝影，紅色以及黃色的範圍最多，代表腦部的神經細胞活性最旺盛。

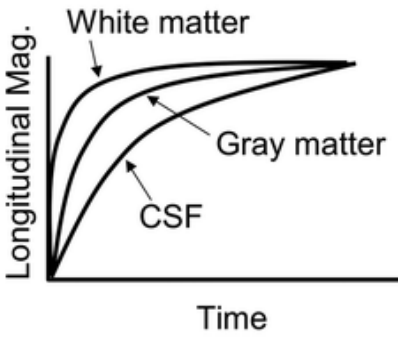




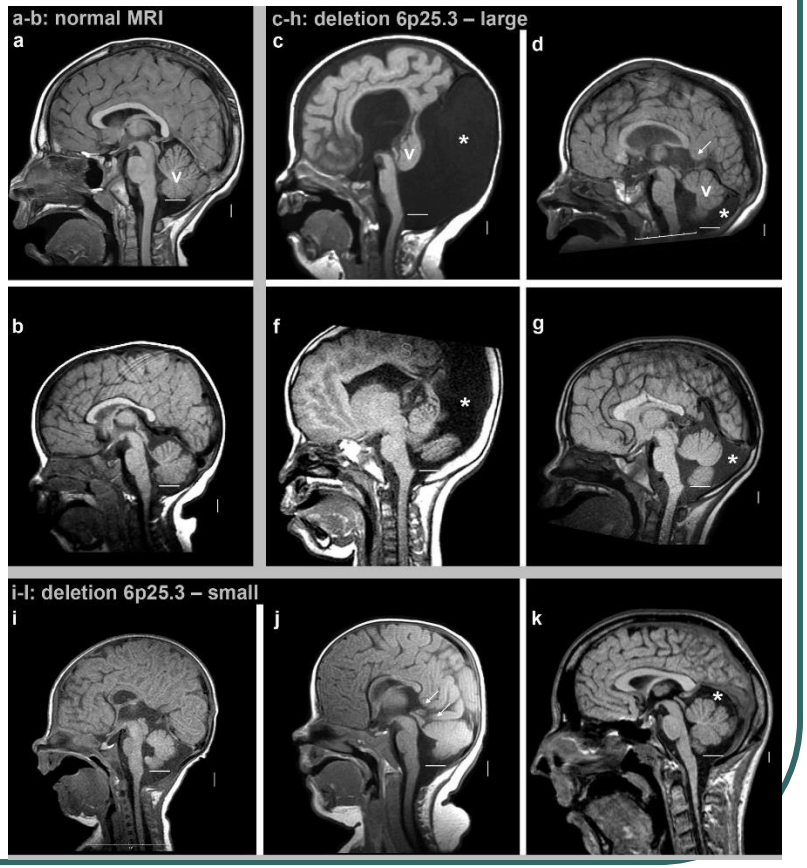
磁振造影(Magnetic Resonance Imaging)

- 利用磁場原理，使儀器改變體內氫原子的旋轉排列方向，原子核就會釋放吸收的能量，能量激發後放出電磁波信號，再經由電腦分析組合成影像，就是一般看到的 MRI 影像，由此所產生的影像可顯示不同的組織密度。

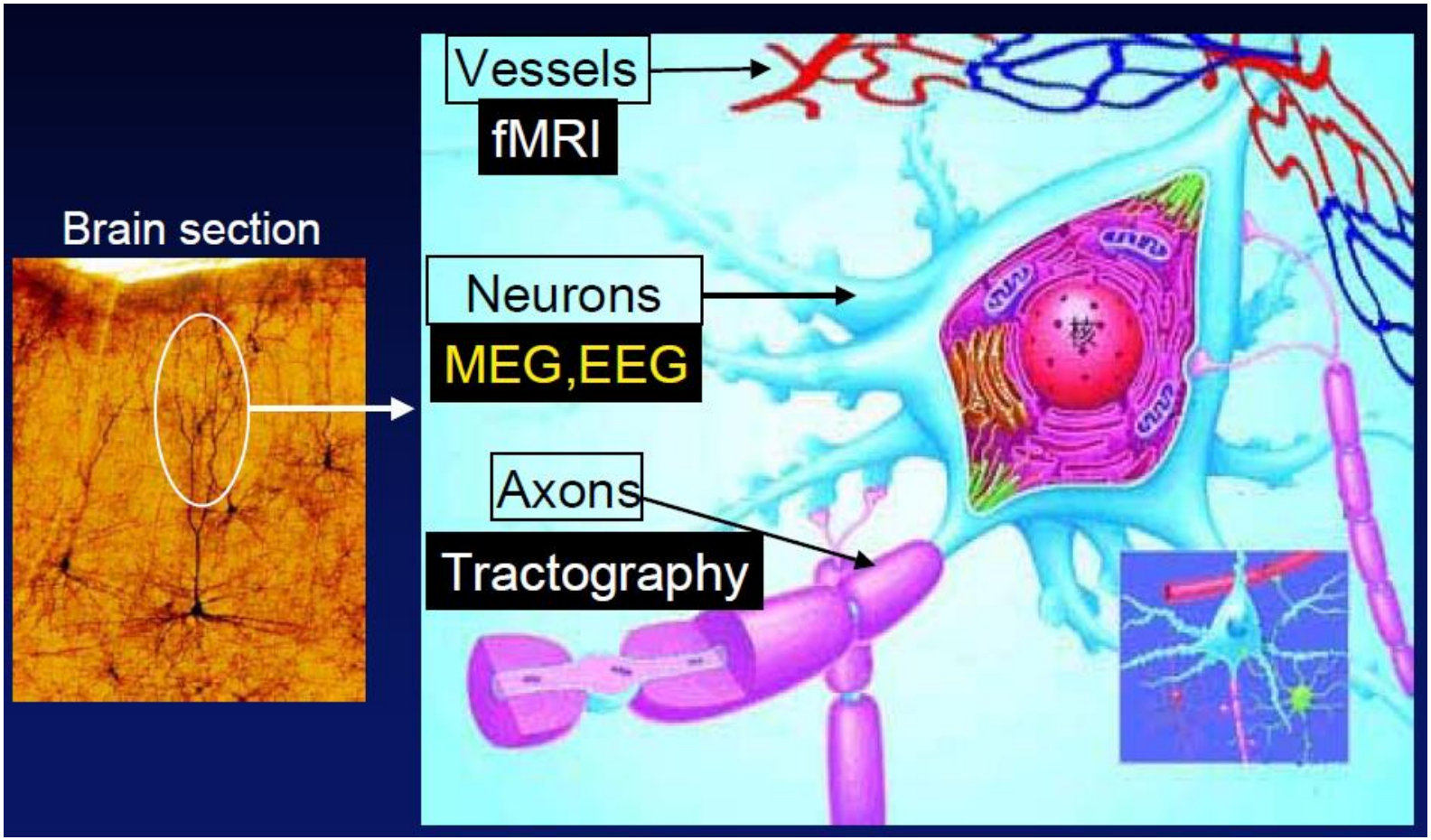








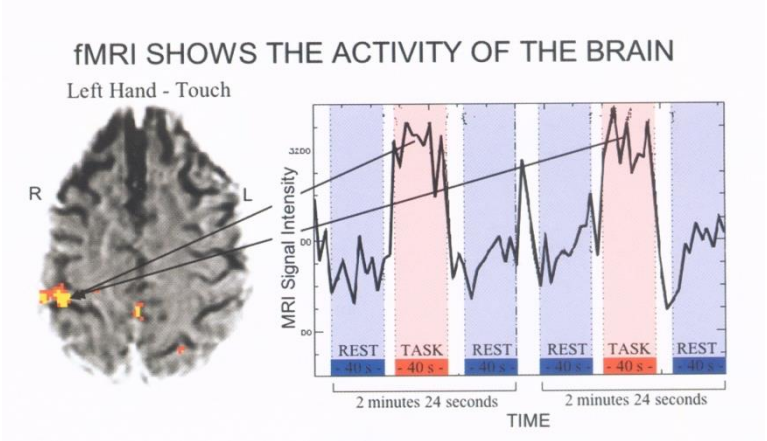
Functional Imaging

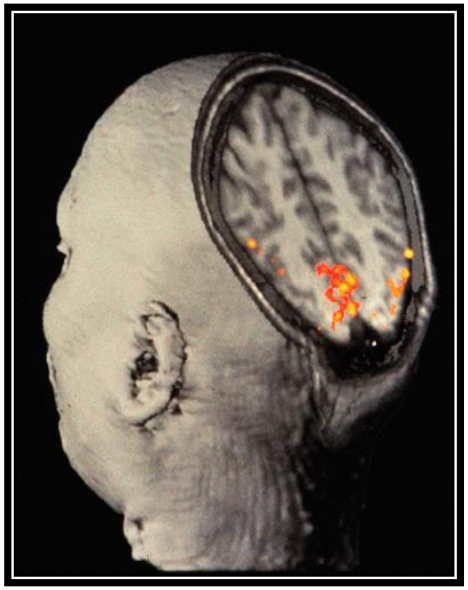


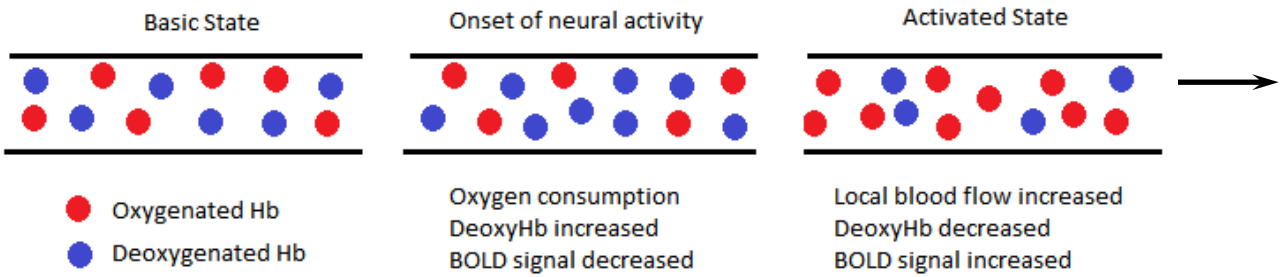
(Kamada, HBM2006)

功能性磁振造影(Functional MRI)









1.5~3.0% 訊號差
1.5 Tesla磁振造影

~6%訊號差
3.0 Tesla磁振造影

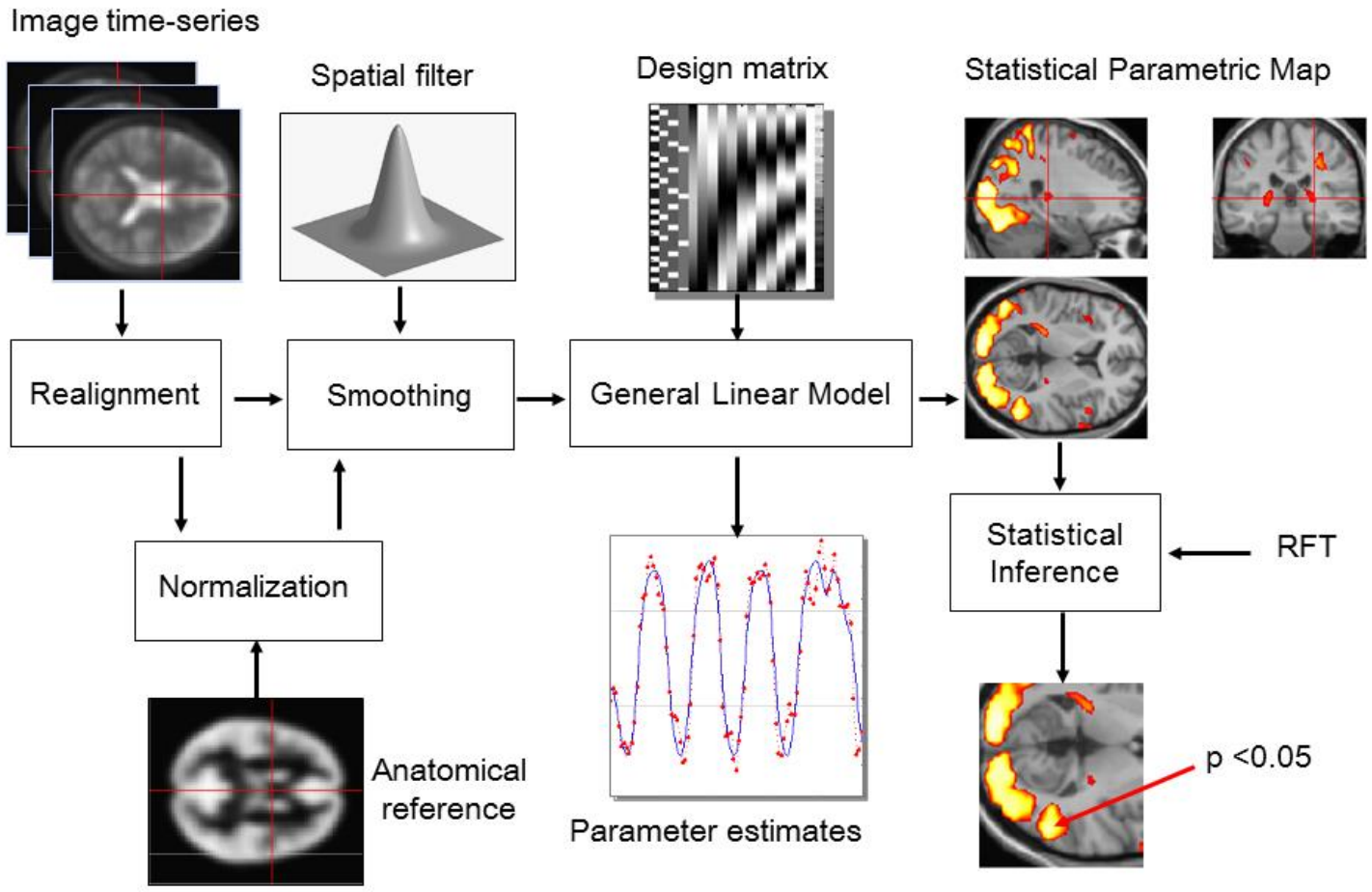
靜止狀態
帶氧血紅素/去氧血紅素
(HbO/Hb)

激發狀態
局部血流增加
(HbO/Hb)

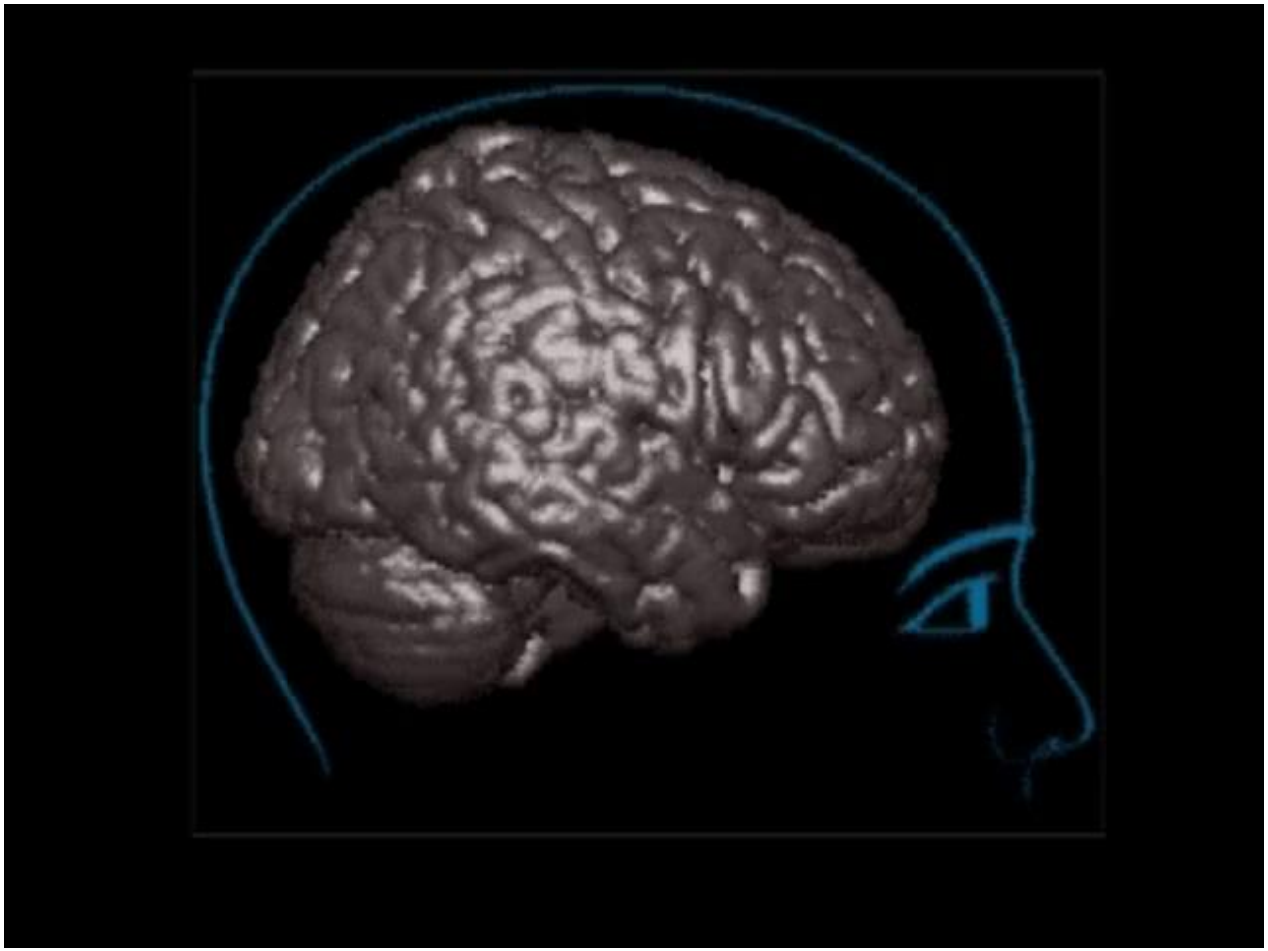
http://bispl.weebly.com/neuro-imaging-and-analysis.html

Statistical Parametric Mapping

SPM



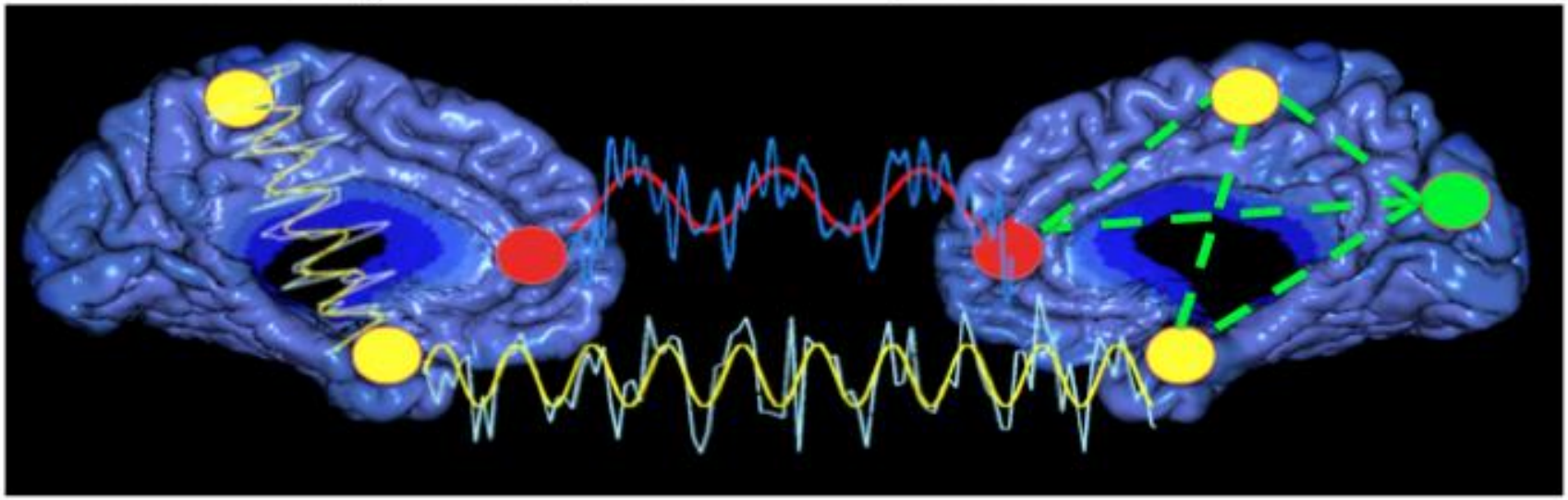
Example



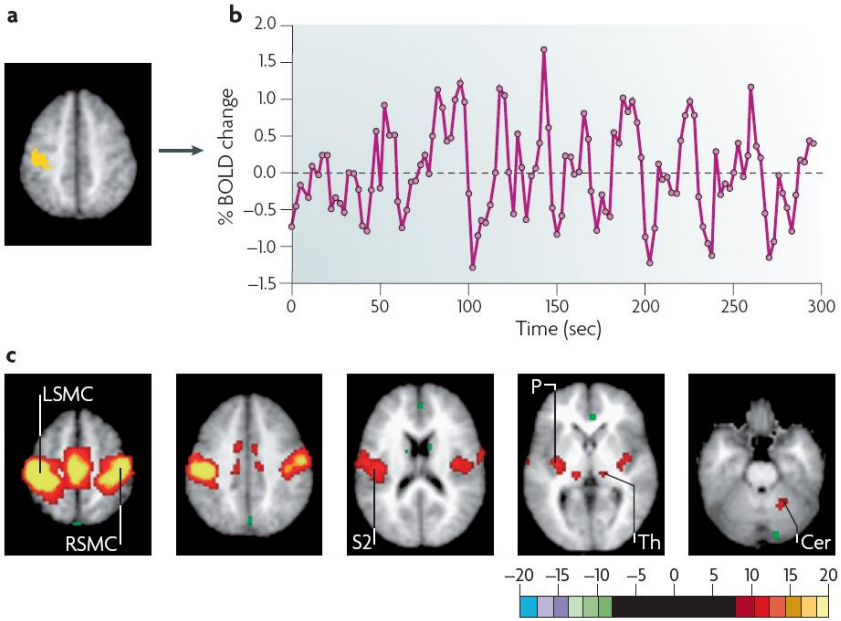
(http://lcni.uoregon.edu/~dow/Space_software/animations.html#Brodmann_brain)

Resting-state fMRI (rsfMRI)

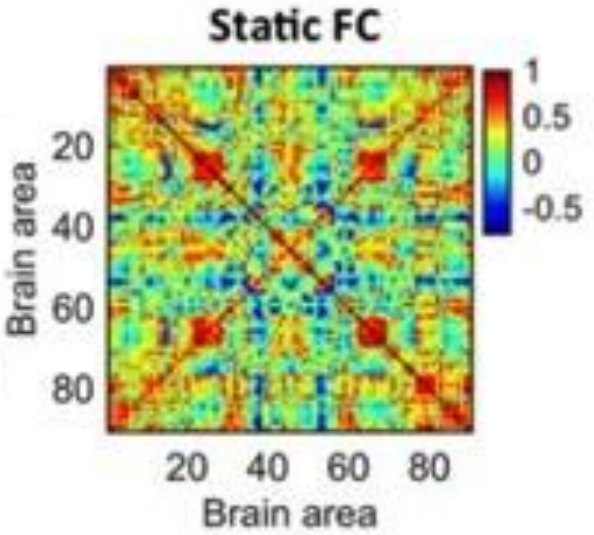
Montreal Neurological Institute, Jeanne Timmins Amphitheater

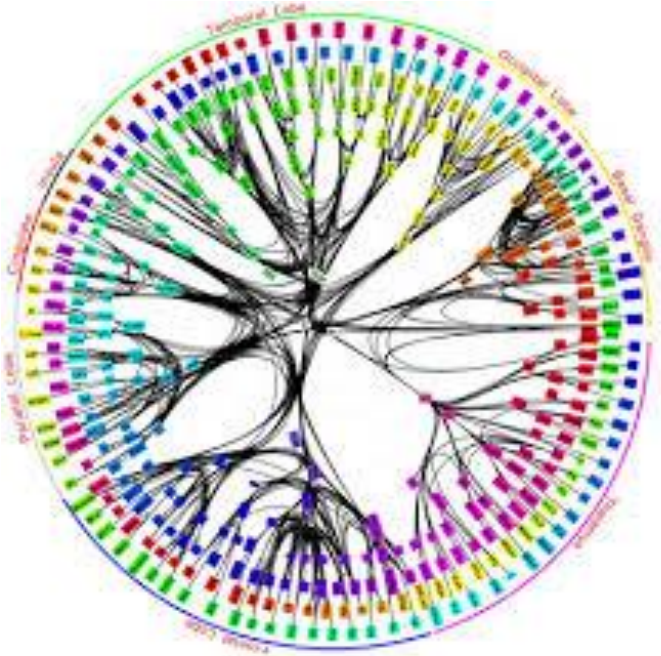


Calculation of rsfMRI



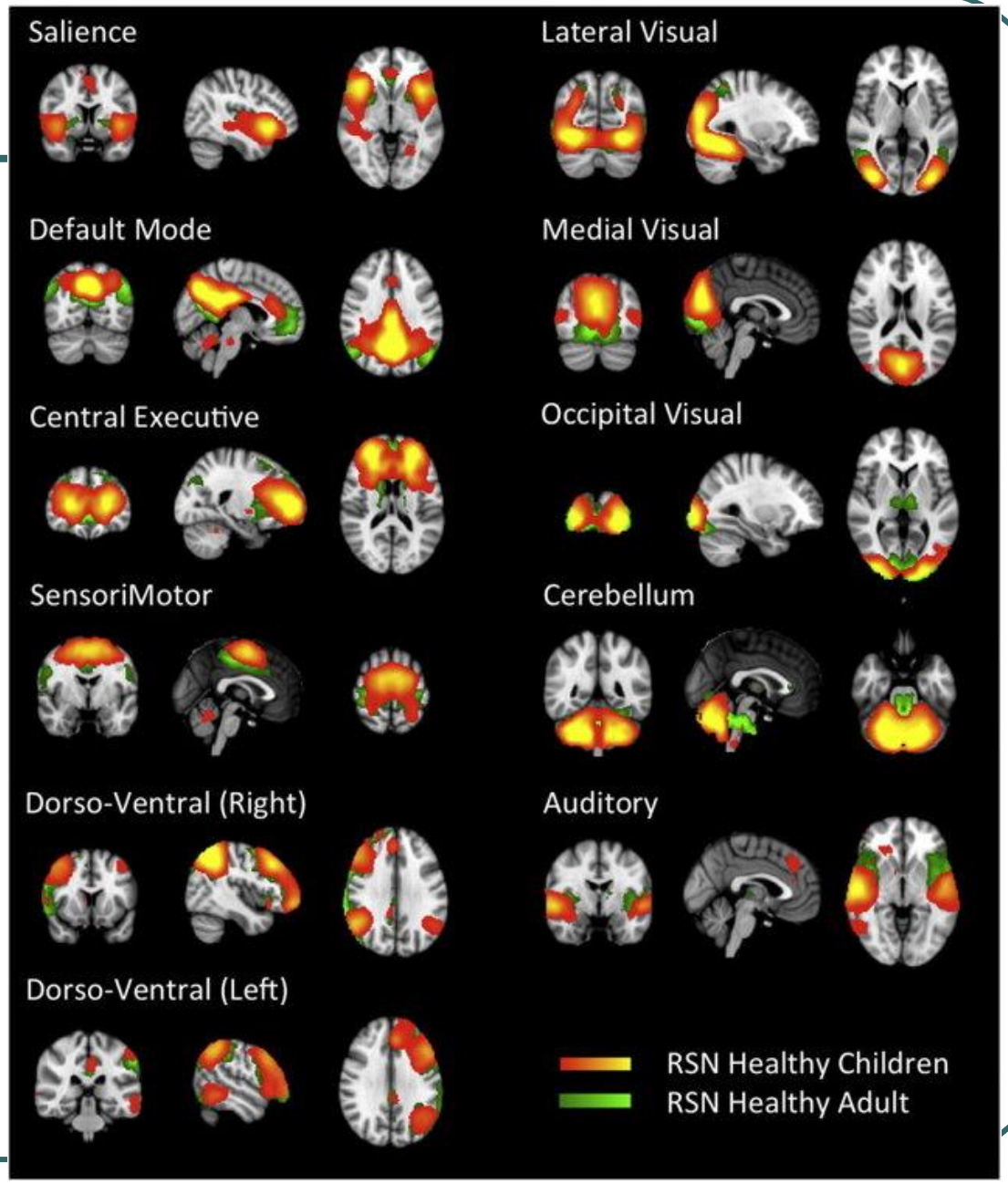
Seed-based correlations





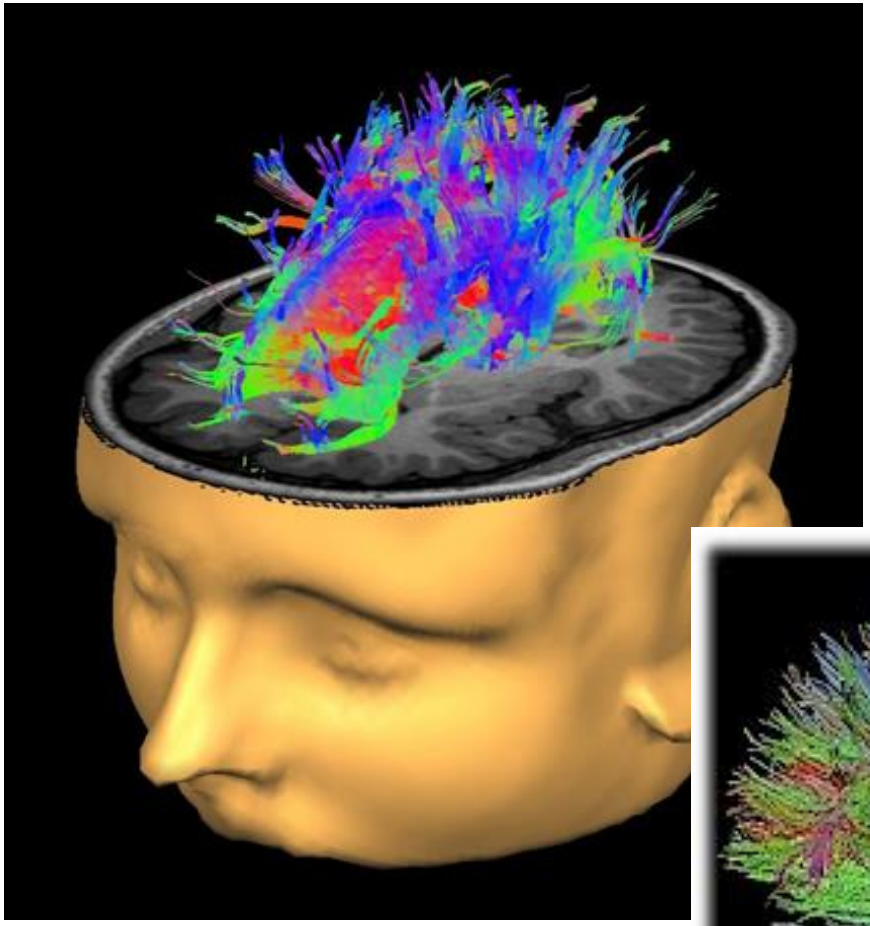
Modha D S , Singh R PNAS 2010;107:13485-13490

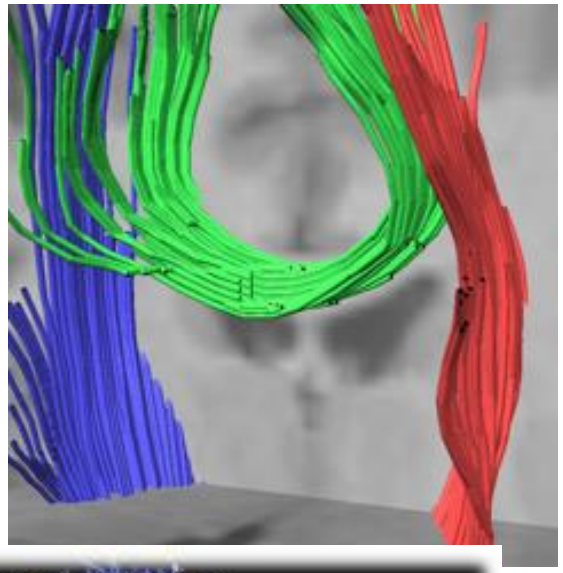
Resting State Networks

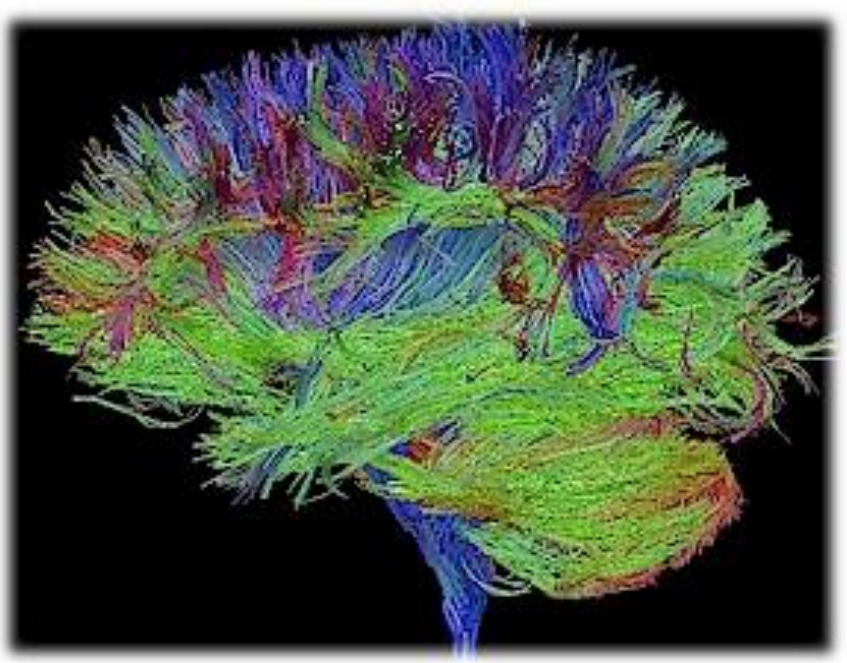


Simons, NeuroImage: Clinical 2014

擴散張量造影(Diffusion Tensor Imaging)



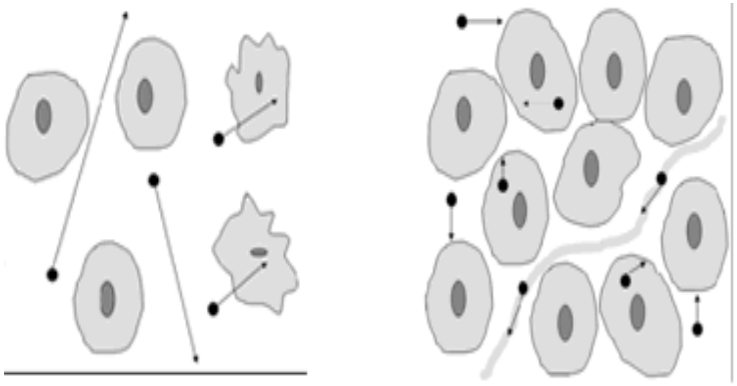




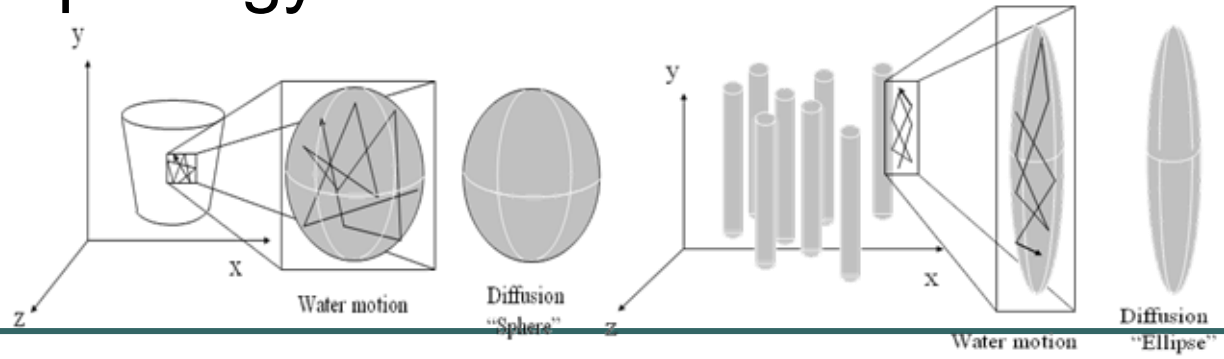
http://mizzoumagarchives.missouri.edu/

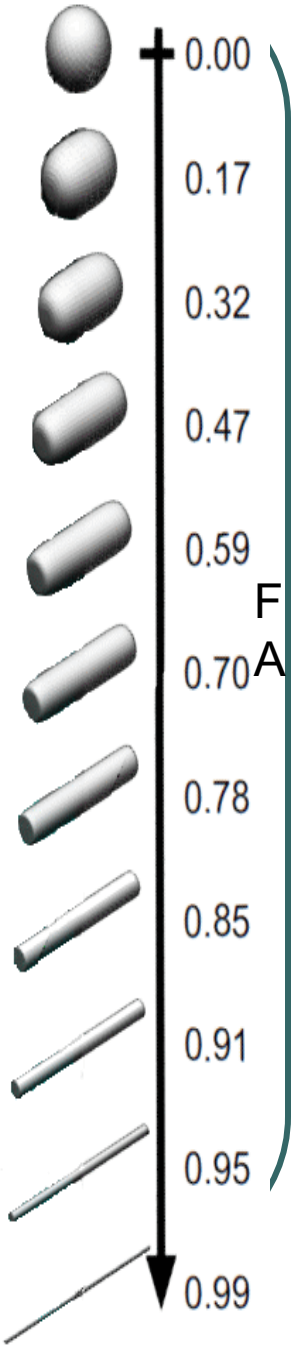
Diffusion Tensor Imaging (DTI)

- Brownian motion



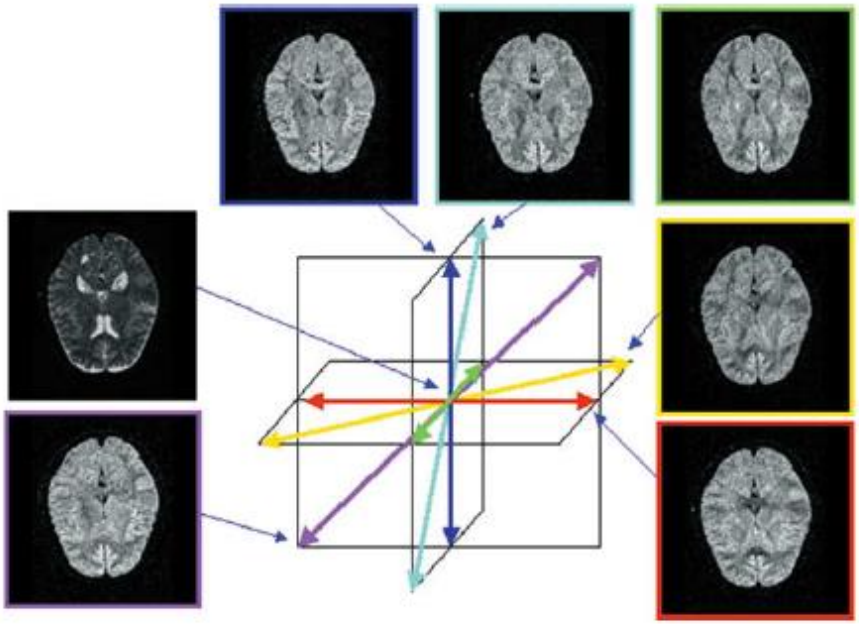
- DTI is most appropriate non-invasive quantification of white matter tract morphology and microstructure in human





(fsl.fmrib.ox.ac.uk/fsl/)

How Anisotropy Is Measured



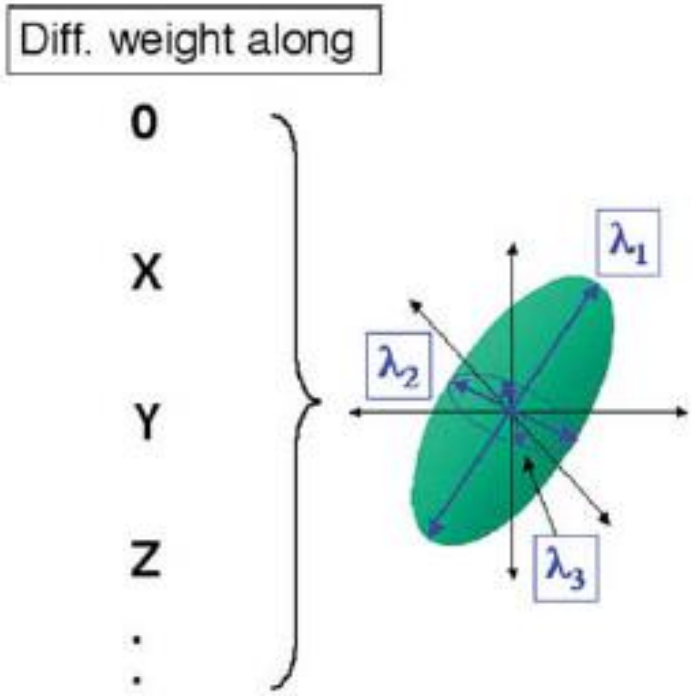


Figure 16 One non-diffusion-weighted image and six diffusion-weighted images along six different axes at least are needed, from which six diffusion constants can be obtained.

$$\overline{\overline{D}} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yz} & D_{yy} & D_{yz} \\ D_{yz} & D_{yz} & D_{zz} \end{bmatrix} \xrightarrow{\textit{diagonalization}} \lambda_1, \lambda_2, \lambda_3, v_1, v_2, v_3,$$

Brain Mapping, Ch15

Data Visualization and Analysis of DTI

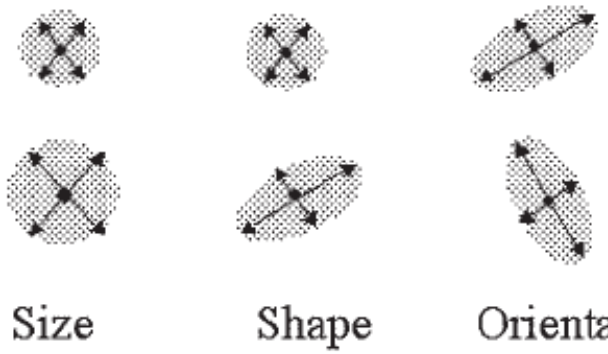
Three parameters to characterize the diffusion ellipsoid

$$FA = \sqrt{\frac{(\lambda_1-\lambda_2)^2+(\lambda_2-\lambda_3)^2+(\lambda_1-\lambda_2)^2}{2(\lambda_1^2+\lambda_2^2+\lambda_3^2)}},$$

$$RA = \frac{\sqrt{(\lambda_1-\lambda_2)^2+(\lambda_2-\lambda_3)^2+(\lambda_1-\lambda_2)^2}}{\lambda_1+\lambda_2+\lambda_3},$$

$$VR = \frac{\lambda_1\lambda_2\lambda_3}{((\lambda_1+\lambda_2+\lambda_3)/3)^3}. \quad (15)$$

Here FA is the fractional anisotropy, RA is the relative anisotropy, and VR is the volume ratio. These parameters all have 0–1 range. Information provided by these parameters is essentially the same. They all indicate how elongated the dif-



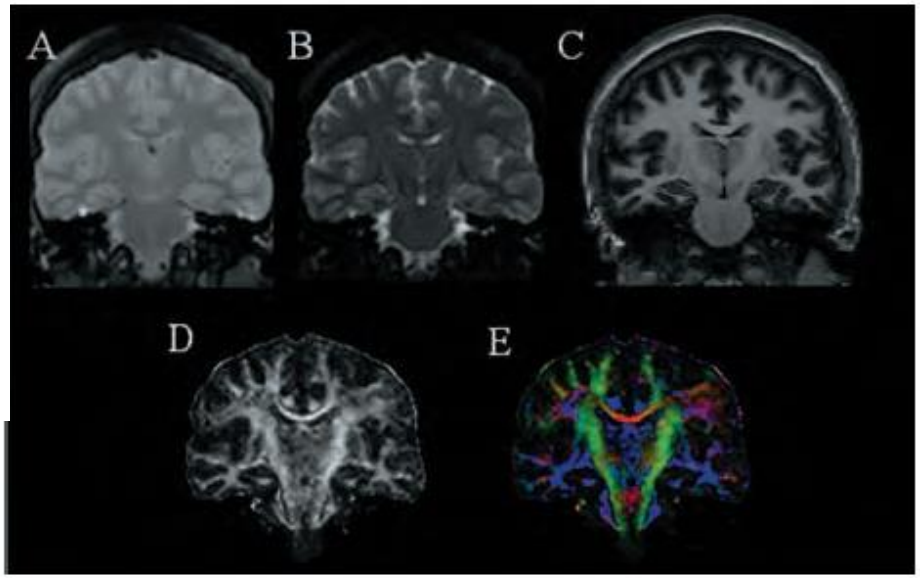
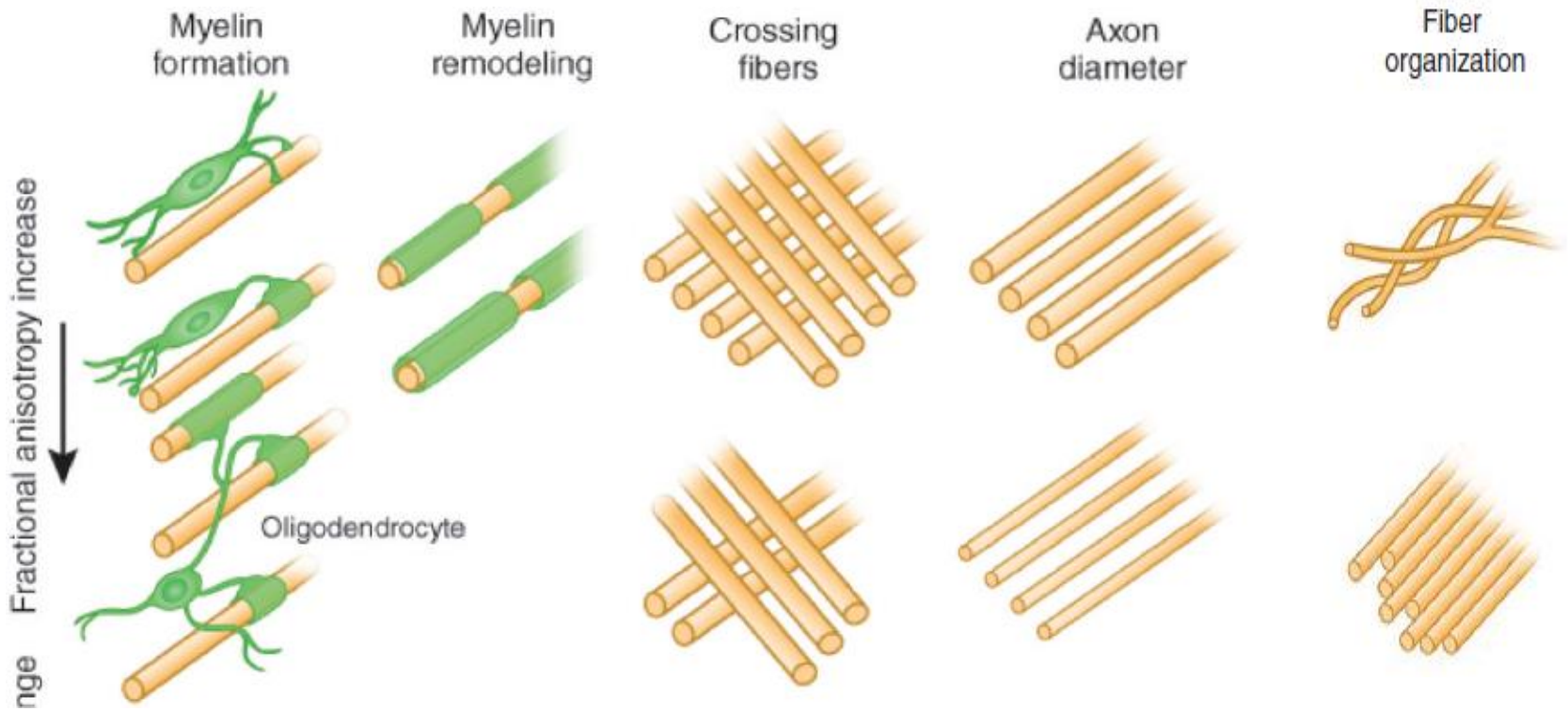


Figure 20 Comparison of proton density (**A**), T_2-weighted (**B**), and T_1-weighted (**C**) images and FA (**D**) and color map (**E**) of a healthy subject.

Brain Mapping, Ch15

Fractional anisotropy (FA)

- FA is sensitive to developmental changes and pathological differences in axonal density, size, myelination, and the coherence of organization of fibers within a voxel, and thus provides an index of the structural integrity of white matter.



Pierpaoli et al., 1996

Tractography

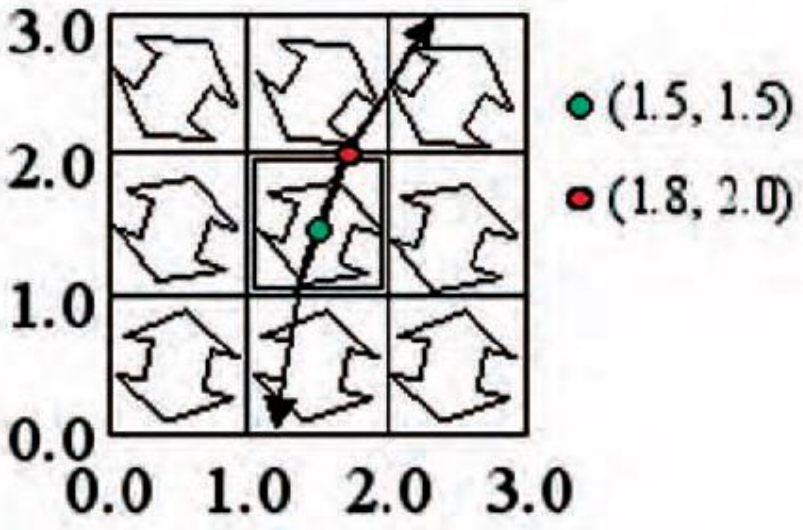
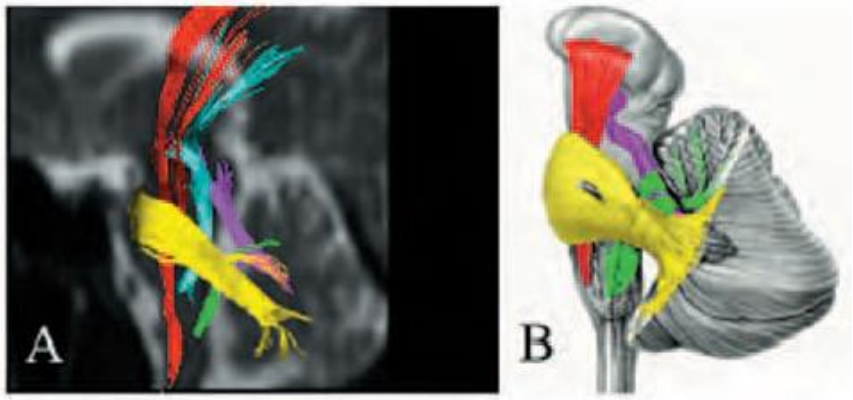
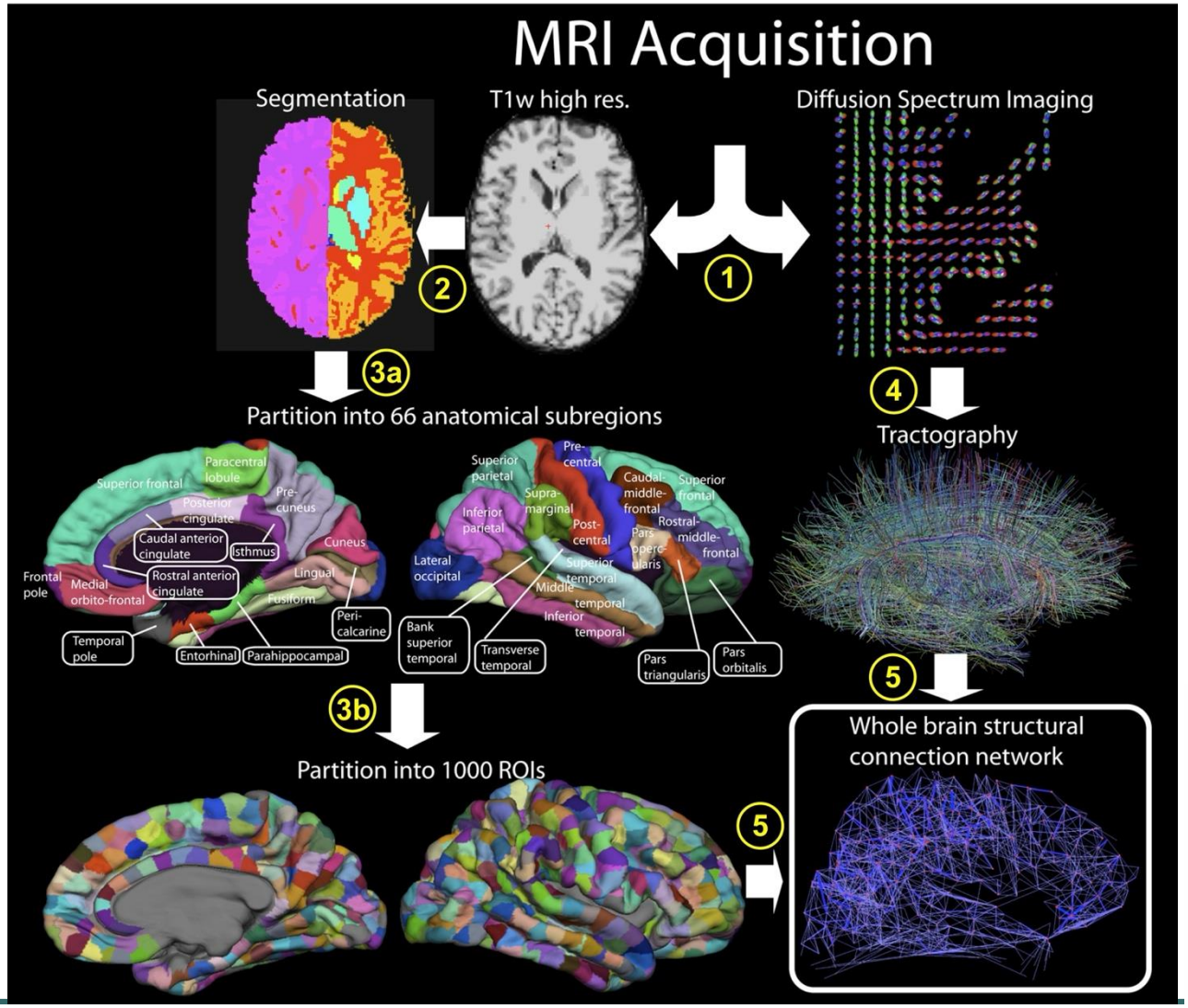


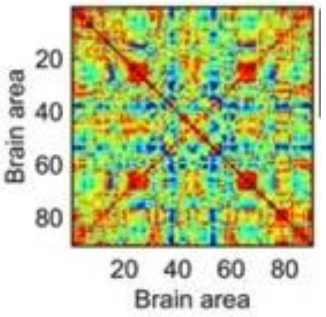
Figure 22 Examples of fiber tracking. Double-headed arrows indicate the orientations of tracts at each pixel, which are obtained from the DTI measurement. Note that the DTI study, which is based on water movement, cannot judge ef fluent and af fluent. Tracking has to be made in both orthograde and retrograde directions.

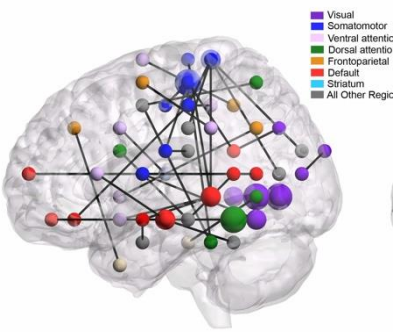


Example of 3D tract reconstruction in the brain-stem region (**A**) and comparison with a postmortem study (**B**).

Connectome extraction procedure

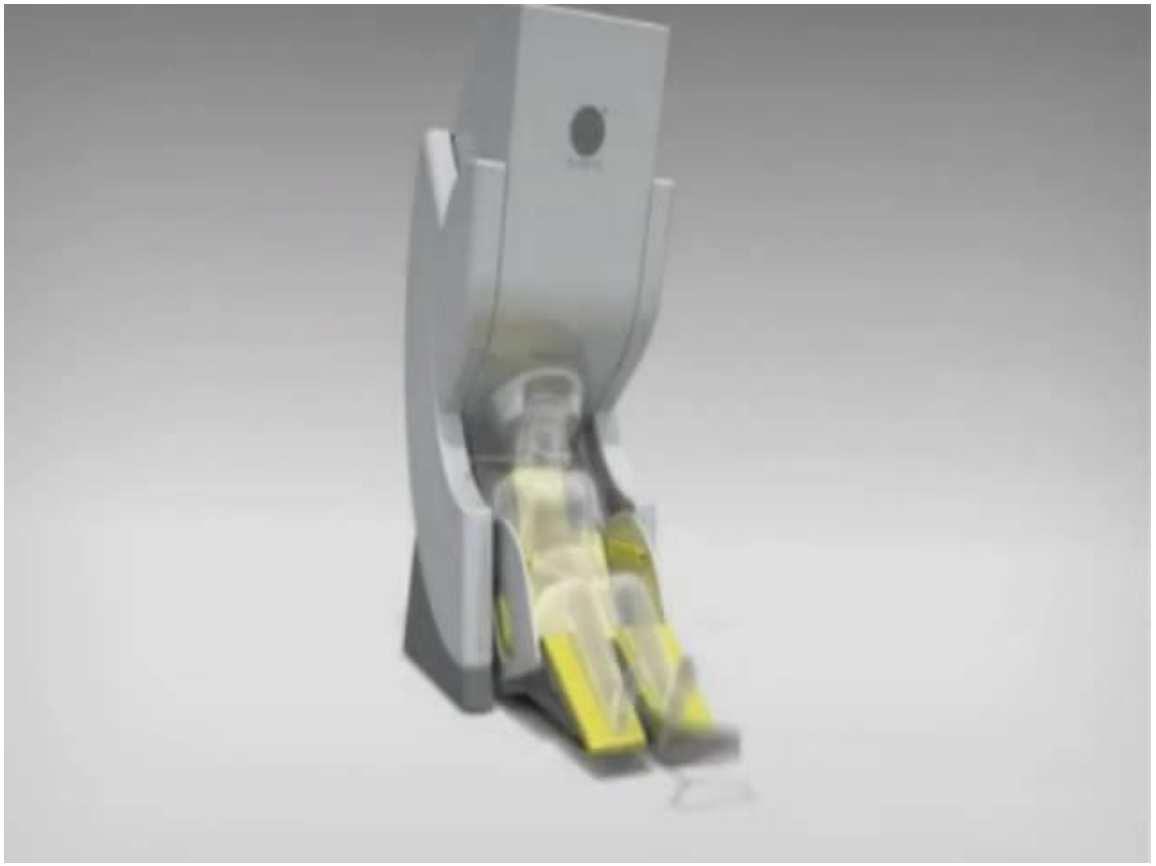






http://commons.wikimedia.org/wiki/File:Connectome_extraction_procedure.jpg

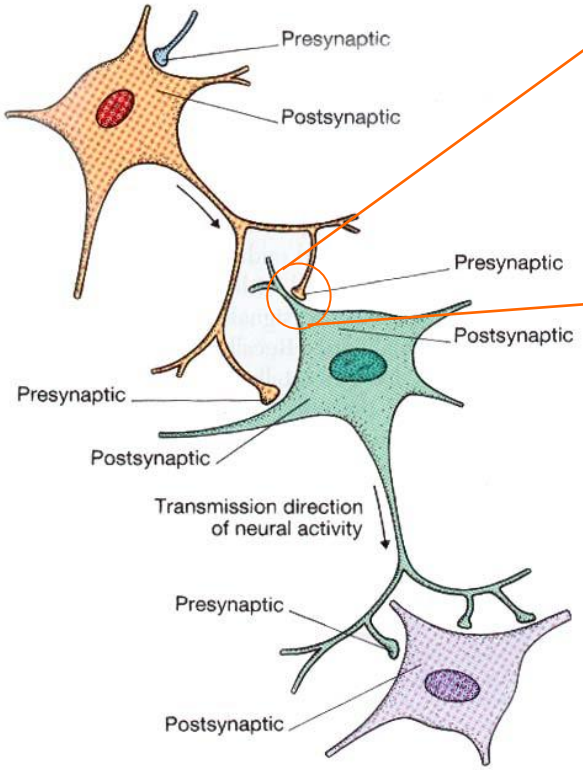
腦電圖/腦磁圖 (Electroencephalography/Magnetoencephalography, EEG/MEG)

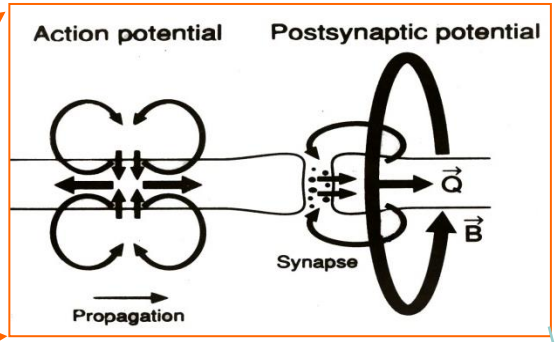


http://www.elekta.com/healthcare-professionals/products/elekta-neuroscience/elekta-triux.html?utm_source=neuromag&utm_medium=redirect&utm_campaign=redirects

Information Propagation in Neurons

- The activity in human brain is an electrophysiological action of small currents.





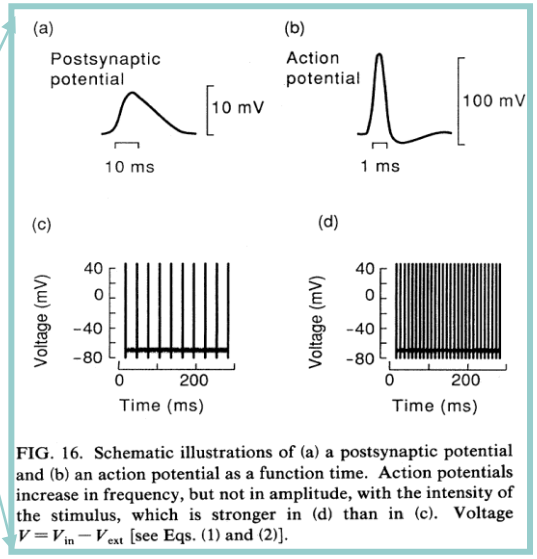
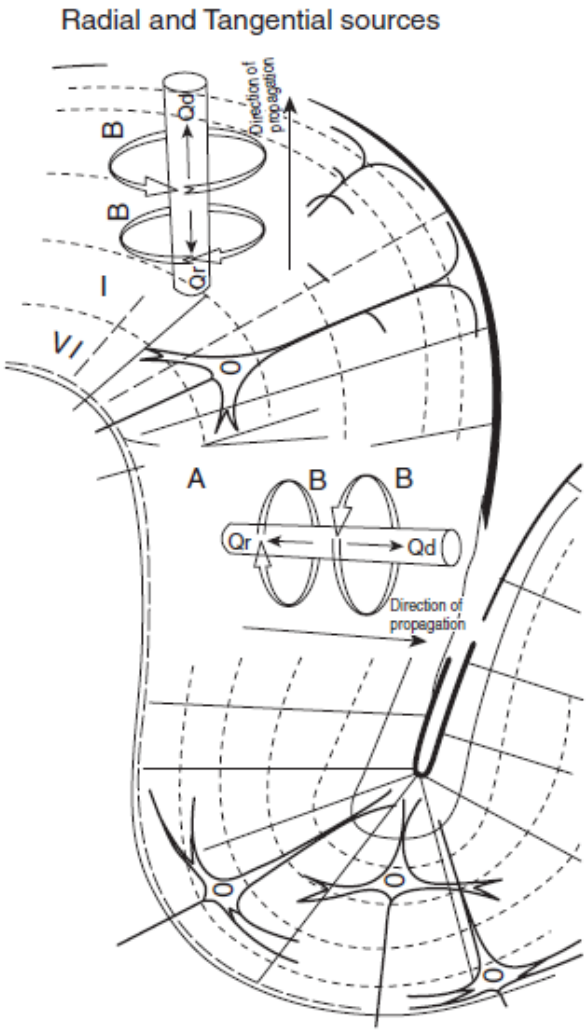


FIG. 16. Schematic illustrations of (a) a postsynaptic potential and (b) an action potential as a function time. Action potentials increase in frequency, but not in amplitude, with the intensity of the stimulus, which is stronger in (d) than in (c). Voltage $V = V_{in} - V_{ext}$ [see Eqs. (1) and (2)].

- Signal emission:
  - *Action potential* is occurred when the neuron is propagating a signal along its axon and the frequency (not amplitude) encodes the neuronal information. (I.e. high frequency 1ms,)
- Signal reception:
  - specific ions rush through the membrane and give rise to a *post-synaptic potential* of about 10mV with a duration of 10ms, what EEG and MEG measured (synaptic current flow).

Source Model of EEG/MEG

- Current dipole: can be viewed as a *battery*



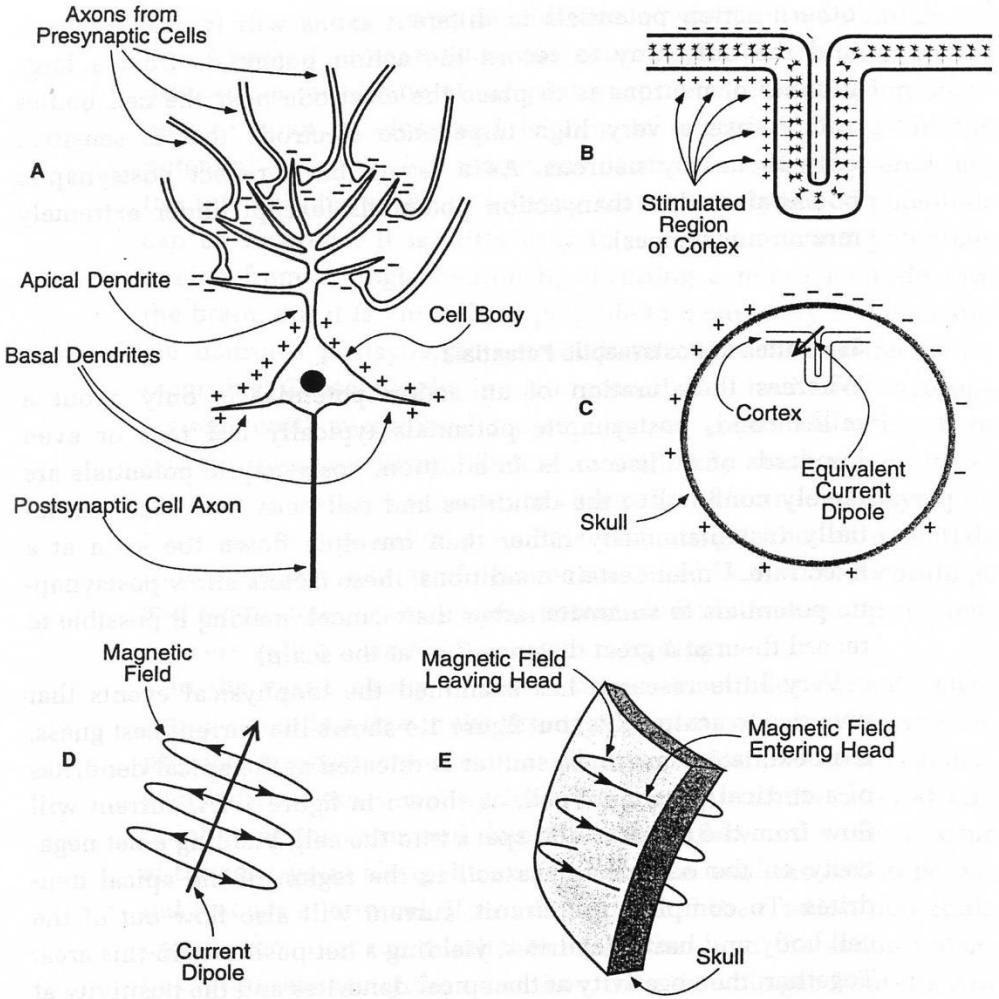
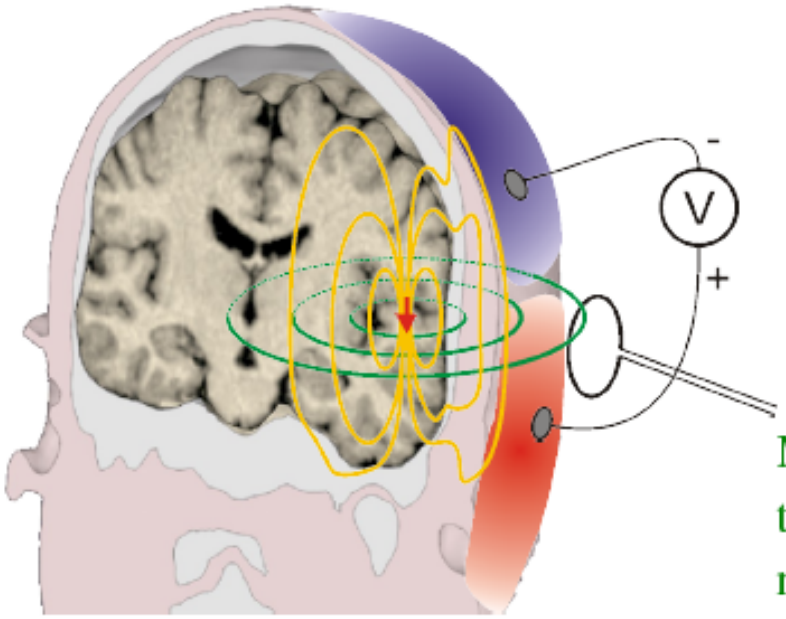


Figure 1.4 Principles of ERP generation.

Neuronal Electromagnetic Signals

- Cortical potentials are generated due to excitatory and inhibitory post-synaptic potentials of pyramidal neurons.

Cellular currents in an active neuron population...



... give rise to extracranial electric potentials and magnetic fields

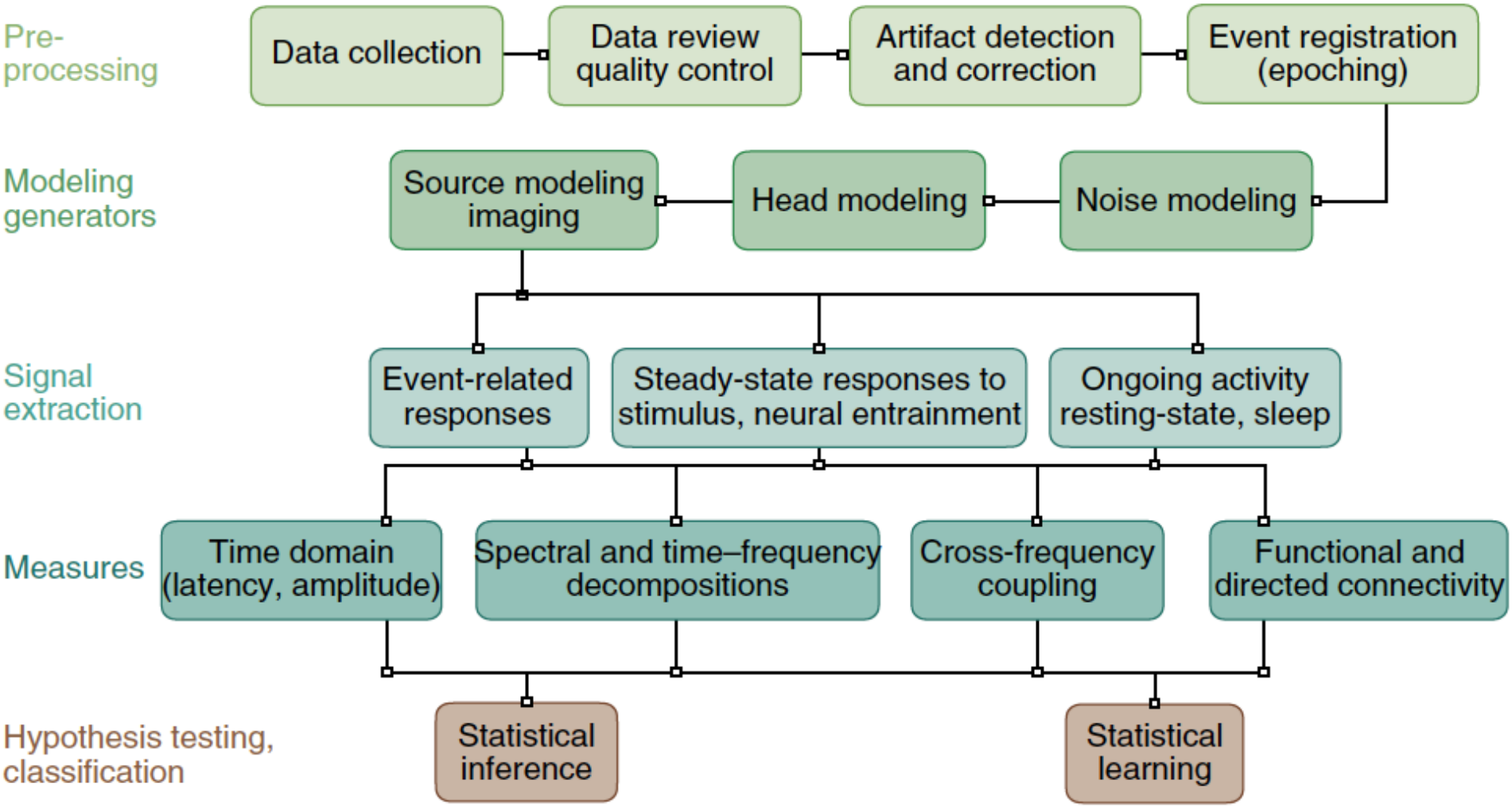
EEG = measuring the potential differences on the scalp

MEG = measuring the extracranial magnetic fields

MEG and EEG are different views of the same neural sources

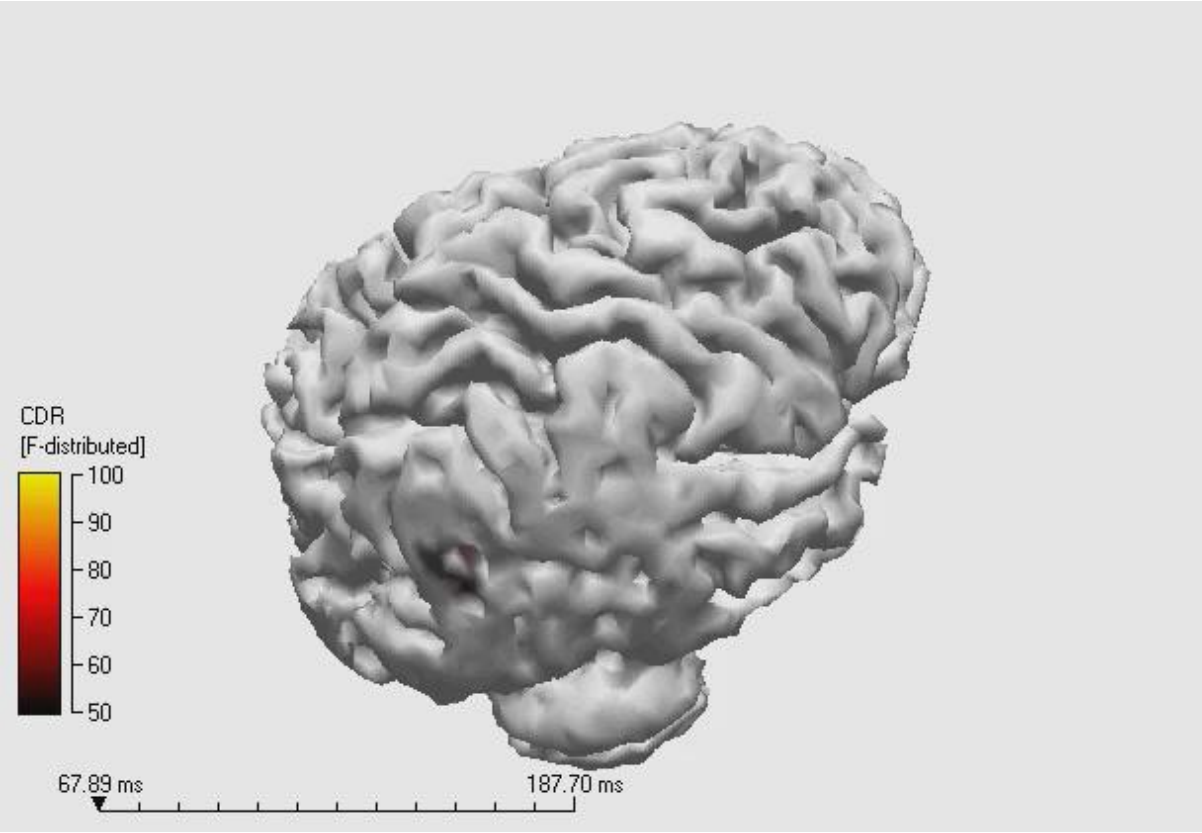
(Lauri Parkkonen, HBM2006)

Major steps required and possibilities offered by EEG/MEG analytics



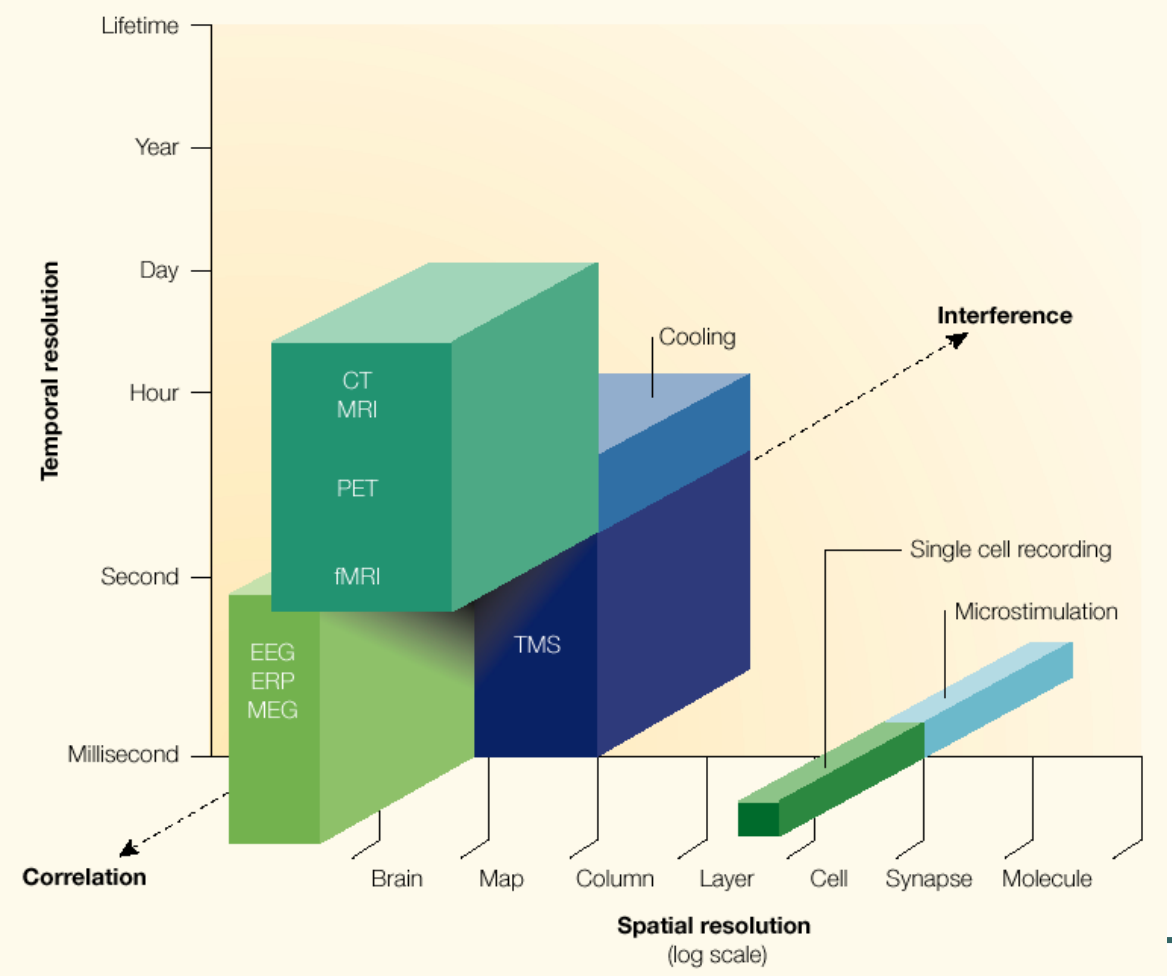
What does the map look like?

Spatiotemporal imaging of face processing in the brain.



Resolution of multimodal brain imaging

- Spatial-temporal resolution illustration of different functional mapping modalities



Different Temporal and Spatial Resolutions between brain mapping techniques

Table 1 Brain Mapping Techniques: Their Temporal and Spatial Resolutions and Sampling Volumes

Method	Energy source	Spatial resolution (mm)	Temporal resolution (s)	Risks	Constraints	Product
Structural MRI	Radio waves	1	N/A	None known	Immobilization, loud	Structure, vasculature, white matter
*f*MRI	Radio waves	4–5	4–10	None known	Immobilization, loud, cooperation	Relative CBF
EEG/MEG	Intrinsic	10	0.01	None known	Artifact, lack of unique localization	Electrophysiology of brain events
MR spectroscopy	Radio waves	10	10–100	None known	Immobilization, loud	Relative chemical concentrations
Optical/near-IR intrinsic signals	Near-IR or visible light	0.05	0.05	None known	some immobilization, surface > depth, limited FOV	Relative CBF, absolute values may be possible
Nuclear (PET and SPECT)	Radiation	5–10	60–1000	Ionizing radiation	Radiation limits immobilization	Physiology, absolute values
Transcranial magnetic stimulation	Magnetic fields	10	0.01	Low risk of seizures	Immobilization, loud	Electrophysiology, conduction times
Post mortem	N/A	0.01	N/A	N/A	Post mortem changes	Microarchitectures, chemoarchitectures

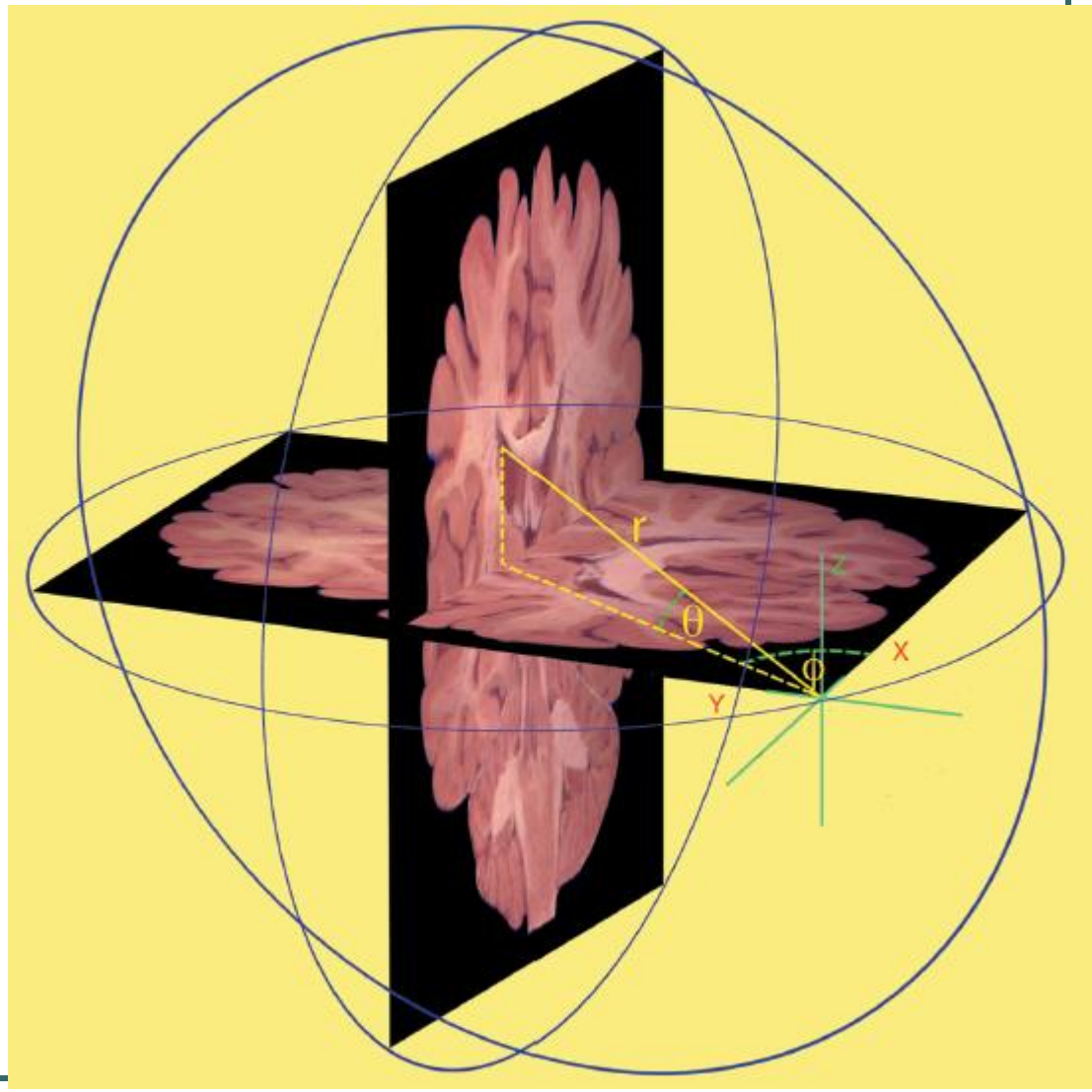
Note: N/A, not applicable

How to Study Brain?

- Q1: How to measure the brain?
  - imaging techniques
- Q2: How to represent the brain?
  - data format
- Q3: How to extract/mine structural and functional information of the brain?
  - analysis techniques
- Q4: How to understand and apply for clinical diseases
  - diagnosis and treatment

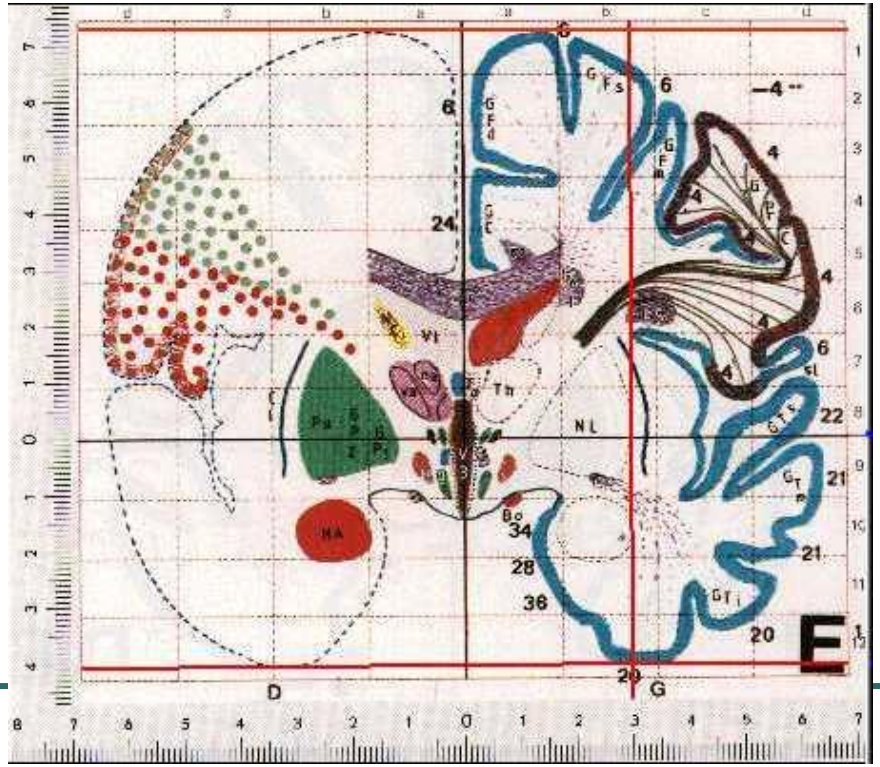
Coordinate referenced maps

- The use of coordinate systems (Cartesian and Polar) to identify any point within the 3D space occupied by the brain data is illustrated.
- Indexing is usually based on distance from a specific anatomic point.

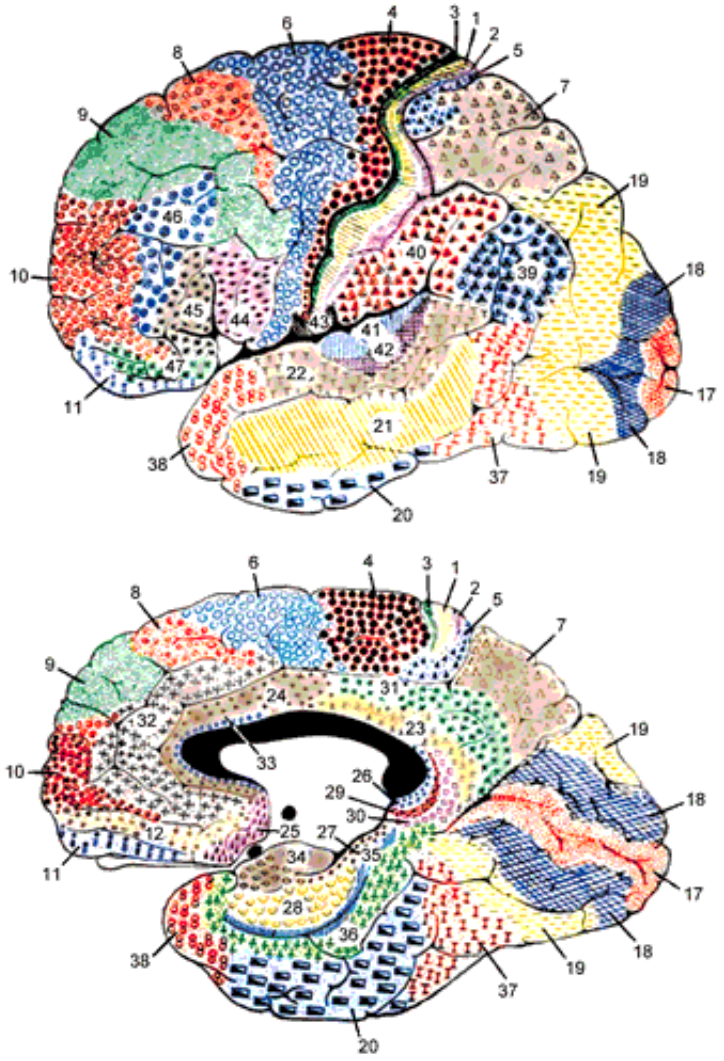


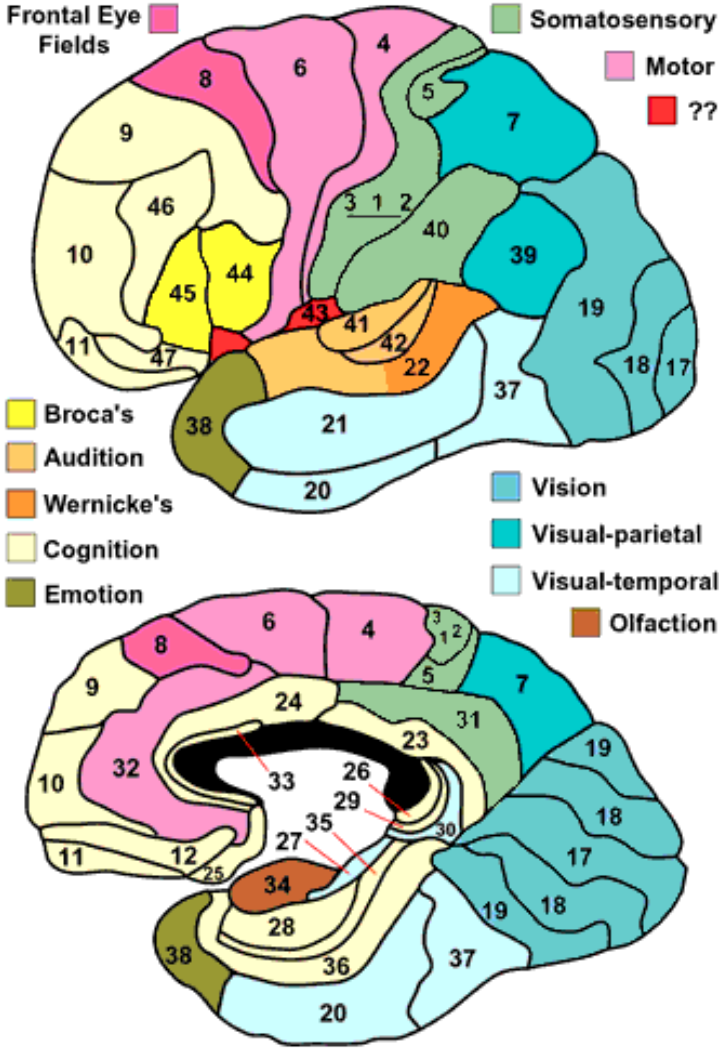
Talairach atlas

- Jean Talairach (January 15, 1911, Perpignan – March 15, 2007, Paris) was a neurosurgeon who practiced at the Centre Hospitalier Ste. Anne in Paris.
- A 61-year-old French woman (In the 1967 atlas the left is with positive coordinates while in the 1988 atlas the left has negative coordinates.)



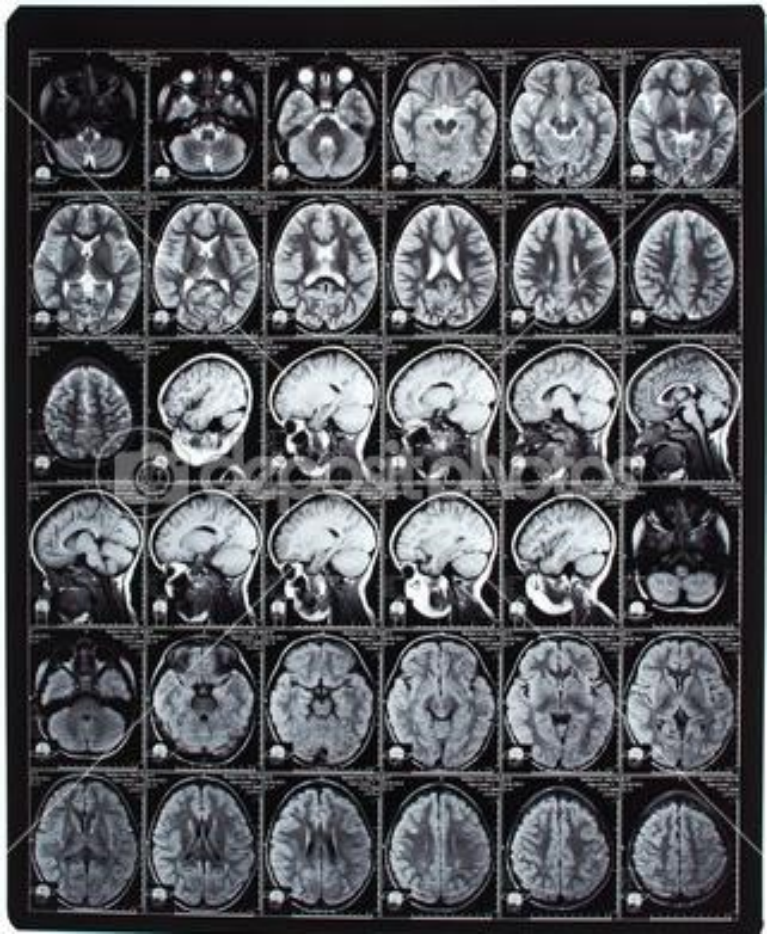
Brain Mapping (Korbinian Brodmann, 1909)



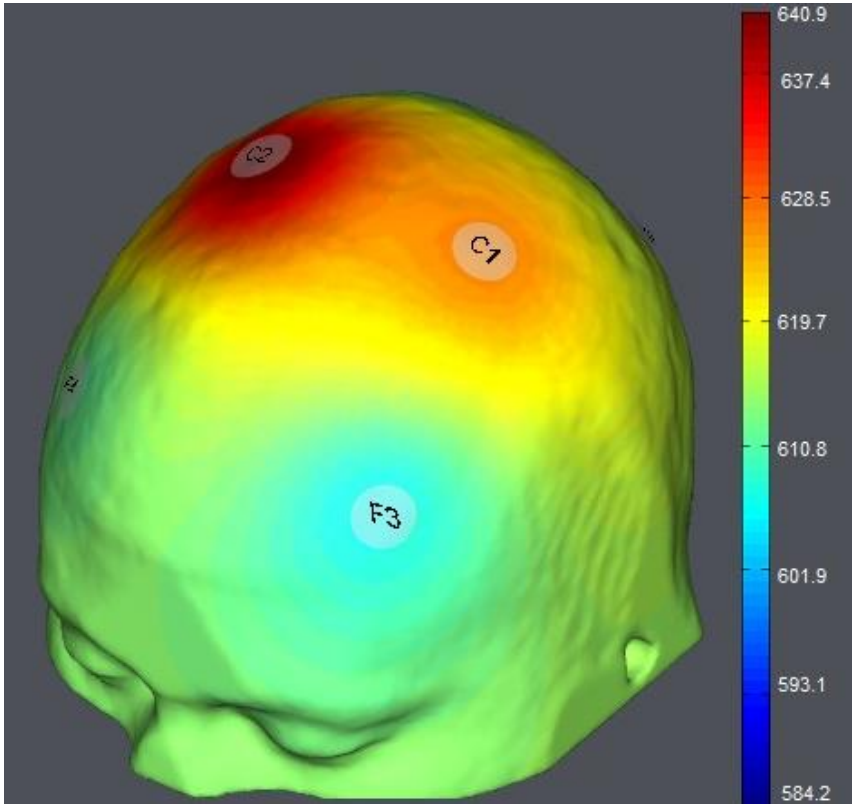


Representing the brain

- Tomographic-based vs. Surface-based



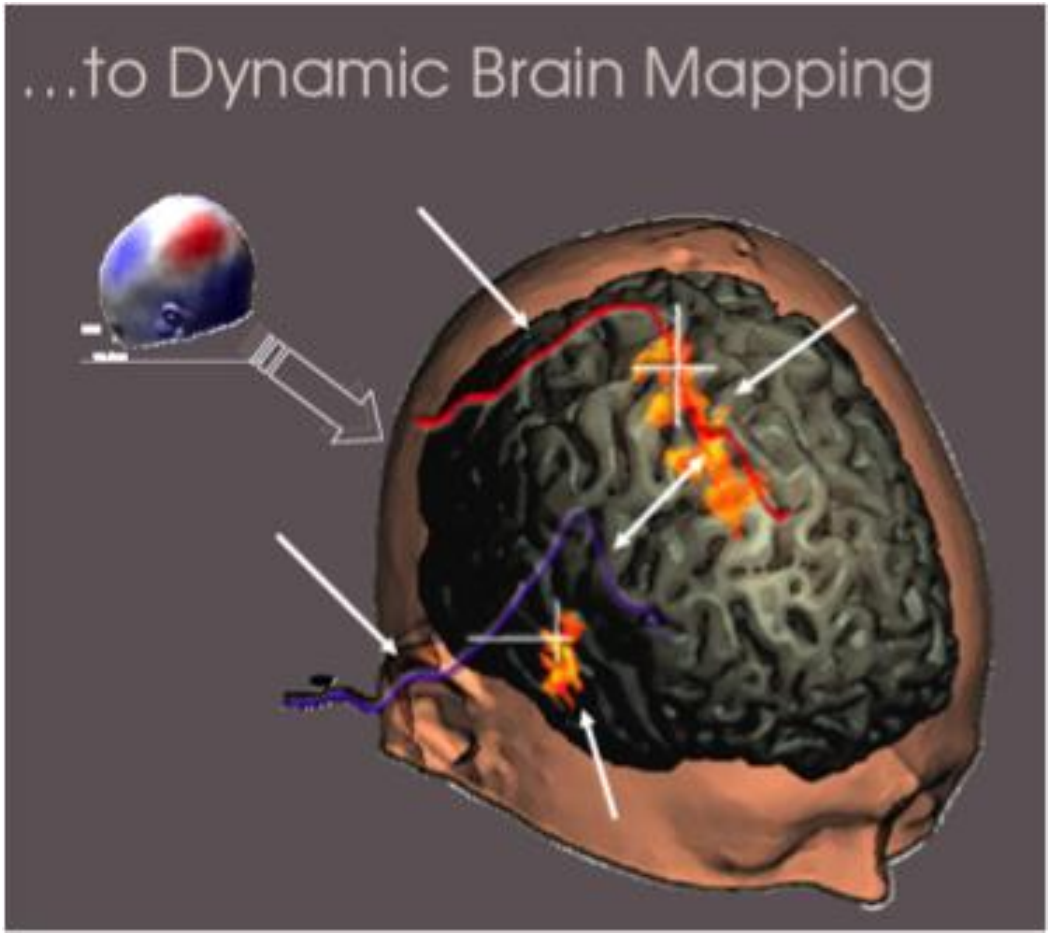
(http://depositphotos.com/13373661/stock-photo-X-ray-image-of-the-brain.html)



(http://blog.neuroelectrics.com/Portals/181943/images/what_is_qeeg.png)

Representing the brain

- static vs. dynamic



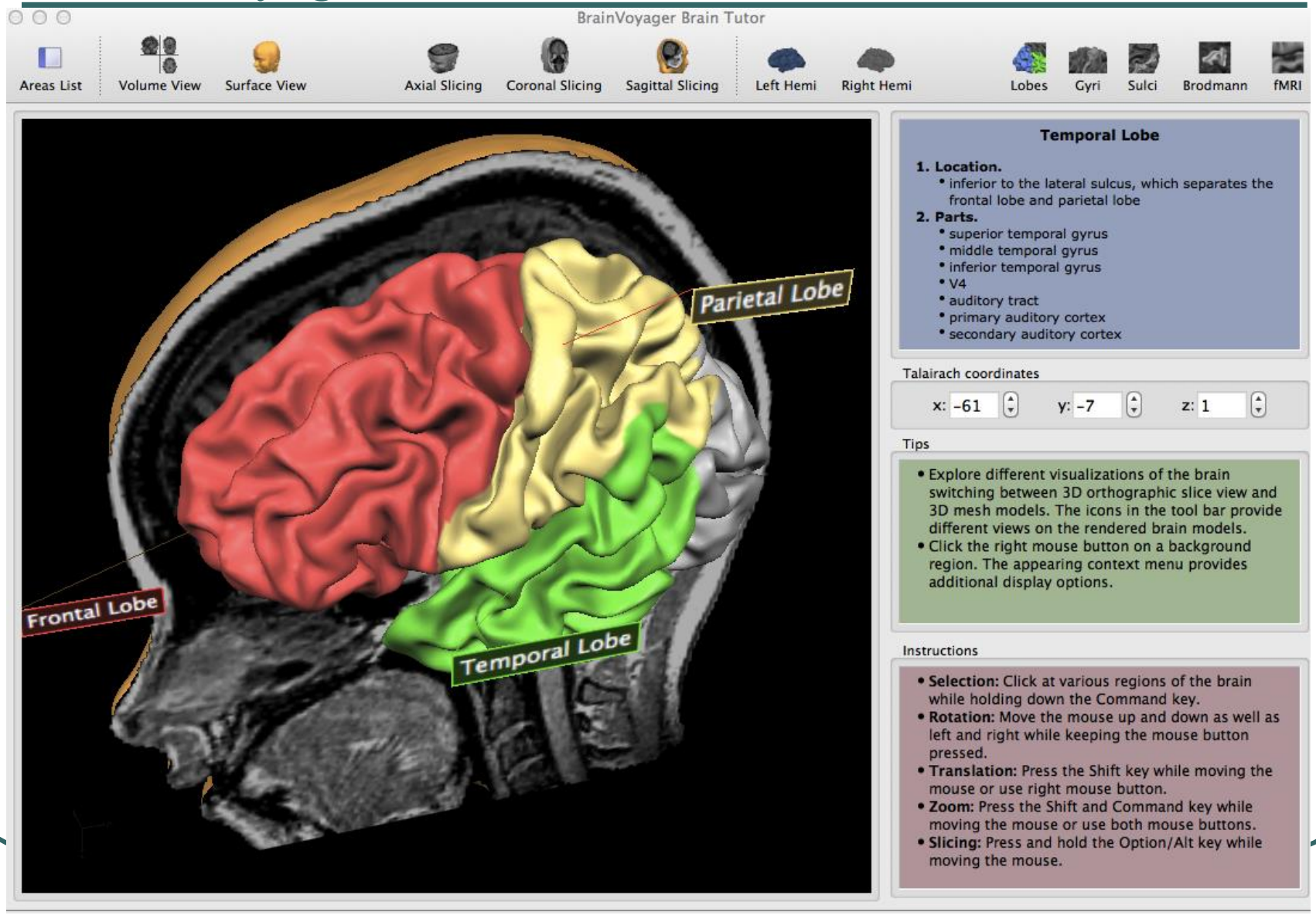
International Consortium for Brain Mapping (ICBM)



Core Research Partners

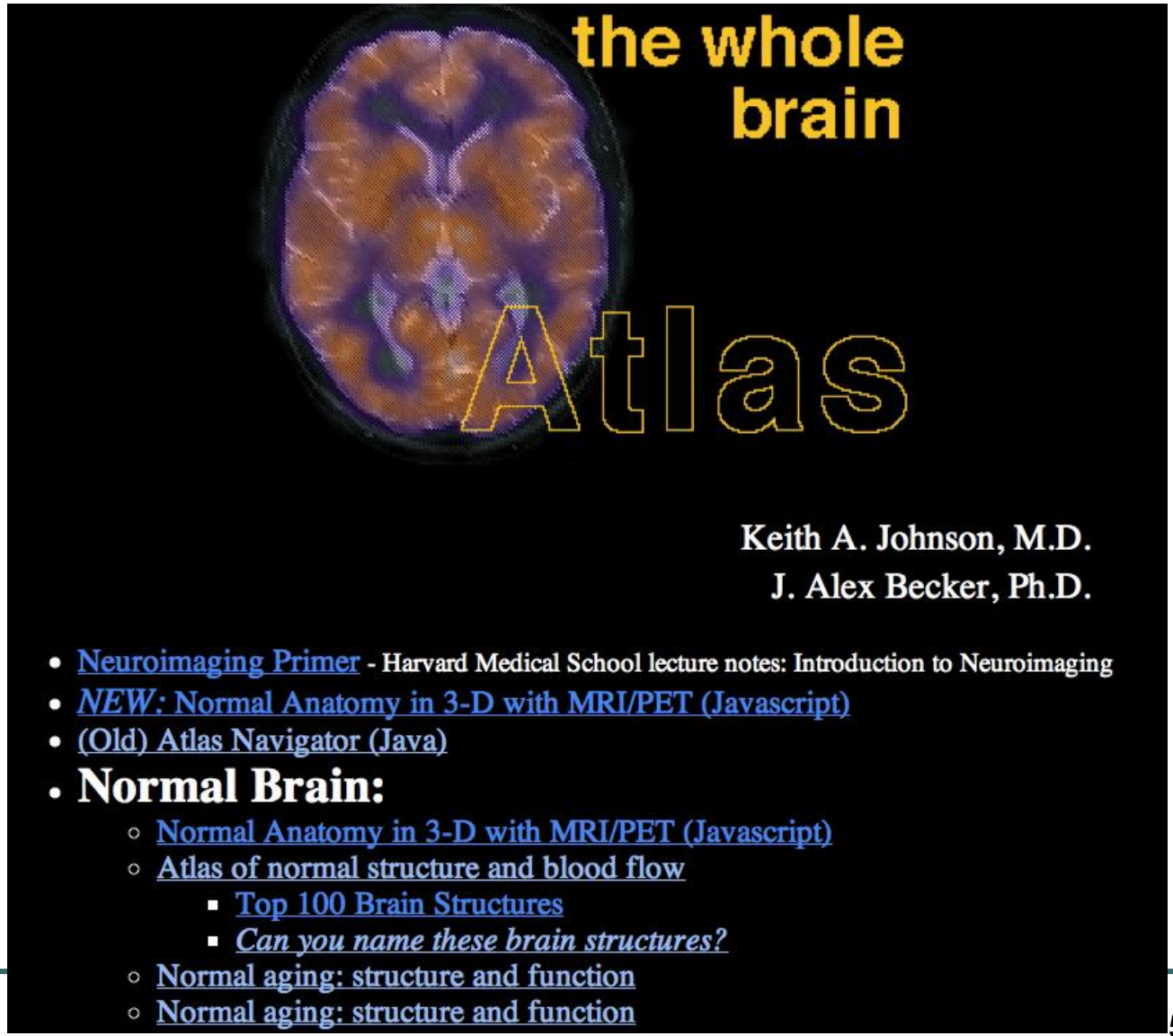
- Laboratory of Neuro Imaging
- McConnell Brain Imaging Centre
- University of Texas at San Antonio Research Imaging Center
- C & O Vogt Institute of Brain Research
- The Insitute of Medicine in Juelich

互動式腦結構與功能資料庫應用軟體 (BrainVoyager Brain Tutor)



線上互動式腦結構與功能影像系統(The Whole Brain Atlas)

http://www.med.harvard.edu/AANLIB/home.html

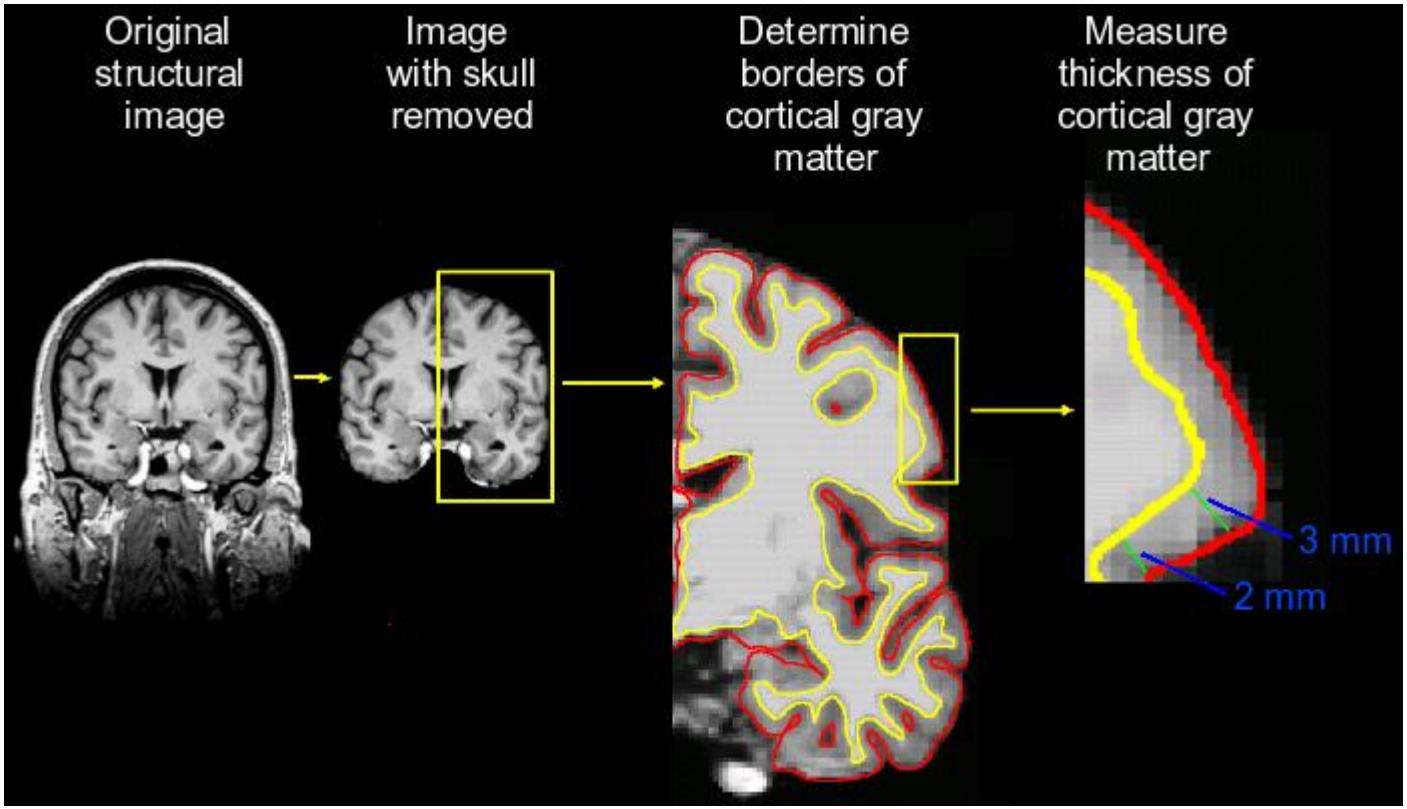


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Anatomical Information

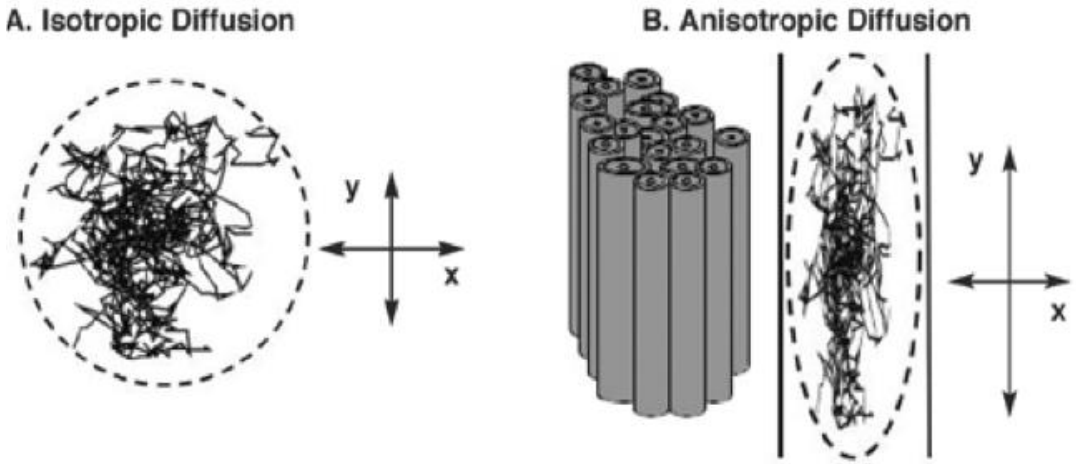
- Ex. 1: Gray matter volumes and cortical thickness extracted from T1/T2-weighted images



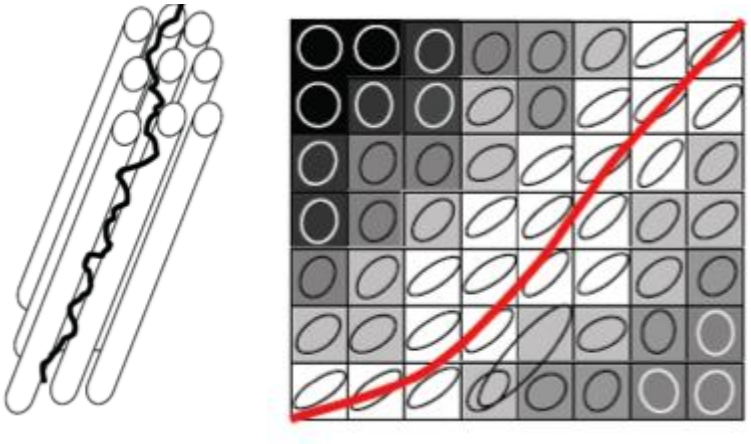
(http://www.martinos.org/neurorecovery/images/MRI_sequence.png)

Anatomical Information

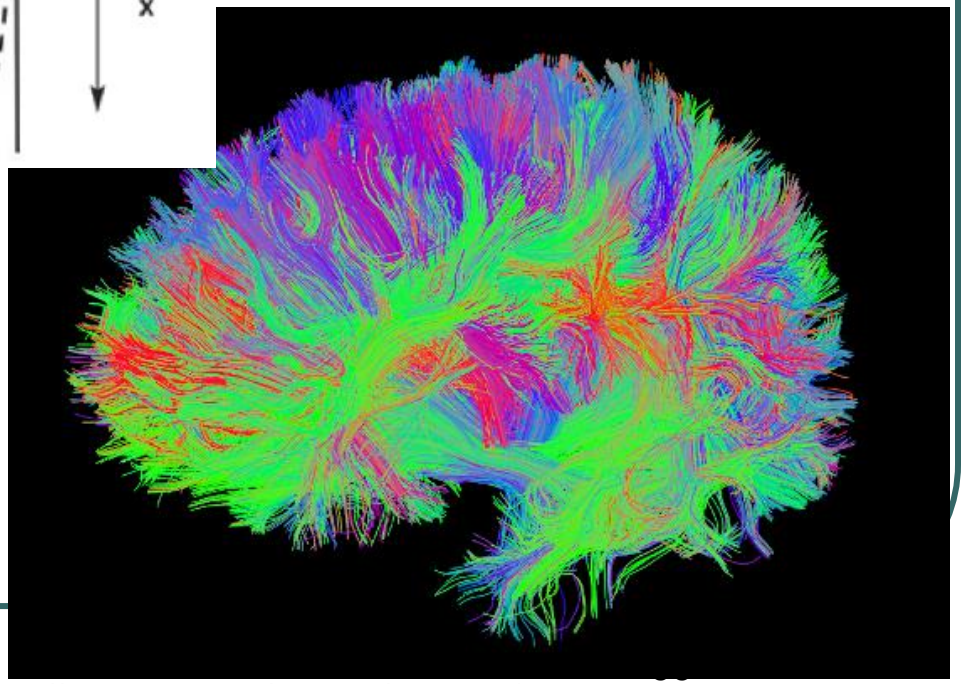
- Ex. 2: White matter tracks extracted from diffusion weighted images (tractography)



Voxels are connected based upon similarities in the maximum diffusion direction.

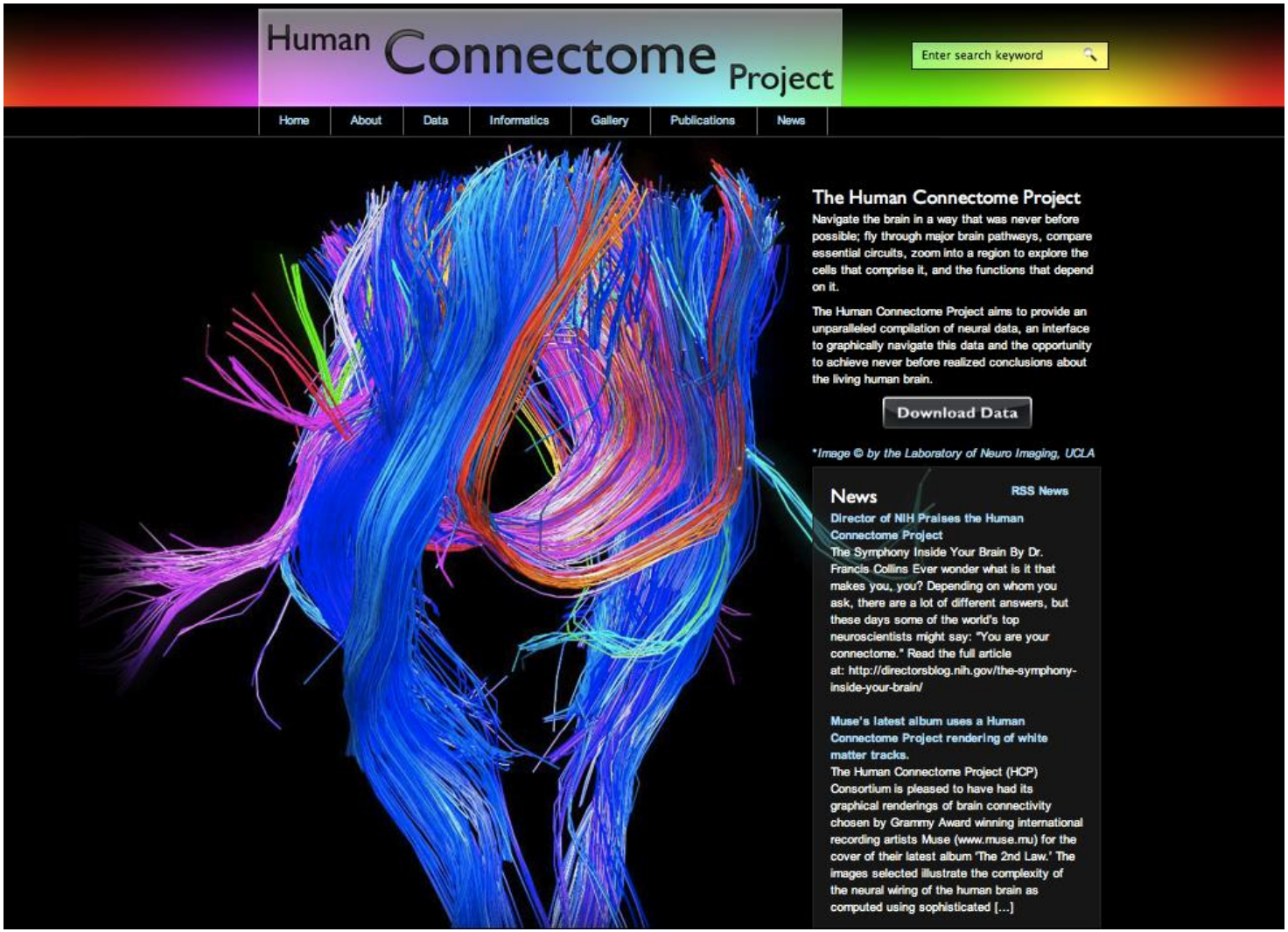


(www.fil.ion.ucl.ac.uk)



http://www.cmtk.org/mapper/_images/ex_tractography1.png

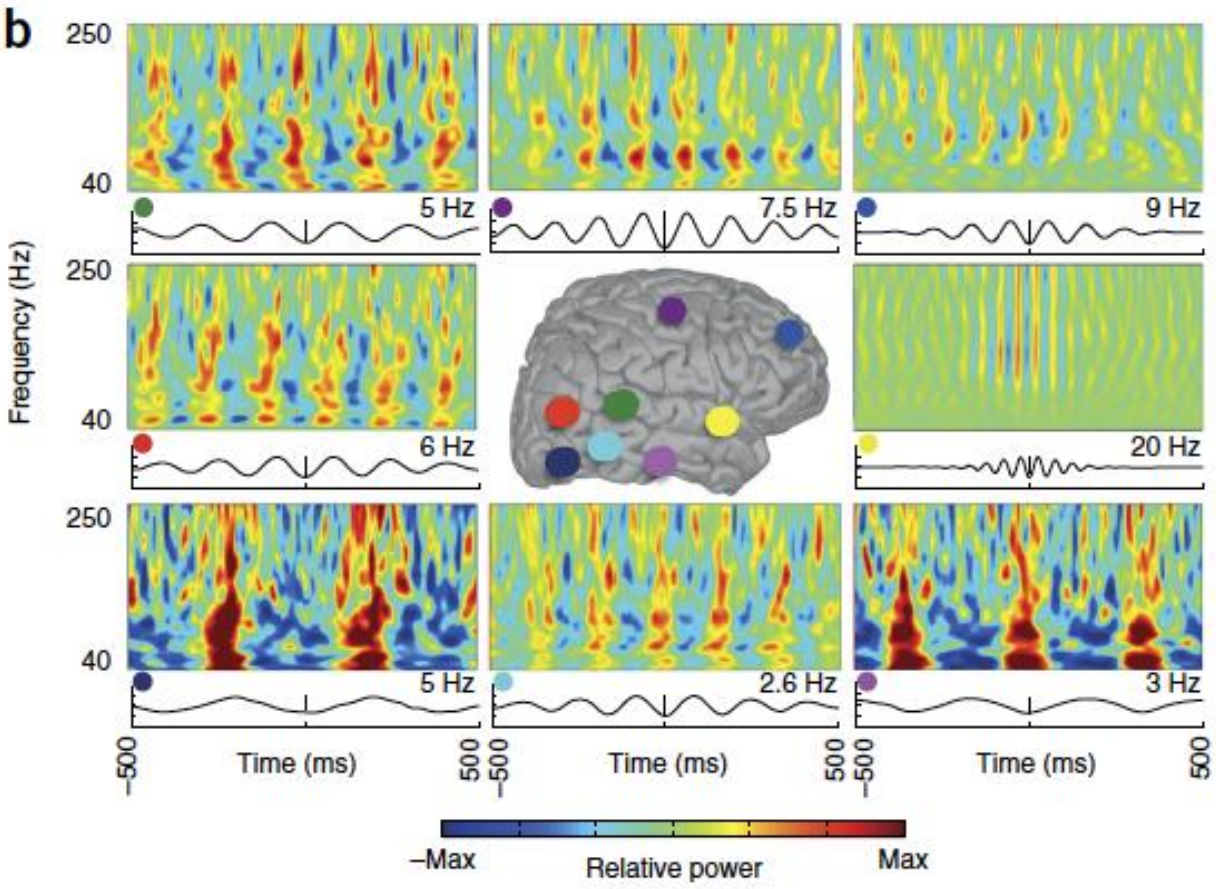
Anatomical Information: Databasing the brain



(http://www.humanconnectomeproject.org)

Functional Information

- Ex. 1: to extract dynamic activity of functional connectivity using time-frequency analysis from EEG/MEG data



Functional Information

- Ex. 2: inter-region connectivity

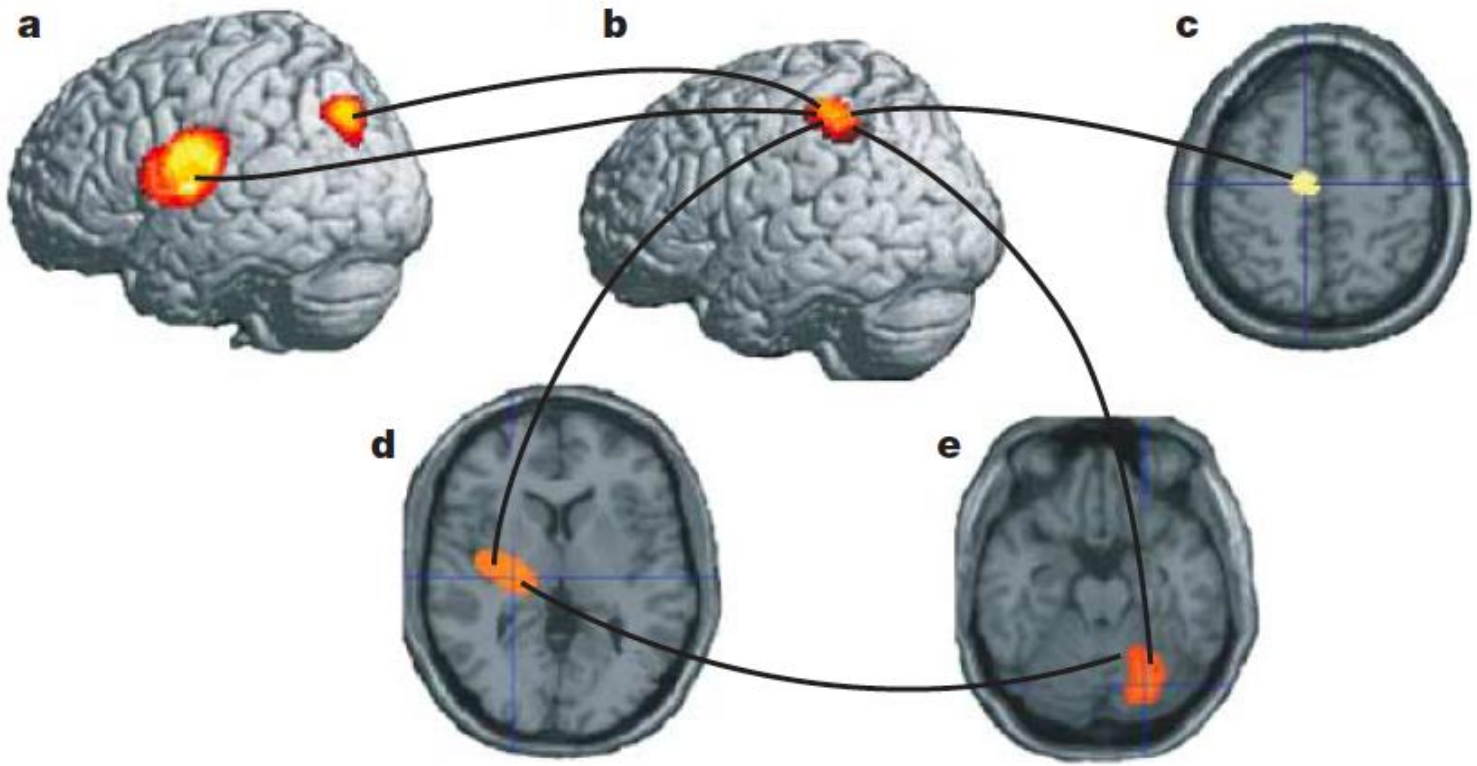
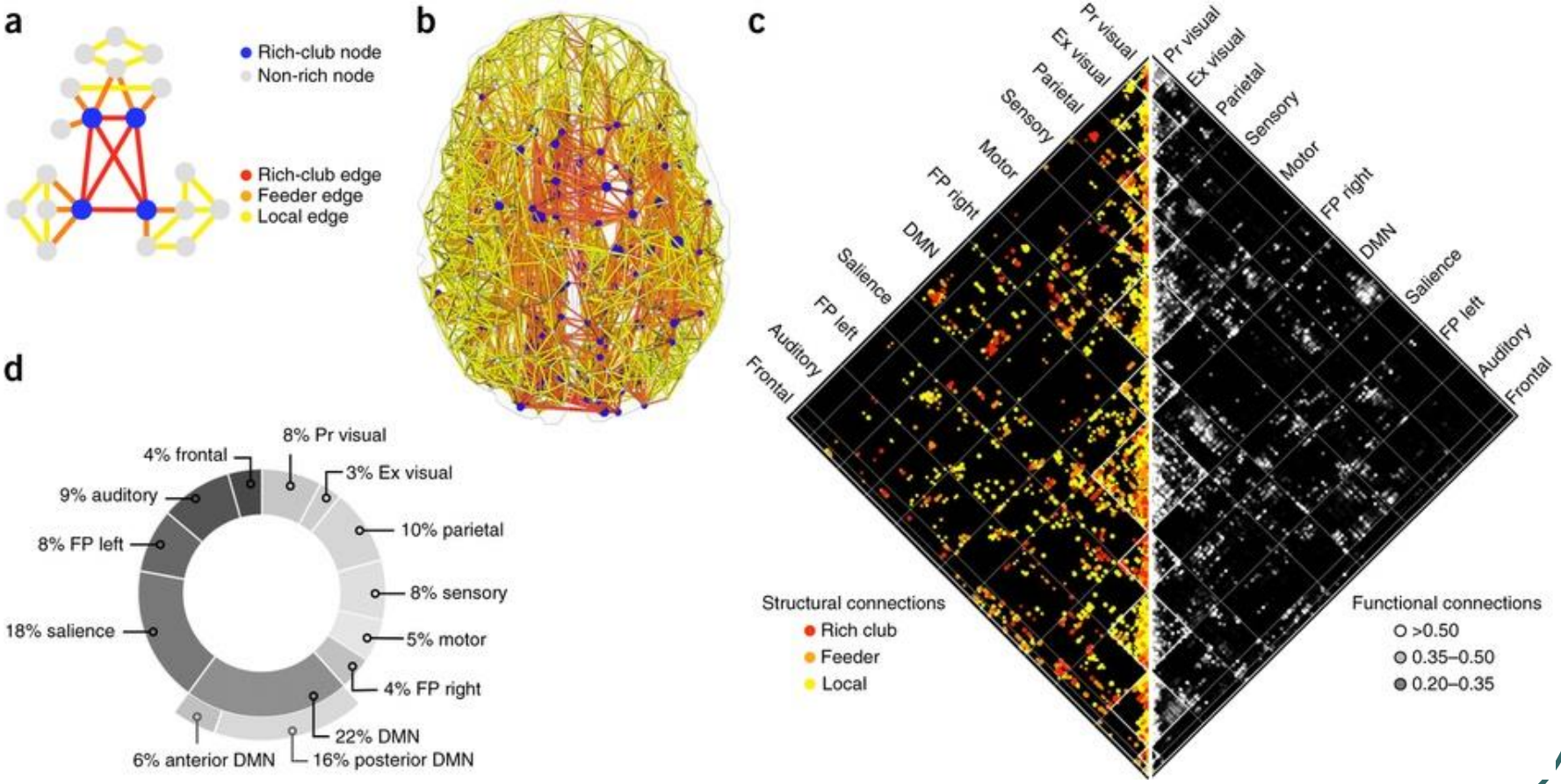


Figure 6 | **Long-range synchronization in Parkinson's disease.** Maps of spatially

(Schnitzler and Gross, Nat. Rev. Neurosci. 2005)

Functional Information

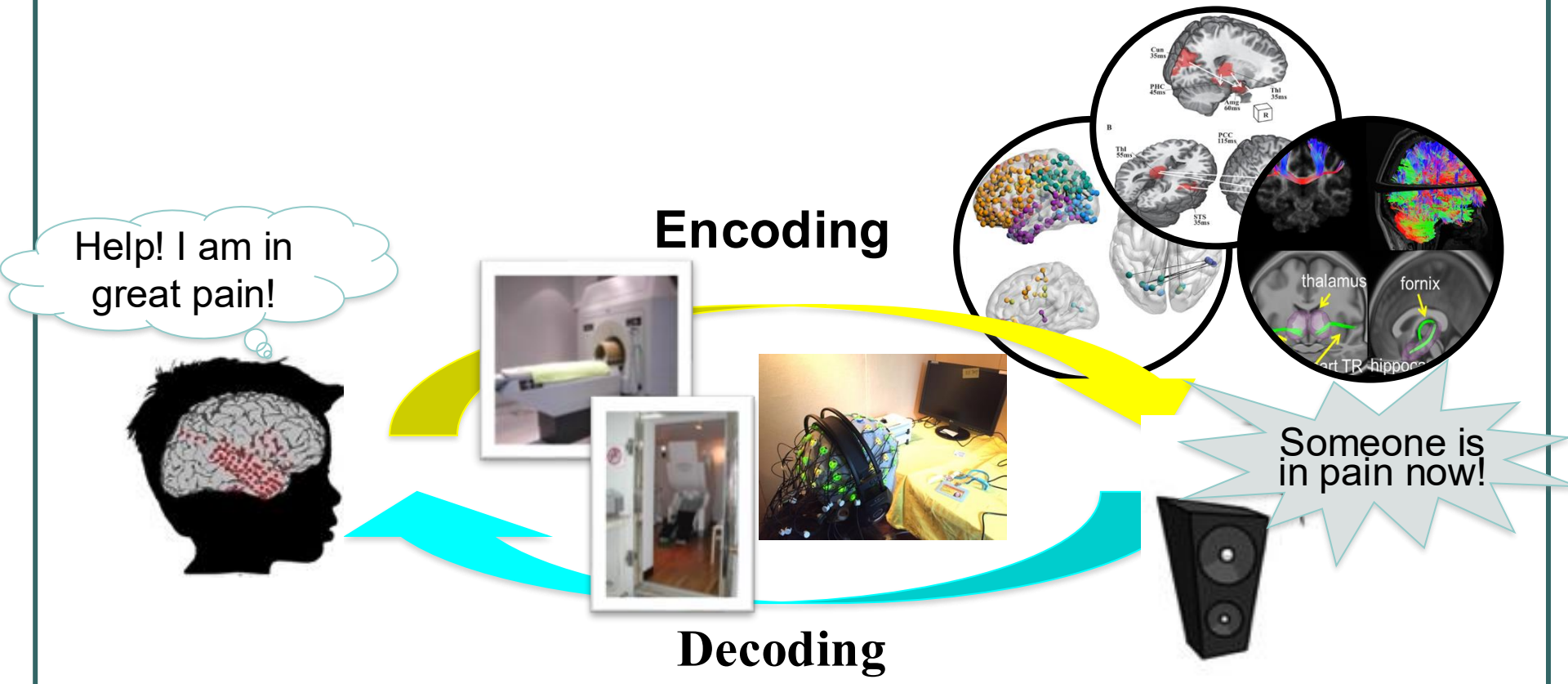
- Ex. 3: connectome analysis



http://bispl.weebly.com/neuro-imaging-and-analysis.html

Functional Information

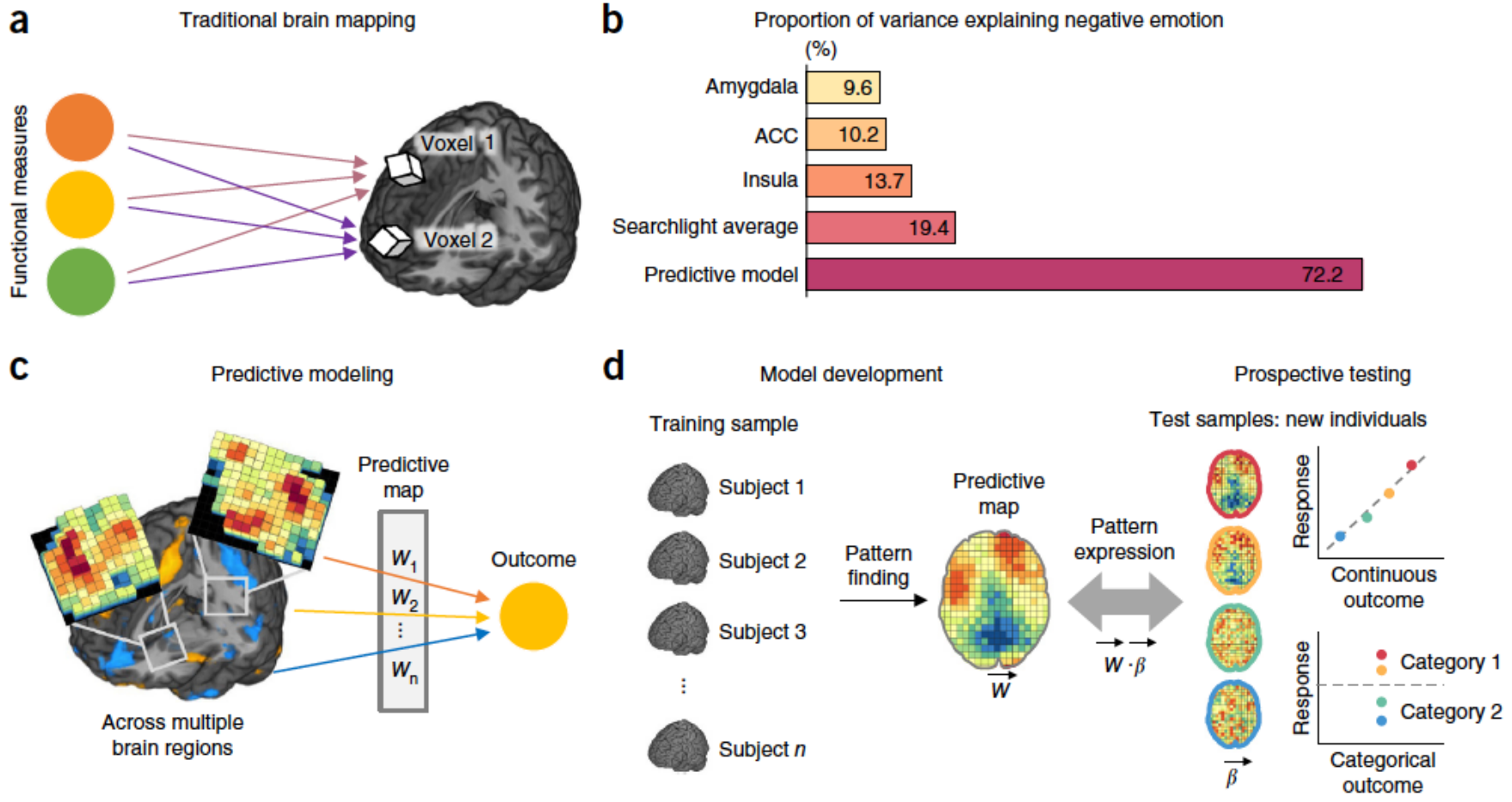
- Ex. 4: Encoding/Decoding of brain activity



How to Study Brain?

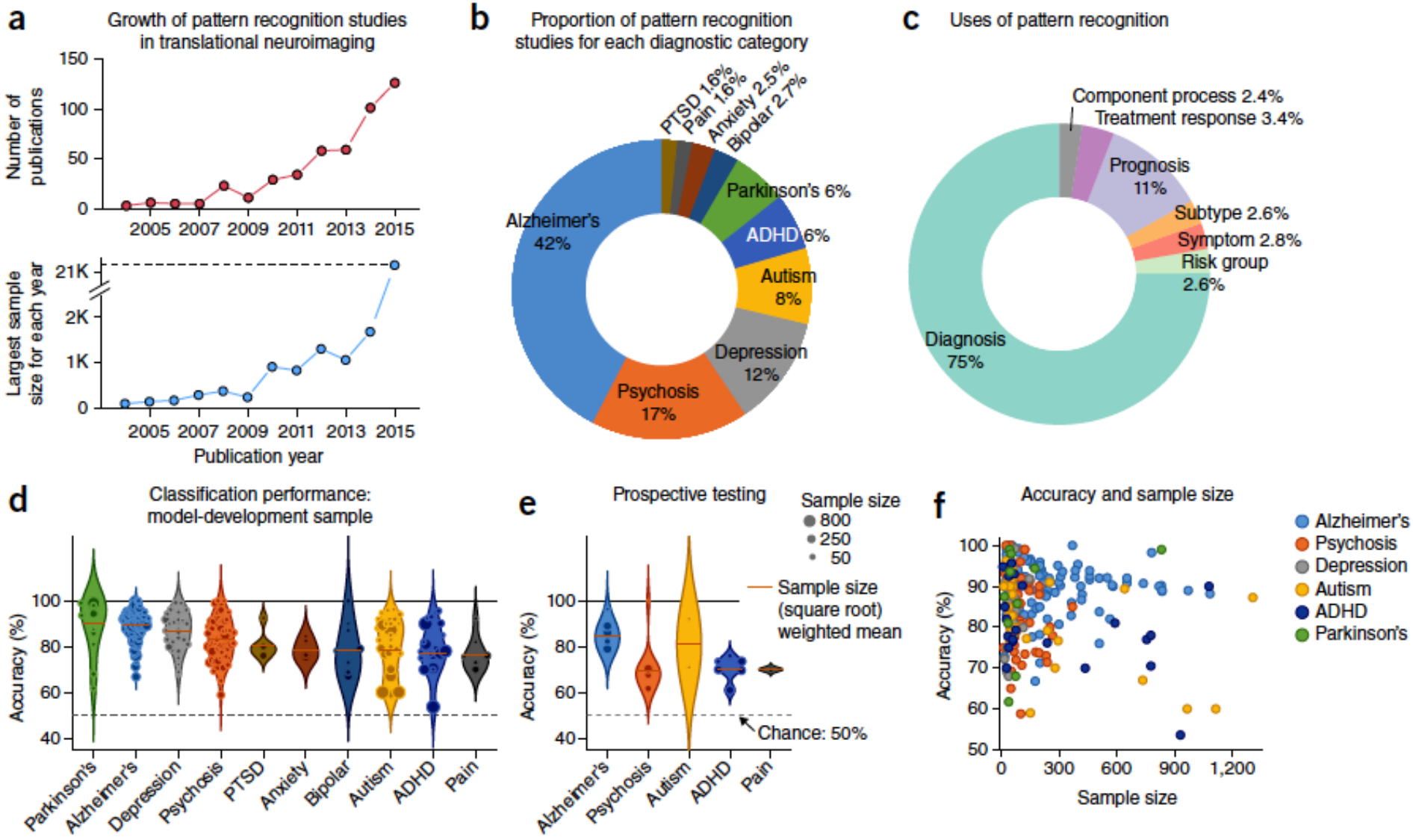
- Q1: How to measure the brain?
  - imaging techniques
- Q2: How to represent the brain?
  - data format
- Q3: How to extract/mine structural and functional information of the brain?
  - analysis techniques
- Q4: How to apply to translational neuroscience?
  - diagnosis and treatment

Multivariate neuroimaging would help understanding how the brain works



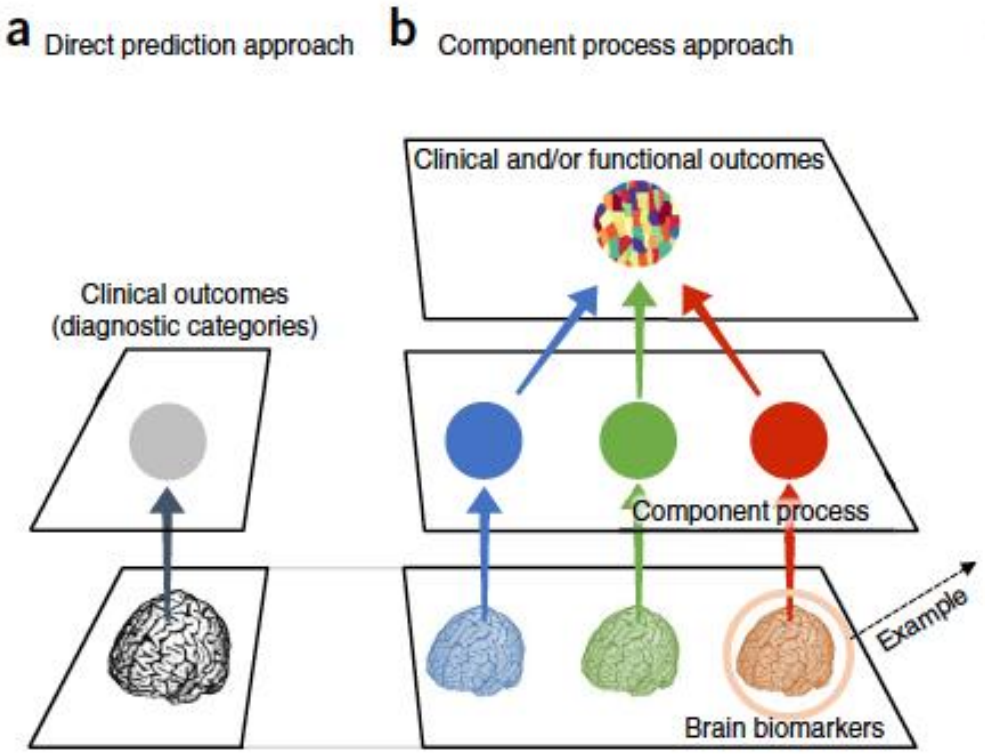
(Woo et al., *Nature Neuroscience*, Feb. 23 2017)

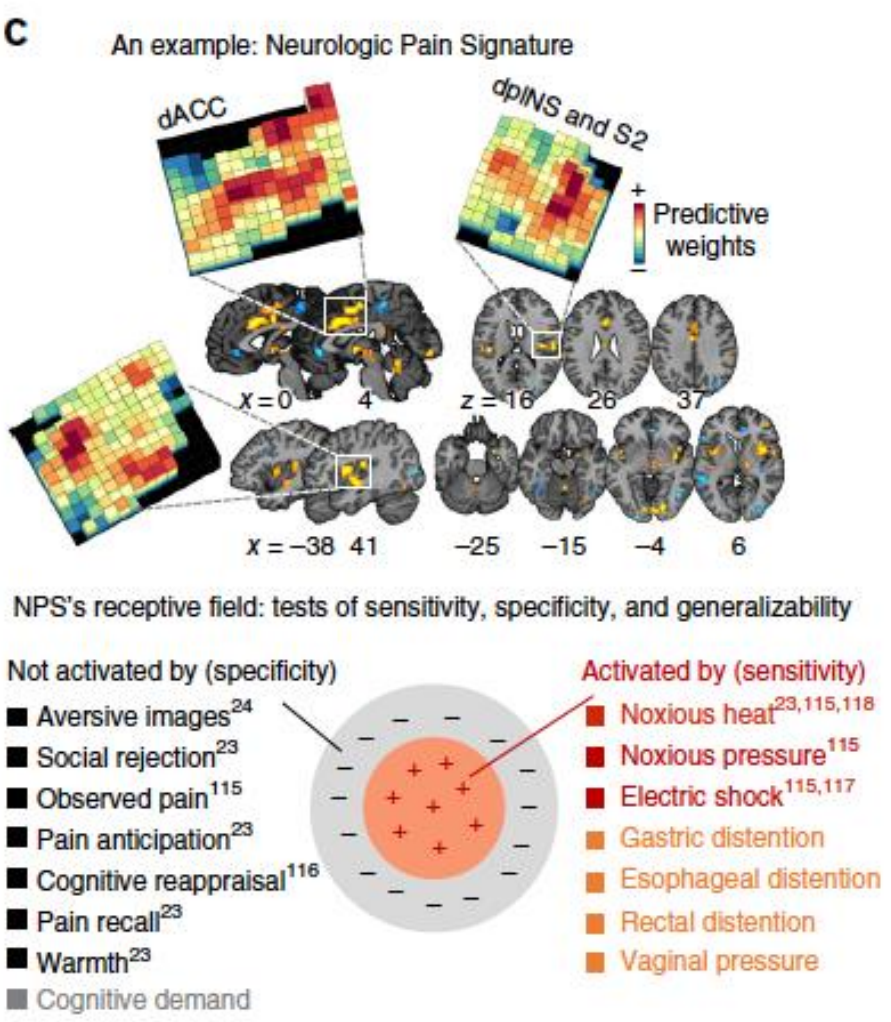
Translational neuroimaging using multivariate predictive models



(Woo et al., *Nature Neuroscience*, Feb. 23 2017)

Future directions for neuroimaging techniques





(Woo et al., *Nature Neuroscience*, Feb. 23 2017)

Image Reconstruction using fMRI

- Reconstructing Visual Experiences from Brain Activity Evoked by Natural Movies



Presented clip



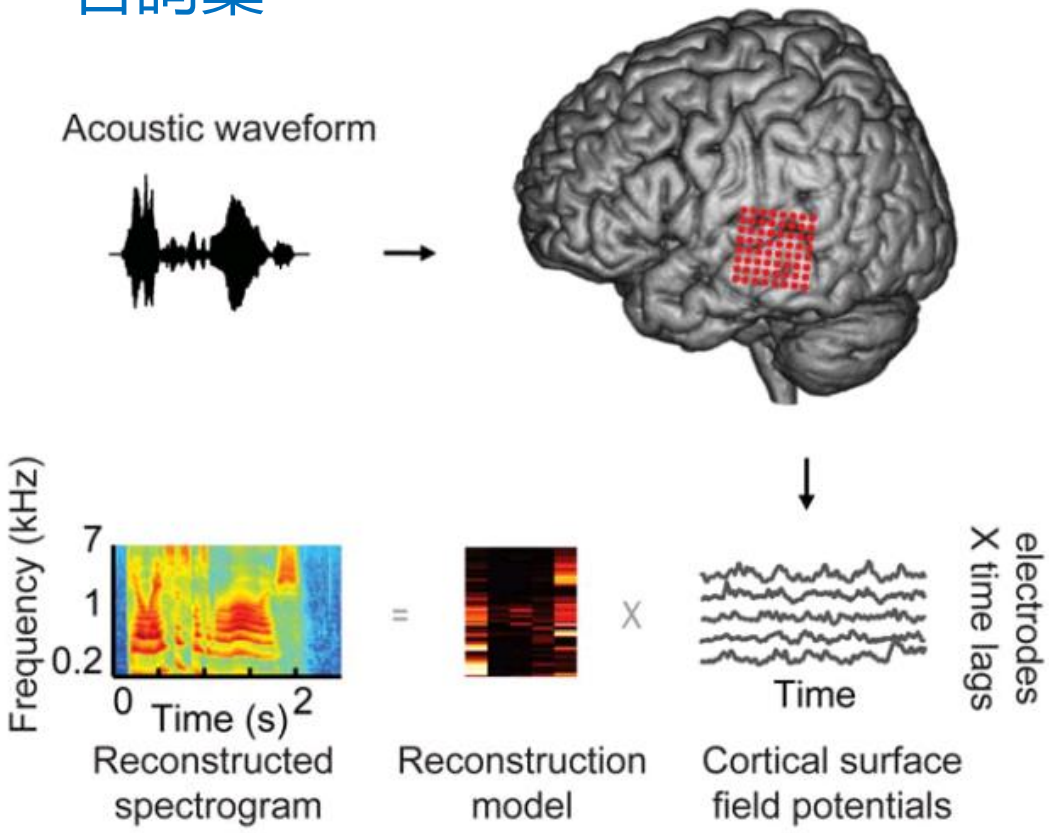
Clip reconstructed from brain activity

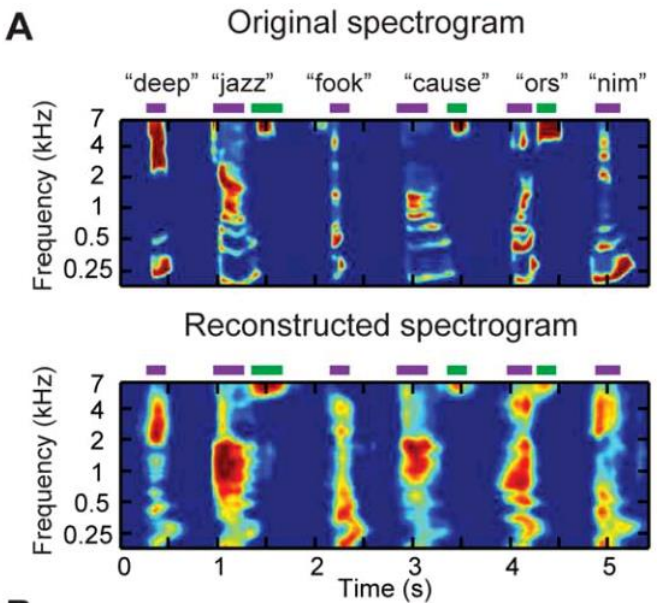


(Current Biology 21, 1–6, October 11, 2011)

我知道你在聽什麼--聲音重現

- 利用顱內腦電圖(ECoG)技術，重現受試者所聽到之語音詞彙



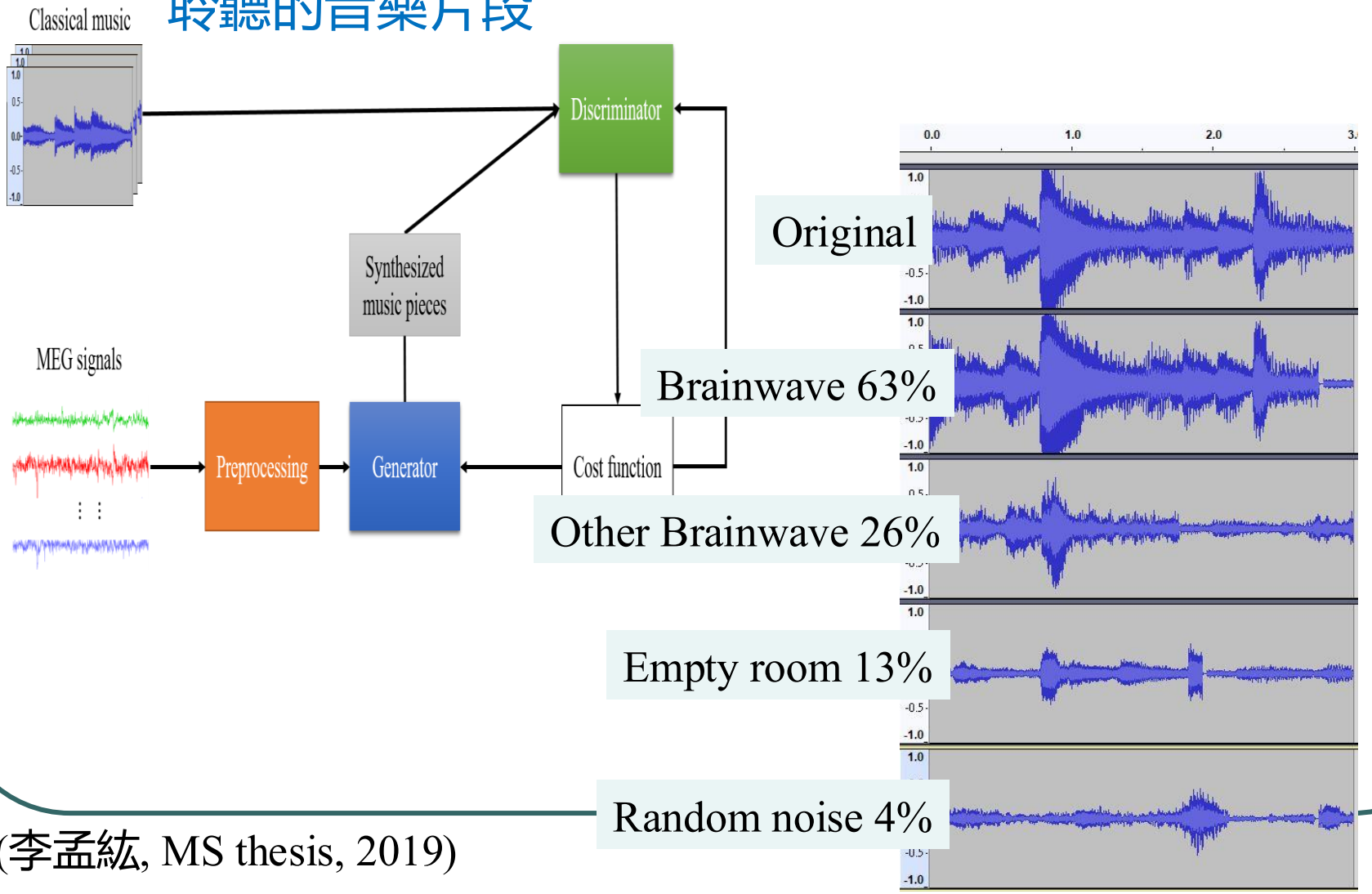


Pasley, PLOS Biology, 2012

我知道你在聽什麼--聲音重現

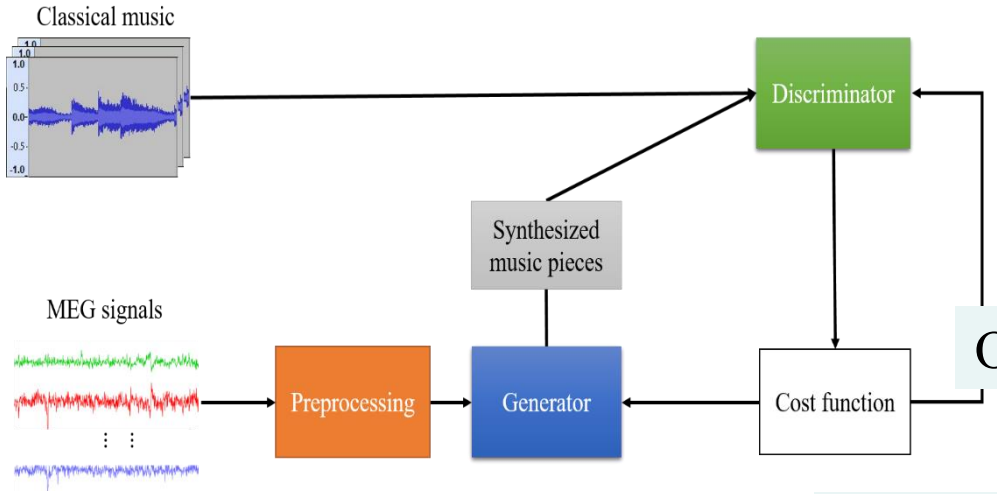


- 利用腦磁圖與深度學習技術，重現受試者所聆聽的音樂片段

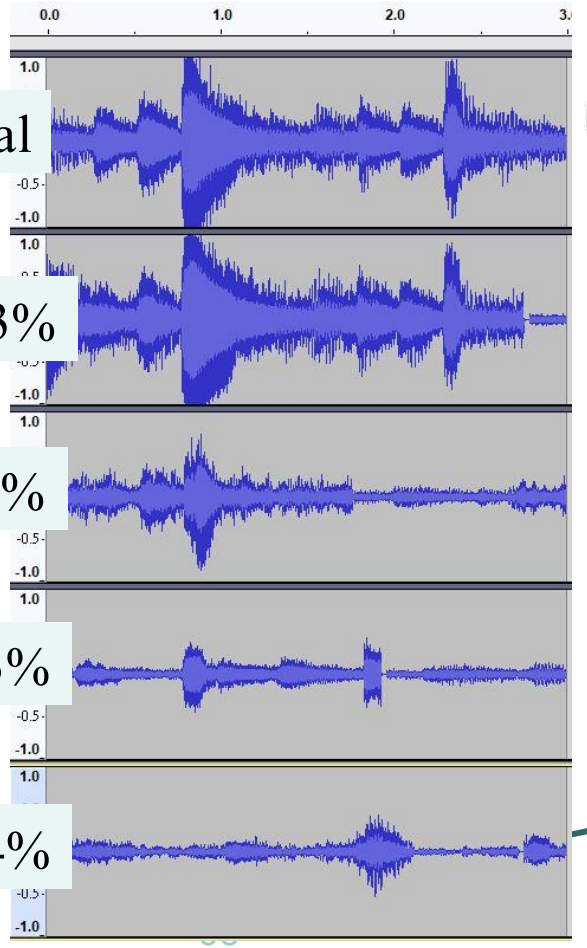


(李孟紘, MS thesis, 2019)

我知道你在聽什麼--聲音重現



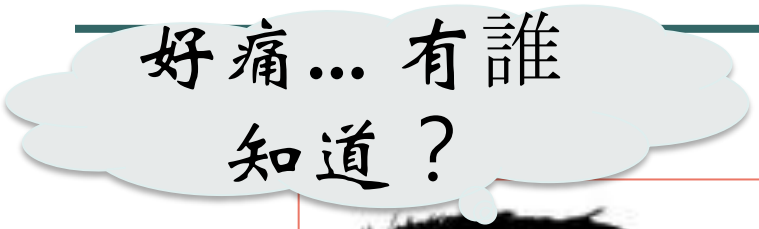
以聆聽該樂曲的腦波所生成的音樂，其與原樂曲之相似度，可代表音樂被編碼於腦中的程度

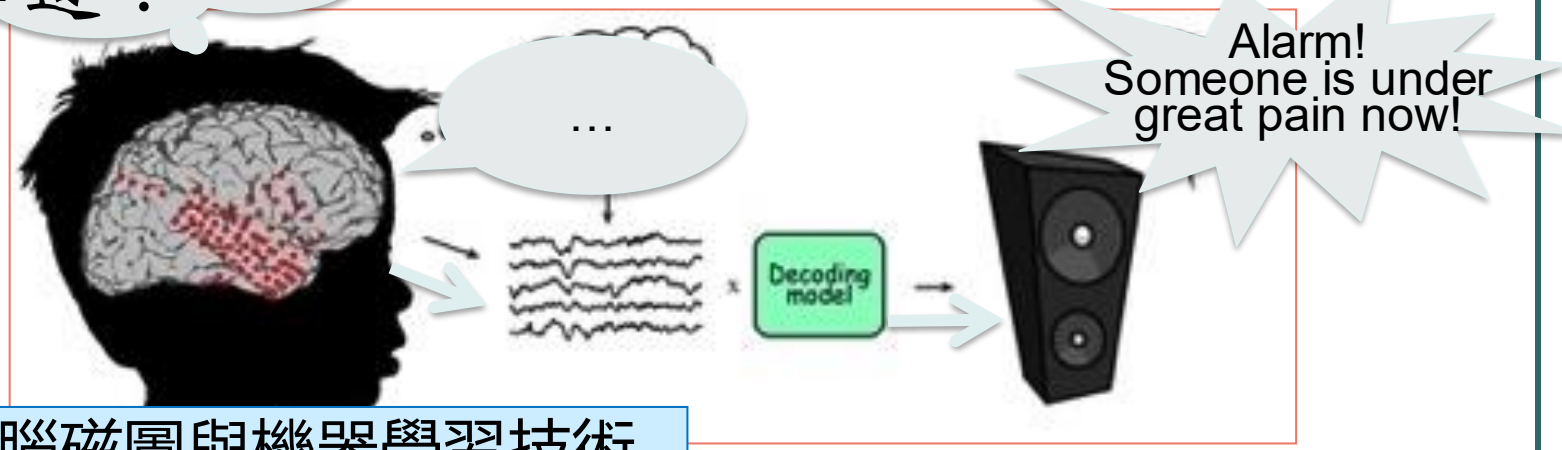


利用腦磁圖與深度學習技術，重現受試者所聆聽的音樂片段

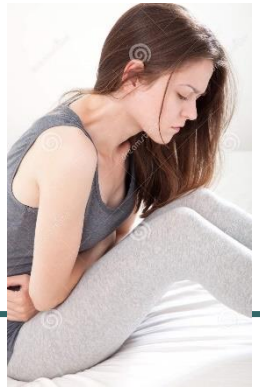
(李孟紘, MS thesis, 2020; Lee et al.)

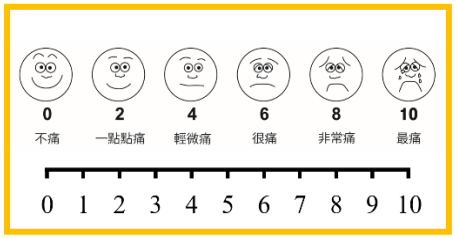
我知道你有多痛....



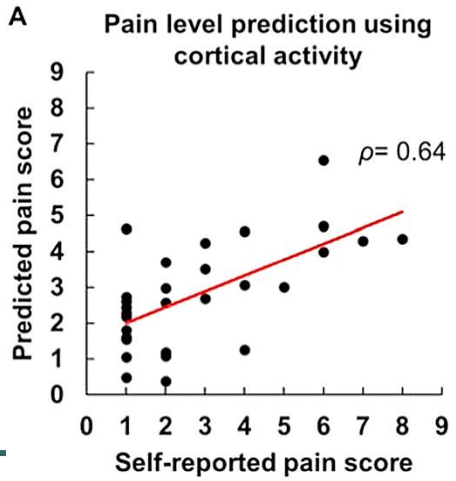


利用靜息態腦磁圖與機器學習技術估測受試者疼痛分數







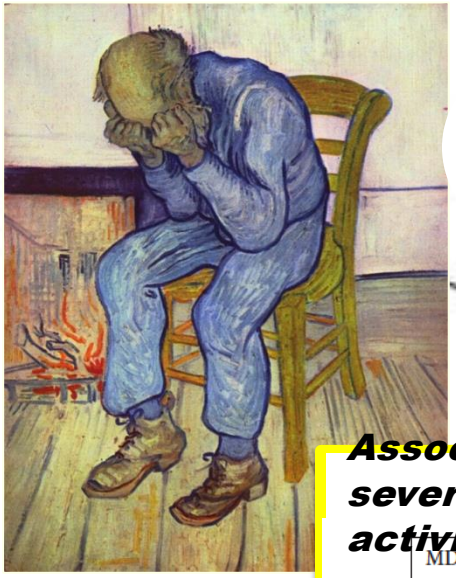


(Kuo et al., Human Brain Mapping, 2018)

我知道你有多憂鬱 …

I am in depressed…

開發深度學習技術，進行**情感性精神疾患分類**，並呈現**腦波相關異常**以找出生物標記



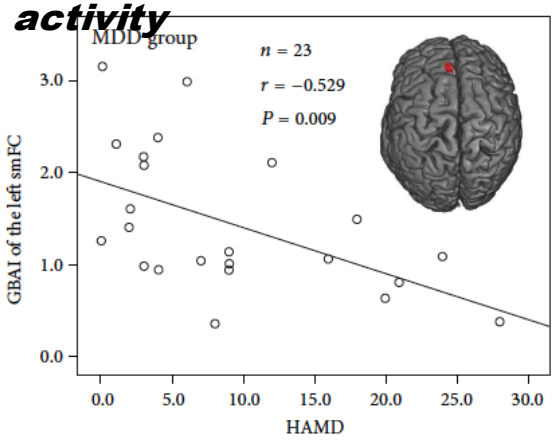


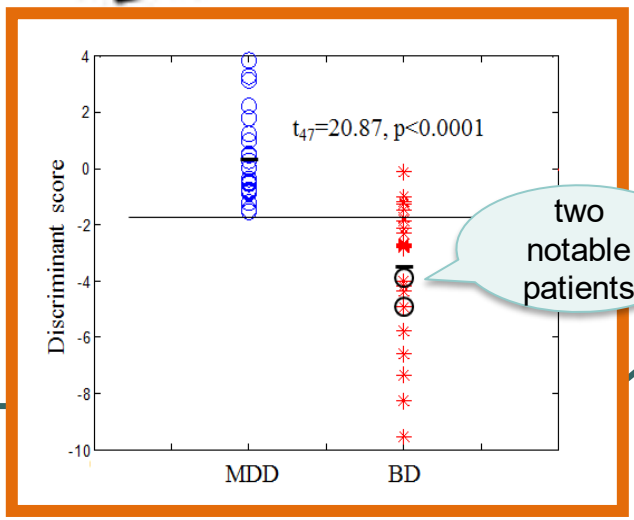


Depression Alarm!

(HAMD = 27)

Association of depressive severity with gamma activity





(Liu et al. *BioMed Research International*, 2014)

我知道你有多憂鬱 …

MEG-based Classification and Grad-CAM Visualization for Major Depressive and Bipolar Disorders with Semi-CNN

特徵擷取

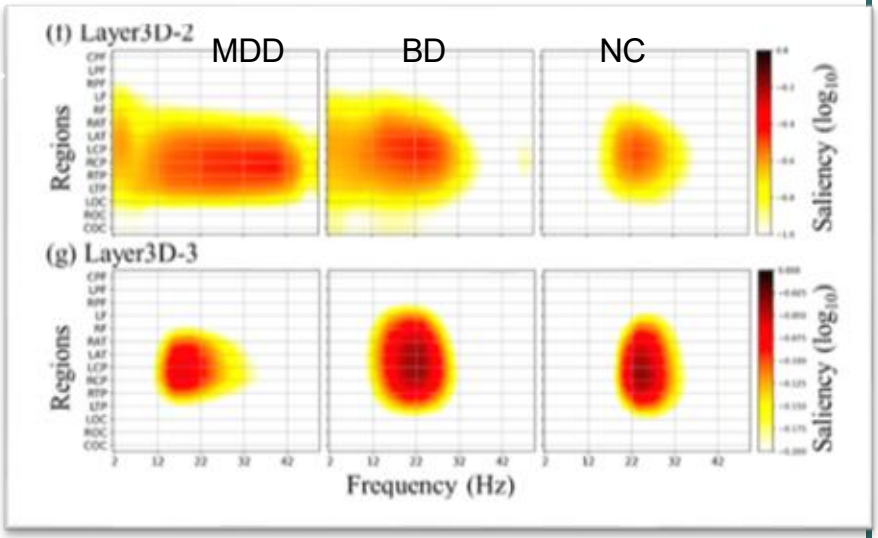
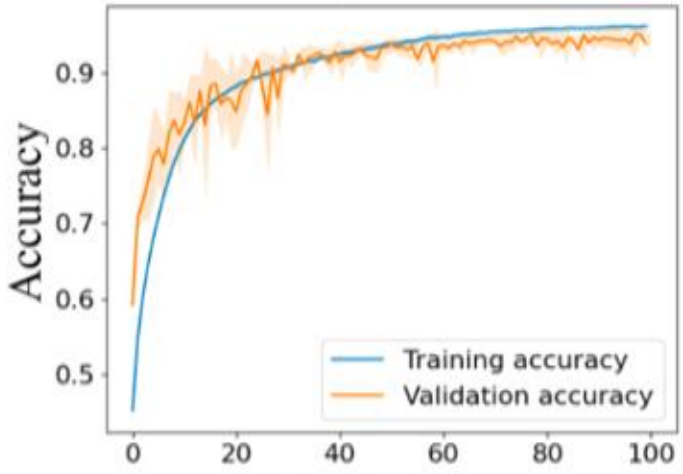


TABLE I. DETAILS OF THE MODEL ARCHITECTURE

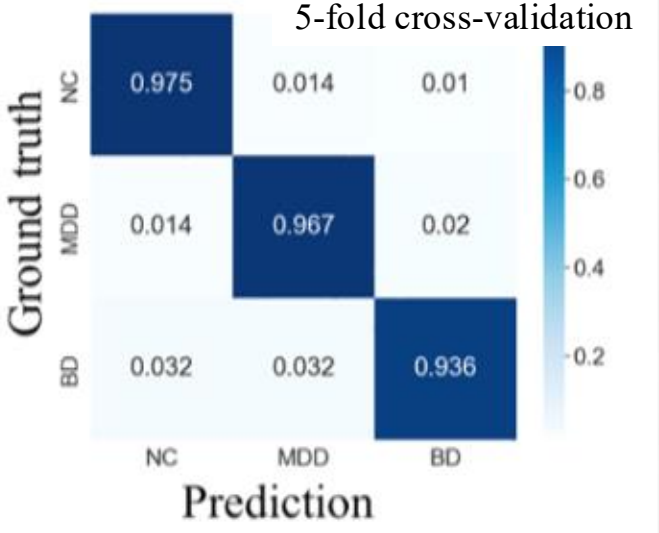
Layers	Output size	Blocks
Input	400×48×14	
Activation	102×26×14	[7×3×3, 64]
Max-pool	51×13×7	[3×3×3], stride 2
Conv2D-1	51×13×7	[Basic[a] 1×3×3, 64] ×3
Conv2D-2	51×13×7	[Basic 1×3×3, 128] ×2
Conv1D	51×13×7	[Basic 3×1×1, 128] ×2
Conv3D-1	51×13×7	[Basic 3×3×3, 128] ×3
Conv3D-2	51×13×7	[Basic 3×3×3, 256] ×3
Conv3D-3	51×13×7	
Adaptive average pooling	512×1	
Dropout	512×1	
Fully connected	100×1	
Dropout	100×1	
Fully connected	3×1	

a. BasicBlock, includes two con

系統表現



5-fold cross-validation



Huang et al. (2022). IEEE EMBC

Brain-Computer Interface
腦機介面 – Robot arm

Thought control of robotic arms using the BrainGate system
(Brown Institute for Brain Science)



Brain-Computer Interface
腦機介面 – Robot arm

- Cortical control of a prosthetic arm for self-feeding



Nature **453**, 1098-1101 (19 June 2008)

EEG-Based Brain-Computer Interactions in Immersive Virtual and Augmented Reality: A Systematic Review

CHUKWUEMEKA NWAGU, Dalhousie University, Canada
ALAA ALSLAITY, Dalhousie University, Canada
RITA ORJI, Dalhousie University, Canada



Fig. 1. Sample images from the reviewed studies. (A) Sublime: four SSVEP targets for hands-free navigation are displayed in the VR application [5] (B) Lucid Loop: an example of the VR experience of simulated lucid dreaming [61] (C) TRAVEE: a patient using the rehabilitation system [84] (D) a P300-based drone control system using a DSI-VR300 integrated EEG electrodes and VR headset [60]. (Images are reproduced with permission of the authors)

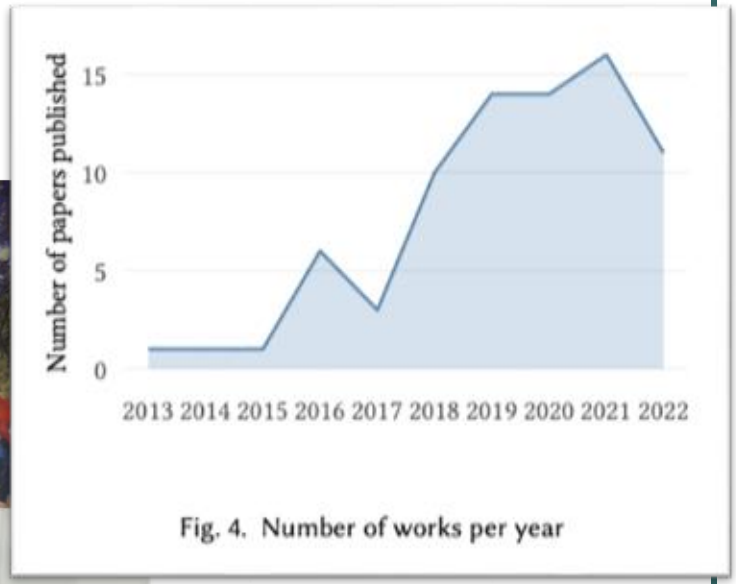


Fig. 4. Number of works per year

C Nwagu, et al. Proceedings of the ACM on Human-Computer Intersaction, 2023

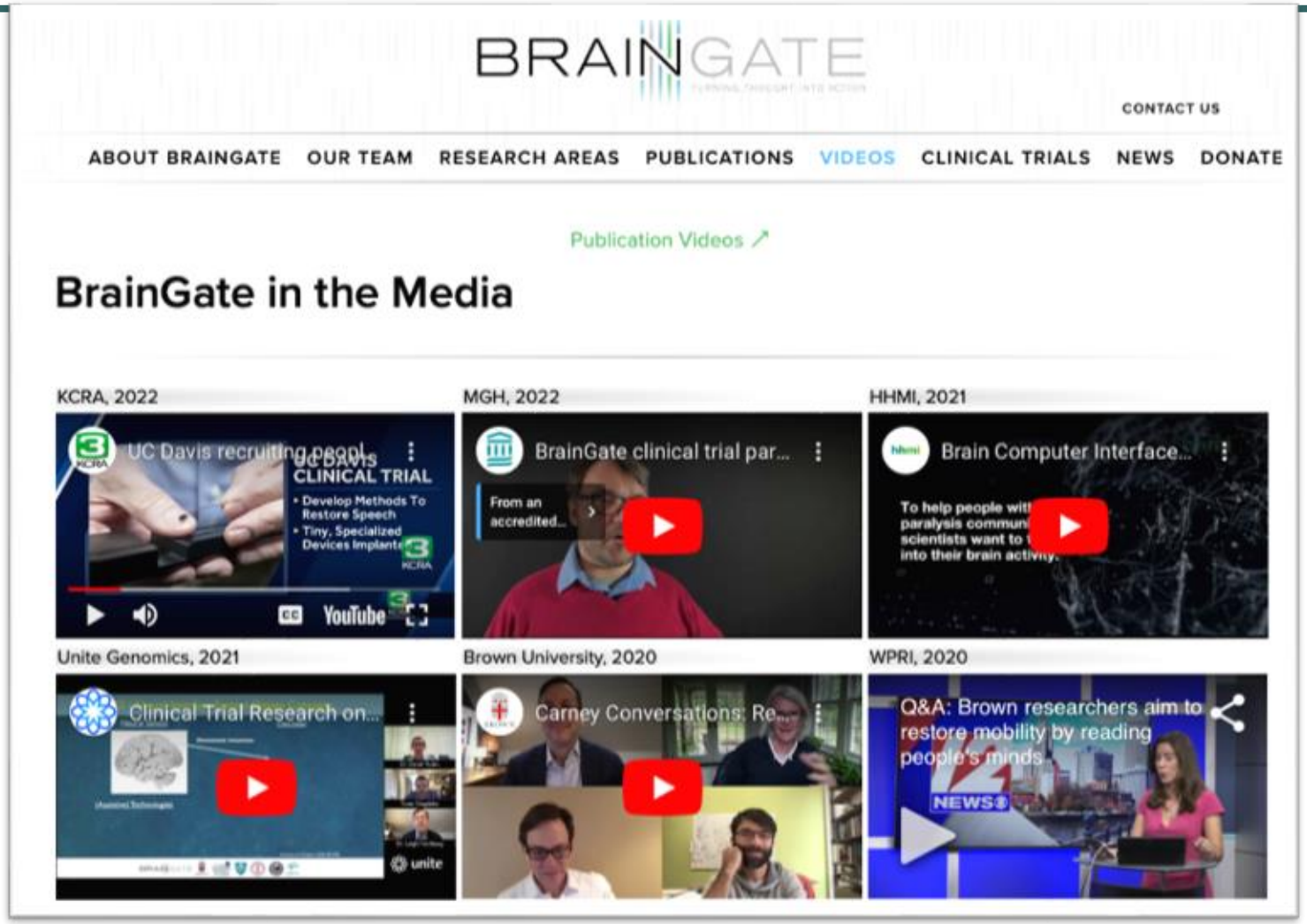
BCI 主要應用範疇：產品、健康、娛樂

Table 2. Domains of application

Category	Domain	Description	Studies
Productivity	Control	Control of real or virtual objects like robots or quad-copters	[27–30, 110, 117
	User experience	Exploring user experience of VR/AR with BCI-based interactions	[5, 74, 7
	IoT	Interaction with smart home appliances	[93, 94,
	Communication	BCI-based interaction between multiple users	[109]
	Simulation	Simulation of real-life or dream scenarios in VR	[48, 61]
	Safety	Explorations in safety	[26]
	Mind training	Systems for mind training	[65]
	Architecture	Neuroarchitecture systems	[13]
	Research	Advancing research in the field	[31, 32, 123]
Health	Rehabilitation	Rehabilitation of patients with motor impairments	[2, 84, 137
	Surgery	Interventions for surgery procedures	[12
	Patient monitoring	Aid medical personnel in monitoring patients	[8]
	Therapy	Therapy systems	[15
	Communication	Alternative communication system for disabled individuals	[38
Entertainment	Games	Explorations of BCI usage in VR gaming	[16
	Art	Creation and appreciation of art works	[79
	Meditation	Meditation systems	[64
	Virtual concert	BCI augmented VR concert applications	[53

C Nwagu, et al. Proceedings of the ACM on Human-Computer Intersaction, 2023

BCI 相關臨床研究成果



https://www.braingate.org

現代 AI 機器人特質 – 專才



運用 “五官” 、具有特定功能，例如

1. 機器手臂：手術醫療、生產線品管
2. 大型語言模型：衛教、服務業
3. 電腦視覺與語音辨識：居家照護



手術機器手臂（醫療）



LOVOT 機器人協助照顧失智症患者（長照）



類家具陪伴機器人（跌倒偵測、病床照護）



掃地機器人（居家生活）



SONY 機器狗（居家陪伴）

結合不同工具，發揮最大效益

不同功能獨立模型
需要大量訓練資料

一個獨立大語言模型
適用各式自然語言情境

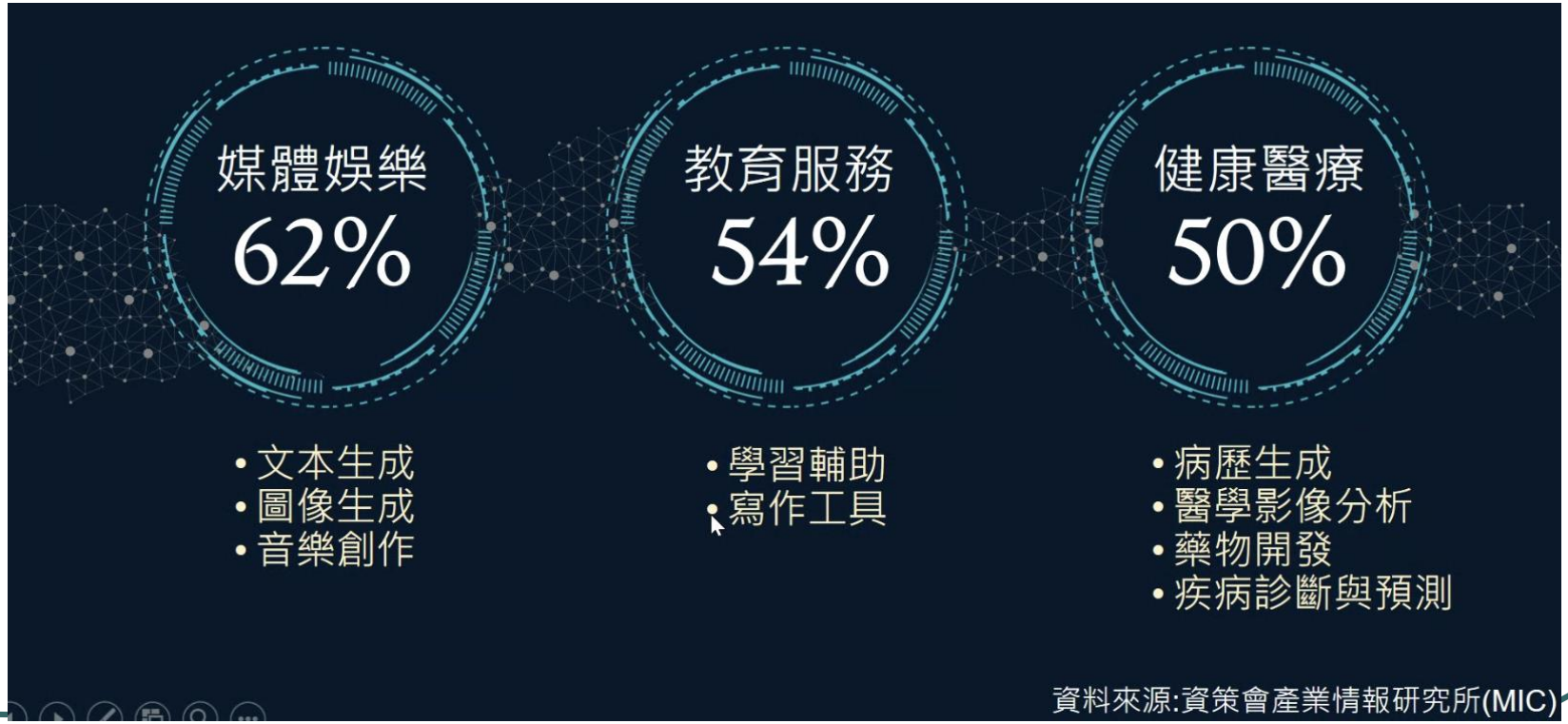


提高效率
改善效果

提高多樣性
增加創造力

不要輸在起跑點 ~
了解生成式AI 是未來必要的基礎能力

在各類領域中，生成式AI應用的比例



How can we make use of these technologies for better life?

Ans: 人工智慧 + 工人智慧 才是王道

透過專家與機器反覆對話，「教導」機器學習人類技能

人類的最終確認
- 答案有可能是AI的幻覺
- 確認回答的準確性

提供正確的詠唱咒語
- 條列式的給予清楚的命令
- 盡量縮限AI回答的範圍、讓ChatGPT追問



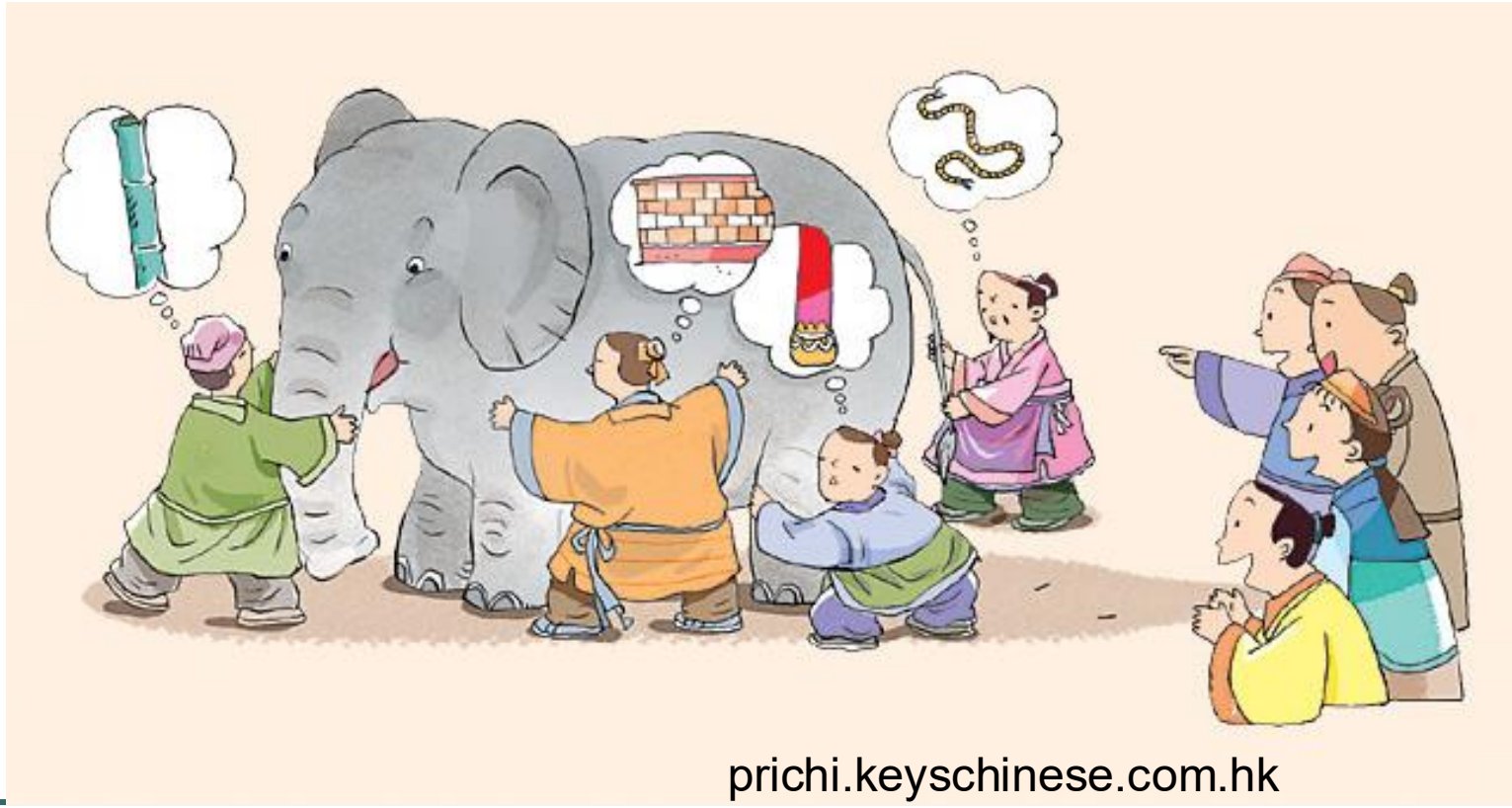
適當運用生成式模型

Tips for data analyses

- Hypothesis-driven vs. data-driven study
  - Region-of-interest (ROI) analysis
  - Whole brain analysis in a voxel-wise manner
- Anatomical vs. functional data
- Within-subject vs. between-subject design
- Choosing appropriate statistical analysis methods
- Be a good story teller!

瞎子摸象

- *The job of a scientist is to listen carefully to nature, not to tell nature how to behave*.
  - By Richard Feynman



Recap

- Three major components for Neuroinfomatics
  - neuroscience data, neuroimaging tools, math. modeling
- Neuroimaging tools
  - MRI, EEG/MEG, fNIRS, TMS,
- Different spatial-temporal information
  - activation regions and timing
  - Connectivity
- Applications

Reading list

- R1. Chiquito Joaqium Crasto (2007), Neuroinformatics, Humana Press Inc.
- R2. Pawel Herman, Introduction to Neuroinformatics, (KTH Royal Institute of Technology, Sweden)
- R3. Redolfi et al. (2022) Italian, European, and international neuroinformatics efforts: An overview, EJN.